

1. 3번	2. 3번	3. 1번	4. 4번	5. 1번
6. 2번	7. 3번	8. 4번	9. 1번	10. 2번
11. 2번	12. 4번	13. 5번	14. 5번	15. 1번
16. 1번	17. 2번	18. 5번	19. 1번	20. 3번
21. 1번	22. 2번	23. 4번	24. 2번	25. 4번
26. 4번	27. 4번	28. 5번	29. 2번	30. 2번

$$\begin{aligned}
 1. \quad k &= \int_0^{\frac{\pi}{4}} \sqrt{1 + \tan^2 x} \, dx = \int_0^{\frac{\pi}{4}} \sqrt{\sec^2 x} \, dx \\
 &= \int_0^{\frac{\pi}{4}} \sec x \, dx = [\ln(\sec x + \tan x)]_0^{\frac{\pi}{4}} \\
 &= \ln(\sqrt{2} + 1) \\
 \therefore e^k - \frac{1}{2} &= e^{\ln(\sqrt{2} + 1)} - \frac{1}{2} = \sqrt{2} + \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad &\int_2^e \left(\frac{1}{x \ln x} \right) dx + [x \ln x - x]_2^e \\
 &= \int_{\ln 2}^1 \frac{1}{t} dt + [x \ln x - x]_2^e \quad (\because \ln x = t) \\
 &= [\ln t]_{\ln 2}^1 + (-2 \ln 2 + 2) \\
 &= -\ln \ln 2 + (-2 \ln 2 + 2) \\
 &= -\ln 4 \ln 2 + 2
 \end{aligned}$$

$$\begin{aligned}
 3. \quad &\int_{\ln 2}^{\ln 5} \frac{e^x}{2e^x - 1} \, dx = \frac{1}{2} [\ln(2e^x - 1)]_{\ln 2}^{\ln 5} \\
 &= \frac{1}{2} (\ln 9 - \ln 3) = \frac{1}{2} \ln 3
 \end{aligned}$$

$$\begin{aligned}
 4. \quad &\int_{-1}^1 3x^2(x^3 + 1)^{\frac{1}{2}} \, dx = \frac{2}{3} [(x^3 + 1)^{\frac{3}{2}}]_{-1}^1 \quad (\because \text{전체적분}) \\
 &= \frac{4\sqrt{2}}{3}
 \end{aligned}$$

$$\begin{aligned}
 5. \quad &\int_0^{\pi/2} \sin 2x \cdot e^{2\sin^2 x} \, dx \\
 &= \int_0^{\pi/2} 2 \sin x \cos x \cdot e^{2\sin^2 x} \, dx \quad (\because 2\text{배각공식}) \\
 &= \frac{1}{2} [e^{2\sin^2 x}]_0^{\pi/2} \quad (\because \text{전체적분}) \\
 &= \frac{1}{2} (e^2 - 1)
 \end{aligned}$$

$$\begin{aligned}
 6. \quad &\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16 - x^2}} \, dx = \int_2^4 16 - t^2 \, dt \quad (\because \sqrt{16 - x^2} = t) \\
 &= \left[16t - \frac{1}{3} t^3 \right]_2^4 = \frac{40}{3}
 \end{aligned}$$

[다른 풀이1]

$$\begin{aligned}
 &\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16 - x^2}} \, dx = \int_0^{\frac{\pi}{3}} 64 \sin^3 t \, dt \quad (x = 4 \sin t) \\
 &= \int_0^{\frac{\pi}{3}} 64 \sin t (1 - \cos^2 t) \, dt \\
 &= \left[-64 \cos t + \frac{64}{3} \cos^3 t \right]_0^{\frac{\pi}{3}} \\
 &= \frac{40}{3}
 \end{aligned}$$

[다른 풀이2]

$$\begin{aligned}
 &\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16 - x^2}} \, dx = \int_0^{\frac{\pi}{3}} 64 \sin^3 t \, dt \text{에서} \\
 &\text{삼각함수 3배각 공식 } \sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha \text{를 활용하면} \\
 &\int_0^{\frac{\pi}{3}} 64 \sin^3 t \, dt = \int_0^{\frac{\pi}{3}} 48 \sin t - 16 \sin 3t \, dt \\
 &= \left[-48 \cos t + \frac{16}{3} \cos 3t \right]_0^{\frac{\pi}{3}} \\
 &= \frac{40}{3}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad &\int_{\frac{1}{2}}^1 \frac{x}{\sqrt{1 - (2x^2)^2}} \, dx = \int_{\frac{1}{2}}^1 \frac{1}{\sqrt{1 - t^2}} \times \frac{1}{4} dt \quad (\because 2x^2 = t) \\
 &= \frac{1}{4} [\sin^{-1} t]_{\frac{1}{2}}^1 = \frac{\pi}{12}
 \end{aligned}$$

$$\begin{aligned}
 8. \quad &\int_{-2}^2 \sqrt{4 - x^2} \, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4 \cos^2 \theta \, d\theta \quad (\because x = 2 \sin \theta) \\
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2(1 + \cos 2\theta) \, d\theta = [2\theta + \sin 2\theta]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 2\pi
 \end{aligned}$$

[참고]

$y = \sqrt{4 - x^2}$ 은 반지름이 2인 반원이므로 $\int_{-2}^2 \sqrt{4 - x^2} \, dx$ 은 반원의 넓이를 이용하여 $\pi \times 2^2 \times \frac{1}{2} = 2\pi$ 로 풀어도 좋다.

$$\begin{aligned}
 9. \quad &\int_0^{1/2} \frac{x^2}{\sqrt{1 - x^2}} \, dx = \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{\cos \theta} \cos \theta \, d\theta \quad (\because x = \sin \theta) \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{6}} 1 - \cos 2\theta \, d\theta = \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}} \\
 &= \frac{1}{2} \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right)
 \end{aligned}$$

$$\begin{aligned}
 10. \quad &\int \frac{1}{\sqrt{x^2 - 16}} \, dx = \cosh^{-1} \frac{x}{4} + C \\
 &= \ln \left(\frac{x}{4} + \sqrt{\frac{x^2}{16} - 1} \right) + C = \ln \left(\frac{x + \sqrt{x^2 - 16}}{4} \right) + C
 \end{aligned}$$

[다른 풀이]

$$\begin{aligned} \int \frac{1}{\sqrt{x^2-16}} dx &= \int \frac{1}{4 \tan t} 4 \sec t \tan t dt \quad (\because x = 4 \sec t) \\ &= \int 4 \sec t dt \\ &= \ln(\sec t + \tan t) + C \\ \text{이때, } x &= 4 \sec t \text{ 이므로 } 1 + \tan^2 t = \sec^2 t \text{ 에서} \\ \tan t &= \frac{\sqrt{x^2-16}}{4} \text{ 이다.} \end{aligned}$$

$$\text{즉, } \ln(\sec t + \tan t) + C = \ln\left(\frac{x + \sqrt{x^2-16}}{4}\right) + C$$

$$\begin{aligned} 11. \int_1^2 \frac{1}{x^2 \sqrt{x^2+4}} dx &= \int_{\tan^{-1} \frac{1}{2}}^{\frac{\pi}{4}} \frac{\cos \theta}{4 \sin^2 \theta} d\theta \quad (\because x = 2 \tan \theta) \\ &= \int_{\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{2}}} \frac{1}{4 t^2} dt = -\left[\frac{1}{4 t}\right]_{\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{2}}} = \frac{\sqrt{5}-\sqrt{2}}{4} \end{aligned}$$

$$\begin{aligned} 12. \int_1^2 \frac{(\ln x)^2}{2} dx &= \frac{1}{2} \left[x(\ln x)^2 - \int 2 \ln x dx \right]_1^2 \quad (\because \text{부분적분}) \\ &= \frac{1}{2} [x(\ln x)^2 - 2(x \ln x - x)]_1^2 = (\ln 2)^2 - 2 \ln 2 + 1 \end{aligned}$$

$$\begin{aligned} 13. \int_0^{\frac{1}{2}} \cos^{-1} x dx &= \left[x \cos^{-1} x - \int \frac{-x}{\sqrt{1-x^2}} dx \right]_0^{\frac{1}{2}} \quad (\because \text{부분적분}) \\ &= [x \cos^{-1} x - \sqrt{1-x^2}]_0^{\frac{1}{2}} = \frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \end{aligned}$$

[다른 풀이]

$\cos^{-1} x = t$ 로 치환하면 $\cos t = x$ 이고,

$$\begin{aligned} \int_0^{\frac{1}{2}} \cos^{-1} x dx &= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} -t \sin t dt = -\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} t \sin t dt \\ &= [-t \cos t + \sin t]_{\frac{\pi}{3}}^{\frac{\pi}{2}} = \frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \quad (\because \text{빠른 부분적분}) \end{aligned}$$

$$\begin{aligned} 14. \int_0^{\frac{1}{2}} \tan^{-1}(2x) dx &= x \tan^{-1}(2x) - \int \frac{2x}{1+4x^2} dx \quad (\because \text{부분적분}) \\ &= \left[x \tan^{-1}(2x) - \frac{1}{4} \ln(1+4x^2) \right]_0^{\frac{1}{2}} = \frac{\pi}{8} - \frac{1}{4} \ln 2 \end{aligned}$$

[다른 풀이]

$\tan^{-1}(2x) = t$ 로 치환하면 $x = \frac{1}{2} \tan t$ 이고

$$\begin{aligned} \int_0^{\frac{1}{2}} \tan^{-1}(2x) dx &= \int_0^{\frac{\pi}{4}} \frac{1}{2} t \sec^2 t dt \\ &= \frac{1}{2} [t \tan t + \ln \cos t]_0^{\frac{\pi}{4}} = \frac{\pi}{4} - \frac{1}{2} \ln 2 \end{aligned}$$

$$\begin{aligned} 15. \int_0^1 (2x-1)e^x dx &= e^x(2x-1) - \int 2e^x dx \quad (\because \text{부분적분}) \\ &= [e^x(2x-1) - 2e^x]_0^1 = -e + 3 \end{aligned}$$

$$\begin{aligned} 16. \int_0^1 x^3 e^{x^2} dx &= \frac{1}{2} \int_0^1 t e^t dt \quad (\because x^2 = t) \\ &= \frac{1}{2} [te^t - \int e^t dt]_0^1 \quad (\because \text{부분적분}) \\ &= \frac{1}{2} [te^t - e^t]_0^1 = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} 17. \int_{\pi}^0 2e^x \cos x dx &= 2 \left[\frac{e^x}{2} (\cos x + \sin x) \right]_{\pi}^0 \quad (\because \text{오일러 적분}) \\ &= 1 + e^{\pi} \end{aligned}$$

[다른 풀이]

$$\begin{aligned} \int_{\pi}^0 e^x \cos x dx &= [e^x \cos x]_{\pi}^0 + \int_{\pi}^0 e^x \sin x dx \quad (\because \text{부분적분}) \\ &= 1 + e^{\pi} + [e^x \sin x]_{\pi}^0 - \int_{\pi}^0 e^x \cos x dx \quad (\because \text{부분적분}) \\ &= 1 + e^{\pi} - \int_{\pi}^0 e^x \cos x dx \text{ 에서} \\ \int_{\pi}^0 e^x \cos x dx &= \frac{e^{\pi} + 1}{2} \text{ 이다.} \end{aligned}$$

$$\begin{aligned} 18. \int_0^1 \frac{5x+7}{x^2+3x+2} dx &= \int_0^1 \frac{5x+7}{(x+1)(x+2)} dx \\ &= \int_0^1 \frac{2}{x+1} + \frac{3}{x+2} dx \\ &= [2 \ln(x+1) + 3 \ln(x+2)]_0^1 \\ &= 2 \ln 2 + 3 \ln 3 - 3 \ln 2 \\ &= 3 \ln 3 - \ln 2 \end{aligned}$$

$$\begin{aligned} 19. \int_0^1 x \sqrt{1-x} dx &= \int_0^1 2t^2 - 2t^4 dt \quad (\because \sqrt{1-x} = t) \\ &= \left[\frac{2}{3} t^3 - \frac{2}{5} t^5 \right]_0^1 = \frac{4}{15} \end{aligned}$$

$$\begin{aligned} 20. \int_0^{\pi^2} \cos \sqrt{t} dt &= \int_0^{\pi} 2x \cos x dx \quad (\because \sqrt{t} = x) \\ &= \left[2x \sin x - \int 2 \sin x dx \right]_0^{\pi} \quad (\because \text{부분적분}) \\ &= [2x \sin x + 2 \cos x]_0^{\pi} = -4 \end{aligned}$$

$$\begin{aligned} 21. \int_0^1 \arctan \sqrt{x} dx &= \int_0^1 2t \tan^{-1} t dt \quad (\because \sqrt{x} = t) \\ &= \left[t^2 \tan^{-1} t - \int \frac{t^2}{1+t^2} dt \right]_0^1 \quad (\because \text{부분적분}) \\ &= \left[t^2 \tan^{-1} t - \int 1 - \frac{1}{1+t^2} dt \right]_0^1 \\ &= [t^2 \tan^{-1} t - (t - \tan^{-1} t)]_0^1 = \frac{\pi}{2} - 1 \end{aligned}$$

$$\begin{aligned} 22. \int_0^3 \frac{x}{\sqrt{x+1}} dx &= \int_1^2 \frac{t^2-1}{t} \times 2t dt \quad (\because \sqrt{x+1} = t) \\ &= 2 \left[\frac{1}{3} t^3 - t \right]_1^2 = \frac{8}{3} \end{aligned}$$

$$\begin{aligned} 23. \int_2^3 \frac{9x+2}{\sqrt{3x-5}} dx &= \int_1^2 \frac{3(t^2+5)+2}{t} \times \frac{2}{3} t dt \quad (\because \sqrt{3x-5} = t) \\ &= \left[\frac{2}{3} t^3 + \frac{34}{3} t \right]_1^2 = 16 \end{aligned}$$

$$\begin{aligned} 24. \int_0^1 \frac{x}{\sqrt{3-2x-x^2}} dx &= \int_0^1 \frac{x}{\sqrt{4-(x+1)^2}} dx \\ &= \int_1^2 \frac{t-1}{\sqrt{4-t^2}} dt \quad (\because x+1 = t) \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{2 \sin \theta - 1}{2 \cos \theta} 2 \cos \theta d\theta \quad (\because t = 2 \sin \theta) \\ &= [-2 \cos \theta - \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = -\frac{\pi}{3} + \sqrt{3} \end{aligned}$$

$$25. \int_0^{\ln \sqrt{3}} \frac{1}{e^x + e^{-x}} dx = \int_1^{\sqrt{3}} \frac{1}{t + \frac{1}{t}} \times \frac{1}{t} dt \quad (\because e^x = t)$$

$$= \int_1^{\sqrt{3}} \frac{1}{t^2 + 1} dt = [\tan^{-1} t]_1^{\sqrt{3}} = \frac{\pi}{12}$$

$$26. \int_0^{\frac{\pi}{2}} \frac{5}{4\sin x + 3\cos x} dx \quad \left(\because \tan \frac{x}{2} = t \right)$$

$$= \int_0^1 \frac{5}{4 \times \frac{2t}{1+t^2} + 3 \times \frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{10}{8t+3-3t^2} dt = \int_0^1 \frac{10}{(3-t)(1+3t)} dt$$

$$= \int_0^1 \frac{1}{3-t} + \frac{3}{1+3t} dt = [-\ln(3-t) + \ln(1+3t)]_0^1 = \ln 6$$

[다른 풀이]

삼각함수의 합성을 활용하면

$$4\sin x + 3\cos x = 5\sin(x+\alpha) \quad \left(\tan \alpha = \frac{3}{4} \right) \text{이므로}$$

$$\int_0^{\frac{\pi}{2}} \frac{5}{4\sin x + 3\cos x} dx = \int_0^{\frac{\pi}{2}} \csc(x+\alpha) dx$$

$$= [-\ln|\csc(x+\alpha) + \cot(x+\alpha)|]_0^{\frac{\pi}{2}}$$

$$= -\ln|\sec \alpha - \tan \alpha| + \ln|\csc \alpha + \cot \alpha|$$

$$= -\ln\left(\frac{5}{4} - \frac{3}{4}\right) + \ln\left(\frac{5}{3} + \frac{4}{3}\right)$$

$$= \ln 2 + \ln 3 = \ln 6$$

[참고]

$$\csc\left(\frac{\pi}{2} + \alpha\right) = \frac{1}{\sin\left(\frac{\pi}{2} + \alpha\right)} = \frac{1}{\cos \alpha} = \sec \alpha$$

$$\cot\left(\frac{\pi}{2} + \alpha\right) = -\tan \alpha \text{이다.}$$

$$27. \int_0^{\frac{\pi}{2}} \frac{1}{2 + \cos 2\theta} d\theta \quad (\because \tan \theta = t)$$

$$= \int_0^{\infty} \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{1}{1+t^2} dt$$

$$= \int_0^{\infty} \frac{1}{t^2 + 3} dt$$

$$= \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^{\infty} = \frac{\pi}{2\sqrt{3}}$$

[다른 풀이]

$$\int_0^{\frac{\pi}{2}} \frac{1}{2 + \cos 2\theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{2 + 2\cos^2 \theta - 1} d\theta \quad (\because \text{반각공식})$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1 + 2\cos^2 \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{\cos^2 \theta + \sin^2 \theta + 2\cos^2 \theta} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{3\cos^2 \theta + \sin^2 \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{\sec^2 \theta}{3 + \tan^2 \theta} d\theta$$

$$= \int_0^{\infty} \frac{1}{3 + t^2} dt \quad (\because \tan \theta = t)$$

$$= \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^{\infty} = \frac{\pi}{2\sqrt{3}}$$

$$28. \int_0^{\frac{1}{2}} \sin^{-1} x dx + \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

$$= [x \sin^{-1} x]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

$$+ \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx \quad (\because \text{부분적분법})$$

$$= \frac{\pi}{12}$$

$$29. \int_0^1 \sqrt{1+x^2} dx = \int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta \quad (\because x = \tan \theta)$$

$$= \frac{1}{2} [\sec \theta \tan \theta + \ln(\sec \theta + \tan \theta)]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} [\sqrt{2} + \ln(\sqrt{2} + 1)]$$

$$30. \int_3^4 \frac{1}{x^2 - 4} dx = \left[\frac{1}{4} \ln \left(\frac{x-2}{x+2} \right) \right]_3^4$$

$$= \frac{1}{4} \ln \left(\frac{1}{3} \right) - \frac{1}{4} \ln \left(\frac{1}{5} \right) = \frac{1}{4} \ln \frac{5}{3}$$

장황수학

1. 4번	2. 3번	3. 3번	4. 2번	5. 4번
6. 3번	7. 1번	8. 4번	9. 1번	10. 3번

1. $\tan^{-1}x = \alpha$ 라 하면 $\tan\alpha = x$, $\cos\alpha = \frac{1}{\sqrt{1+x^2}}$ 이다.

$$\begin{aligned} & \int_0^1 \sin\left(\tan^{-1}x + \cos^{-1}\frac{1}{\sqrt{1+x^2}}\right) dx \\ &= \int_0^1 \sin(2\alpha) dx = 2 \int_0^1 \sin(\alpha) \cos(\alpha) dx \\ &= 2 \int_0^1 \frac{x}{\sqrt{1+x^2}} \times \frac{1}{\sqrt{1+x^2}} dx = 2 \int_0^1 \frac{x}{1+x^2} dx \\ &= [\ln(1+x^2)]_0^1 = \ln 2 \end{aligned}$$

2. $f(2) - f(0) = \int_0^2 f'(x) dx = \int_0^2 \frac{1}{\sqrt{x^2+4}} dx = \left[\sinh^{-1} \frac{x}{2} \right]_0^2$
 $= \sinh^{-1} 1 = \ln(1 + \sqrt{2})$
 즉, $f(2) = \ln(1 + \sqrt{2})$

3. $\int_{\frac{1}{2}}^1 (\cos^{-1}x)^2 dx = - \int_{\frac{\pi}{3}}^0 t^2 \sin t dt \quad (\because \cos^{-1}x = t)$
 $= - \left[-t^2 \cos t + 2t \sin t + 2 \cos t \right]_{\frac{\pi}{3}}^0 = \frac{\sqrt{3}\pi}{3} - 1 - \frac{\pi^2}{18}$

4. $\int_0^\pi e^x \sin^2 x dx = \int_0^\pi e^x \left(\frac{1 - \cos 2x}{2} \right) dx$
 $= \frac{1}{2} \int_0^\pi e^x - e^x \cos 2x dx$
 $= \frac{1}{2} [e^x]_0^\pi - \frac{1}{2} \left[\frac{e^x}{5} (\cos 2x + 2 \sin 2x) \right]_0^\pi$
 $= \frac{1}{2} (e^\pi - 1) - \frac{1}{10} (e^\pi - 1) = \frac{2}{5} e^\pi - \frac{2}{5}$

5. $\int_1^{e^{\frac{\pi}{2}}} \sin(\ln x) dx \quad (\because \ln x = t)$
 $= \int_0^{\frac{\pi}{2}} e^t \sin t dt = \left[\frac{e^t}{2} (\sin t - \cos t) \right]_0^{\frac{\pi}{2}} = \frac{e^{\frac{\pi}{2}}}{2} + \frac{1}{2}$

6. $\int_0^1 \left(2x \sin \frac{1}{x^2} - \frac{2}{x} \cos \frac{1}{x^2} \right) dx = \left[x^2 \sin \frac{1}{x^2} \right]_0^1 = \sin 1$

[다른 풀이]

$$\begin{aligned} & \int_0^1 \left(2x \sin \frac{1}{x^2} - \frac{2}{x} \cos \frac{1}{x^2} \right) dx \\ &= \int_0^1 2x \sin \frac{1}{x^2} dx - \int_0^1 \frac{2}{x} \cos \frac{1}{x^2} dx \\ &= \left[x^2 \sin \frac{1}{x} \right]_0^1 + \int_0^1 \frac{2}{x} \cos \frac{1}{x^2} dx - \int_0^1 \frac{2}{x} \cos \frac{1}{x^2} dx \quad (\text{부분 적분}) \\ &= \left[x^2 \sin \frac{1}{x} \right]_0^1 = \sin 1 \end{aligned}$$

7. $\int \frac{15x^2}{(x^2+1)(x^2+4)} dx = \int \frac{-5}{x^2+1} + \frac{20}{x^2+4} dx$
 $= -5 \tan^{-1}x + 10 \tan^{-1} \frac{x}{2} + C$

8. $\int \frac{1+2x^2}{x(1+x^2)^2} dx = \int \frac{x^2+1+x^2}{x(1+x^2)^2} dx = \int \frac{x}{(1+x^2)^2} + \frac{1}{x(1+x^2)} dx$
 $= \frac{-\frac{1}{2}}{1+x^2} + \int \frac{1}{x} + \frac{-x}{1+x^2} dx = \frac{1}{-2(1+x^2)} + \ln x - \frac{1}{2} \ln(1+x^2) + C$

[다른 풀이]

$$\int \frac{1+2x^2}{x(1+x^2)^2} dx = \int \frac{1}{x} + \frac{ax+b}{1+x^2} + \frac{cx+d}{(1+x^2)^2} dx$$

이때, 분자를 비교해보면

$$1+2x^2 = (1+x^2)^2 + (ax+b)x(1+x^2) + (cx+d)x$$

$$\Leftrightarrow 1+a=0, b=0, a+c=0, b+d=0$$

이므로 $a=-1, b=0, c=1, d=0$ 이다.

$$\int \frac{1}{x} + \frac{-x}{1+x^2} + \frac{x}{(1+x^2)^2} dx = \ln x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2(1+x^2)} + C$$

9. $\int \frac{1}{x^3+1} dx = \int \frac{1}{(x+1)(x^2-x+1)} dx$
 $= \int \frac{\frac{1}{3}}{x+1} + \frac{-\frac{1}{3}x + \frac{2}{3}}{x^2-x+1} dx$
 $= \int \frac{\frac{1}{3}}{x+1} + \frac{-\frac{1}{6}(2x-1) + \frac{1}{2}}{x^2-x+1} dx$
 $= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2-x+1) + \int \frac{\frac{1}{2}}{x^2-x+1} dx$
 $= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2-x+1) + \frac{1}{2} \int \frac{1}{\left(x-\frac{1}{2}\right)^2 + \frac{3}{4}} dx$
 $= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2-x+1) + \frac{1}{2} \times \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}} + C$
 $= \frac{1}{3} \ln(x+1) - \frac{1}{6} \ln(x^2-x+1) + \frac{1}{\sqrt{3}} \tan^{-1} \frac{2x-1}{\sqrt{3}} + C$

10. $\int \frac{f(x)}{x^2(x+1)^3} dx = \int \frac{a_1}{x} + \frac{a_2}{x^2} + \frac{a_3}{x+1} + \frac{a_4}{(x+1)^2} + \frac{a_5}{(x+1)^3} dx$
 의 부정적분이 유리함수가 되려면 $a_1 = a_3 = 0$ 이다.

$$\text{또한 } \int \frac{A}{x^2} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3} dx$$

$$= \int \frac{A(x+1)^3 + Bx^2(x+1) + Cx^2}{x^2(x+1)^3} dx \text{ 이므로}$$

$$f(x) = ax^2 + bx + c = A(x+1)^3 + Bx^2(x+1) + Cx^2 \text{ 이다.}$$

$$f(0) = 1 \text{ 이므로 } c = 1 \text{ 이다.}$$

$$\text{이때, 계수 비교를 하면 } A+B=0, 3A+B+C=a,$$

$$3A=b, A=c=1 \text{ 이다.}$$

$$A=1, B=-1, b=3, c=1 \text{ 이므로 } f(x) = ax^2 + 3x + 1 \text{ 이다.}$$

$$\text{즉, } f'(x) = 2ax + 3 \text{ 이므로 } f'(0) = 3 \text{ 이다.}$$

1. 4번	2. 4번	3. 1번	4. 1번	5. 2번
6. 4번	7. 4번	8. 1번	9. 18	10. 3번
11. 1번				

$$\begin{aligned}
 1. \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc^4 x dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc^2 x \cdot \csc^2 x dx \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (1 + \cot^2 x) \csc^2 x dx = \int_1^0 -(1+t^2) dt \quad (\because \cot x = t) \\
 &= -\left[t + \frac{1}{3}t^3\right]_1^0 = \frac{4}{3}
 \end{aligned}$$

$$\begin{aligned}
 2. \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} [(\tan x)^{-2} \sec^4 x + \tan^3 x (\sec x)^{-1}] dx \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left((\tan x)^{-2} (1 + \tan^2 x) \sec^2 x + \frac{\sin^3 x \times \cos x}{\cos^3 x} \right) dx \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{\tan^2 x} + 1 \right) \sec^2 x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{1}{\cos^2 x} - 1 \right) \sin x dx \\
 &= \int_1^{\sqrt{3}} \left(\frac{1}{t^2} + 1 \right) dt + \int_{\frac{1}{2}}^1 \left(\frac{1}{u^2} - 1 \right) du \quad (\because \tan x = t, \cos x = u) \\
 &= \left[-\frac{1}{t} + t \right]_1^{\sqrt{3}} + \left[-\frac{1}{u} - u \right]_{\frac{1}{2}}^1 \\
 &= \frac{2}{3}\sqrt{3} - \frac{3}{2}\sqrt{2} + \frac{5}{2}
 \end{aligned}$$

$$\begin{aligned}
 3. \int_{\frac{1}{2}}^1 \sqrt{\frac{1}{x} - 1} dx &= \int_{\frac{1}{2}}^1 \sqrt{\frac{1-x}{x}} dx = \int_{\frac{1}{2}}^1 \frac{\sqrt{1-x}}{\sqrt{x}} dx \\
 &= \int_{\frac{1}{\sqrt{2}}}^1 \frac{\sqrt{1-t^2}}{t} 2t dt \quad (\because \sqrt{x} = t) \\
 &= 2 \int_{\frac{1}{\sqrt{2}}}^1 \sqrt{1-t^2} dt = 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 \theta d\theta \quad (\because t = \sin \theta) \\
 &= 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1 + \cos 2\theta}{2} d\theta = \left[\theta + \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 4. \int_0^2 \tan^{-1} \frac{2-x}{1+2x} dx &= \int_0^2 \tan^{-1} 2 - \tan^{-1} x dx \\
 &= 2 \tan^{-1} 2 - \int_0^2 \tan^{-1} x dx \\
 &= 2 \tan^{-1} 2 - \left[x \tan^{-1} x - \int \frac{x}{1+x^2} dx \right]_0^2 \quad (\because \text{부분적분}) \\
 &= 2 \tan^{-1} 2 - \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^2 = \frac{1}{2} \ln 5
 \end{aligned}$$

$$\begin{aligned}
 5. \int_0^{2a} x \sin^{-1} \left(\frac{\sqrt{2a-x}}{2\sqrt{a}} \right) dx \\
 &= \left[\frac{1}{2} x^2 \sin^{-1} \left(\frac{\sqrt{2a-x}}{2\sqrt{a}} \right) \right]_0^{2a} + \frac{1}{4} \int_0^{2a} \frac{x^2}{\sqrt{4a^2-x^2}} dx \quad (\because \text{부분적분}) \\
 &= \frac{1}{4} \int_0^{2a} \frac{x^2}{\sqrt{4a^2-x^2}} dx = a^2 \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta \quad (\because x = 2a \sin \theta) \\
 &= \frac{a^2}{2} \int_0^{\frac{\pi}{2}} 1 - \cos 2\theta d\theta
 \end{aligned}$$

$$= \frac{a^2}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4} a^2$$

[다른 풀이]

$$\sin^{-1} \left(\frac{\sqrt{2a-x}}{2\sqrt{a}} \right) = \theta \text{ 라 하면 } \sin \theta = \frac{\sqrt{2a-x}}{2\sqrt{a}} \text{ 이고}$$

양변을 제곱하면 $4a \sin^2 \theta = 2a - x$

$$\Rightarrow x = 2a - 4a \sin^2 \theta = 2a - 4a(1 - \cos^2 \theta)$$

$$= 4a \cos^2 \theta - 2a = 2a \cos 2\theta \text{ 이므로}$$

$$\int_0^{2a} x \sin^{-1} \left(\frac{\sqrt{2a-x}}{2\sqrt{a}} \right) dx$$

$$= \int_{\frac{\pi}{4}}^0 2a \cos 2\theta \cdot (-4a \sin 2\theta) d\theta$$

$$= 4a^2 \int_0^{\frac{\pi}{4}} \theta \sin 4\theta d\theta$$

$$= 4a^2 \left[-\frac{\theta}{4} \cos 4\theta + \frac{1}{16} \sin 4\theta \right]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{4} a^2 \text{ 이다.}$$

6. $(e^x f(x))' = e^x (f(x) + f'(x))$ 임을 이용하자.

$$\int_0^1 \frac{x^3 e^{x^2}}{(x^2+1)^2} dx = \int_0^1 \frac{x x^2 e^{x^2}}{(x^2+1)^2} dx \quad (\because x^2+1=t)$$

$$= \int_1^2 \frac{(t-1)e^{t-1}}{t^2} \frac{1}{2} dt$$

$$= \frac{1}{2e} \int_1^2 \frac{(t-1)e^t}{t^2} dt$$

$$= \frac{1}{2e} \int_1^2 e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt$$

$$= \frac{1}{2e} \left[e^t \frac{1}{t} \right]_1^2 \quad (\because \text{곱미분})$$

$$= \frac{1}{4} e - \frac{1}{2}$$

$$\begin{aligned}
 7. \int_1^\infty \frac{2x^3+3x^2+3}{x^3(x^2+1)} dx &= \int_1^\infty \frac{2x^3+3(x^2+1)}{x^3(x^2+1)} dx \\
 &= \int_1^\infty \frac{2}{x^2+1} + \frac{3}{x^3} dx = \left[2 \tan^{-1} x - \frac{3}{2x^2} \right]_1^\infty = \frac{\pi}{2} + \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 8. \int_0^1 \frac{9}{1+x^3} dx &= \int_0^1 \frac{9}{(x+1)(x^2-x+1)} dx \\
 &= \int_0^1 \left(\frac{3}{x+1} + \frac{-3x+6}{x^2-x+1} \right) dx \\
 &= [3 \ln(1+x)]_0^1 - \frac{3}{2} \int_0^1 \frac{(2x-1)-3}{x^2-x+1} dx \\
 &= 3 \ln 2 - \frac{3}{2} \int_0^1 \frac{2x-1}{x^2-x+1} dx + \frac{9}{2} \int_0^1 \frac{1}{x^2-x+1} dx \\
 &= 3 \ln 2 - \frac{3}{2} [\ln(x^2-x+1)]_0^1 + \frac{9}{2} \int_0^1 \frac{1}{\left(x-\frac{1}{2}\right)^2 + \frac{3}{4}} dx \\
 &= 3 \ln 2 + 3\sqrt{3} \left[\tan^{-1} \left(\frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) \right]_0^1 \\
 &= \ln 8 + \sqrt{3} \pi
 \end{aligned}$$

$$\begin{aligned}
9. & \int_{1+\sqrt{3}}^{\infty} \left(\frac{12}{x-1} - \frac{12(x+2)}{x^2+x+1} \right) dx \\
&= \int_{1+\sqrt{3}}^{\infty} \left(\frac{12}{x-1} - \frac{6(2x+1)+18}{x^2+x+1} \right) dx \\
&= \int_{1+\sqrt{3}}^{\infty} \left(\frac{12}{x-1} - \frac{6(2x+1)}{x^2+x+1} - \frac{18}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \right) dx \\
&= \left[12 \ln |x-1| - 6 \ln |x^2+x+1| \right]_{1+\sqrt{3}}^{\infty} \\
&\quad - \left[18 \times \frac{2}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}} \right]_{1+\sqrt{3}}^{\infty} \\
&= \left[6 \ln \frac{(x-1)^2}{x^2+x+1} - 12 \sqrt{3} \tan^{-1} \frac{2x+1}{\sqrt{3}} \right]_{1+\sqrt{3}}^{\infty} \\
&= 6 \ln(2+\sqrt{3}) - 12 \sqrt{3} \left(\frac{\pi}{2} - \tan^{-1} \frac{3+2\sqrt{3}}{\sqrt{3}} \right) \\
&= 6 \ln(2+\sqrt{3}) - 12 \sqrt{3} \left(\frac{\pi}{2} - \tan^{-1}(2+\sqrt{3}) \right) \\
&= 6 \ln(2+\sqrt{3}) - 6 \sqrt{3} (\pi - 2 \tan^{-1}(2+\sqrt{3})) \\
&= 6 \ln(2+\sqrt{3}) - 6 \sqrt{3} \times \frac{\pi}{6} \quad (\because \text{아래부분 참고}) \\
&= -\sqrt{3} \pi + 6 \ln(2+\sqrt{3}) \\
&\text{즉, } a=3, b=6 \text{ 이므로 } ab=18
\end{aligned}$$

[참고]

$\tan^{-1}(2+\sqrt{3})=X$ 라 하면 $\tan X=2+\sqrt{3}$ 이다.

$$\Rightarrow \tan(\pi - 2 \tan^{-1}(2+\sqrt{3}))$$

$$= \tan(\pi - 2X) = -\tan 2X$$

$$= -\frac{2 \tan X}{1 - \tan^2 X} = -\frac{2(2+\sqrt{3})}{1 - (2+\sqrt{3})^2} = \frac{1}{\sqrt{3}}$$

$$\text{즉, } \pi - 2 \tan^{-1}(2+\sqrt{3}) = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

$$\Rightarrow \tan^{-1}(2+\sqrt{3}) = \frac{5}{12} \pi$$

$$\tan(15^\circ) = 2 - \sqrt{3}, \tan(75^\circ) = 2 + \sqrt{3}$$

$$\begin{aligned}
10. & I_7 + I_9 = \int_0^{\frac{\pi}{4}} \tan^7 x dx + \int_0^{\frac{\pi}{4}} \tan^9 x dx \\
&= \int_0^{\frac{\pi}{4}} \tan^7 x (1 + \tan^2 x) dx = \int_0^{\frac{\pi}{4}} \tan^7 x \sec^2 x dx \\
&= \frac{1}{8} [\tan^8 x]_0^{\frac{\pi}{4}} = \frac{1}{8}
\end{aligned}$$

[참고]

$$I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx \text{ 일 때, } I_{n+2} + I_n = \frac{1}{n+1}$$

$$I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x dx \text{ 일 때, } I_{n+2} + I_n = \frac{1}{n+1}$$

$$\begin{aligned}
11. & \int_0^1 \frac{\sec^2 x - \tan x}{e^x} dx \\
&= \int_0^1 e^{-x} \sec^2 x dx - \int_0^1 e^{-x} \tan x dx \\
&= [e^{-x} \tan x]_0^1 + \int_0^1 e^{-x} \tan x dx - \int_0^1 e^{-x} \tan x dx \quad (\because \text{부분적분}) \\
&= \tan(1) e^{-1}
\end{aligned}$$

[다른 풀이]

$$(e^{-x} \tan x)' = -e^{-x} \tan x + e^{-x} \sec^2 x \text{ 이므로}$$

$$\int_0^1 \frac{\sec^2 x - \tan x}{e^x} dx = [e^{-x} \tan x]_0^1 = \tan(1) e^{-1}$$

1. 1번	2. 3번	3. 4번	4. 1번	5. 1번
6. 1번	7. 2번	8. 1번	9. 2번	10. 250
11. 1번	12. 3번	13. 4번	14. 2번	15. 3번
16. 4번	17. 1번	18. 4번	19. 4번	20. 4번
21. 3번	22. 2번	23. 2번	24. 4번	25. 3번
26. 1번	27. 3번	28. 5번	29. 4번	30. 2번
31. 1번	32. 3번	33. 4번	34. 4번	

$$1. U_f(P) = \frac{1}{2} \times 1 + \frac{1}{2} \times \frac{2}{3} + 2 \times \frac{1}{2} = \frac{11}{6}$$

$$2. n = 4 \text{ 이므로 } \Delta x = \frac{1}{2} \text{ 이다. 또한, } f(x) = \frac{1}{1+x} \text{ 이므로}$$

$$\begin{aligned} & \frac{\Delta x}{3} (1 \times y_0 + 4 \times y_1 + 2 \times y_2 + 4 \times y_3 + 1 \times y_4) \\ &= \frac{1}{3} \left(1 \times 1 + 4 \times \frac{2}{3} + 2 \times \frac{1}{2} + 4 \times \frac{2}{5} + 1 \times \frac{1}{3} \right) \\ &= \frac{11}{10} \text{ 이다.} \end{aligned}$$

3. 그래프를 그려보면 ①②③은 직선 $x = 2$ 에 대칭이므로

$$\int_{2-a}^2 f(x) dx = \int_2^{2+a} f(x) dx \text{ 를 만족한다.}$$

그러나 ④ $f(x) = (x-2)^3 - 1$ 은 직선 $x = 2$ 에 대칭이 아니고

따라서 $\int_{2-a}^2 f(x) dx = \int_2^{2+a} f(x) dx$ 을 만족하지 못한다.

4. $y = x - x^2$ 은 $x = 0$, $x = 1$ 에서 x 축과 만난다.
또한 그래프 개형을 그려보면 $0 \leq x \leq 1$ 일 때, S 는 최댓값을 갖는다.
즉, $a = 0$, $b = 1$ 이다.

5. 미적분학의 제 2 기본정리에 의하여

$$f\left(\frac{3}{2}\right) - f(0) = \int_0^{\frac{3}{2}} f'(x) dx = \frac{5}{8} \text{ 이다.}$$

$$\begin{aligned} 6. \int_0^2 |x^2 - x| dx &= - \int_0^1 x^2 - x dx + \int_1^2 x^2 - x dx \\ &= - \left[\frac{1}{3} x^3 - \frac{1}{2} x^2 \right]_0^1 + \left[\frac{1}{3} x^3 - \frac{1}{2} x^2 \right]_1^2 \\ &= - \left(\frac{1}{3} - \frac{1}{2} \right) + \left(\frac{8}{3} - 2 - \frac{1}{3} + \frac{1}{2} \right) \\ &= 1 \end{aligned}$$

$$7. \int_0^a \frac{\sqrt{a-x}}{\sqrt{x} + \sqrt{a-x}} dx = \frac{a}{2} \text{ 이므로}$$

$$\int_0^{2014} \frac{\sqrt{2014-x}}{\sqrt{x} + \sqrt{2014-x}} dx = 1007 \text{ 이다.}$$

$$8. \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4} \text{ 이므로}$$

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \sum_{k=1}^{100} \frac{\sin^k x}{\sin^k x + \cos^k x} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} + \dots + \frac{\sin^{100} x}{\sin^{100} x + \cos^{100} x} dx \\ &= \frac{\pi}{4} + \frac{\pi}{4} + \dots + \frac{\pi}{4} = \frac{\pi}{4} \times 100 = 25\pi \text{ 이다.} \end{aligned}$$

$$9. \int_{-a}^a f(x) dx = \int_0^a f(x) + f(-x) dx \text{ 이므로}$$

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(x) dx &= \int_0^{\frac{\pi}{2}} f(x) + f(-x) dx \\ &= \int_0^{\frac{\pi}{2}} \cos x dx = 1 \text{ 이다.} \end{aligned}$$

$$\begin{aligned} 10. -100 \int_e^{e^2} f(x) dx \\ &= -100 \left[[x f(x)]_e^{e^2} - \int_e^{e^2} x f'(x) dx \right] (\because \text{부분적분}) \\ &= -100 \left\{ [e^2 f(e^2) - e f(e)] - \int_e^{e^2} \frac{\ln x}{x} dx \right\} \\ &= -100 \left(-1 - \int_1^{e^2} t dt \right) (\because \ln x = t \text{로 치환}) \\ &= -100 \left(-1 - \frac{1}{2} [t^2]_1^{e^2} \right) \\ &= 250 \end{aligned}$$

$$11. \int_a^b [f(x)]^2 dx = 1$$

$$\Leftrightarrow \left[x f^2(x) - \int x 2f(x) f'(x) dx \right]_a^b = 1 (\because \text{부분적분})$$

$$\Leftrightarrow b f^2(b) - a f^2(a) - \int_a^b x 2f(x) f'(x) dx = 1$$

$$\Leftrightarrow \int_a^b x f(x) f'(x) dx = -\frac{1}{2} (\because f(a) = f(b) = 0)$$

$$12. \int_0^2 g(2-x) dx \text{ 에서 } 2-x = t \text{로 치환하면}$$

$$\int_0^2 g(2-x) dx = \int_0^2 g(t) dt \text{ 이다.}$$

$$\text{또한, } 4 \int_0^1 g(2x) dx \text{ 에서 } 2x = \theta \text{로 치환하면}$$

$$4 \int_0^1 g(2x) dx = 2 \int_0^2 g(\theta) d\theta \text{ 이다.}$$

$$\text{그리고 } \int_0^4 g(x) dx = -3, \int_2^4 g(x) dx = \frac{2}{3} \text{ 이므로}$$

$$\int_0^2 g(x) dx = \int_0^4 g(x) dx - \int_2^4 g(x) dx = -\frac{11}{3} \text{ 이다.}$$

$$\begin{aligned} \text{즉, } \int_0^2 g(2-x) dx + 4 \int_0^1 g(2x) dx \\ &= \int_0^2 g(t) dt + 2 \int_0^2 g(\theta) d\theta = 3 \int_0^2 g(x) dx = -11 \text{ 이다.} \end{aligned}$$

13. $\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$ 이므로
 $\int_1^3 f(x)dx + \int_2^5 f^{-1}(x)dx = 13$ 이고
 $\Rightarrow \int_2^5 f^{-1}(x)dx = 13 - \int_1^3 f(x)dx = \frac{17}{2}$ 이다.

[다른 풀이]

$$\begin{aligned} \int_2^5 f^{-1}(y)dy &= \int_1^3 xf'(x)dx \quad (\because f^{-1}(y) = x) \\ &= [xf(x)]_1^3 - \int_1^3 f(x)dx \quad (\because \text{부분적분}) \\ &= 3f(3) - f(1) - \frac{9}{2} = \frac{17}{2} \text{ 이다.} \end{aligned}$$

14. $\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$ 이므로
 $\int_0^1 f(x)dx + \int_0^2 f^{-1}(x)dx = 2$ 이고
 $\Rightarrow \int_0^2 f^{-1}(x)dx = 2 - \int_0^1 f(x)dx$
 $= 2 - \int_0^1 x^3 + x dx = 2 - \left[\frac{1}{4}x^4 + \frac{1}{2}x^2 \right]_0^1 = \frac{5}{4}$ 이다.

[다른 풀이]

$$\begin{aligned} f(0) = 0, f(1) = 2 \text{ 이므로} \\ \int_0^2 f^{-1}(x)dx &= \int_0^1 tf'(t)dt \quad (\because f^{-1}(x) = t) \\ &= \int_0^1 3t^3 + t dt = \left[\frac{3}{4}t^4 + \frac{1}{2}t^2 \right]_0^1 = \frac{5}{4} \text{ 이다.} \end{aligned}$$

15. $\lim_{n \rightarrow \infty} \frac{2}{n} \sum_{k=1}^n \left[\left(1 + \frac{2k}{n}\right)^2 - \left(1 + \frac{2k}{n}\right) \right]$
 $= \int_0^2 (1+x)^2 - (1+x) dx \quad \left(\because \frac{2k}{n} = x, \frac{2}{n} = dx \right)$
 $= \left[\frac{1}{3}(1+x)^3 - \frac{1}{2}(1+x)^2 \right]_0^2 = \frac{14}{3}$

16. $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}\sqrt{n+n}} \right)$
 $= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n}\sqrt{n+k}} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{1+\frac{k}{n}}} \times \frac{1}{n}$
 $= \int_0^1 \frac{1}{\sqrt{1+x}} dx \quad \left(\because \frac{k}{n} = x, \frac{1}{n} = dx \right)$
 $= 2 \left[\sqrt{1+x} \right]_0^1 = 2\sqrt{2} - 2$

17. $\lim_{n \rightarrow \infty} \frac{(1+2^2+3^2+\dots+n^2)^2}{n(1+2^4+3^4+\dots+n^4)}$
 $= \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n k^2 \sum_{k=1}^n k^2}{n \sum_{k=1}^n k^4} = \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \left(\frac{k}{n}\right)^2 \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^2 \frac{1}{n}}{\sum_{k=1}^n \left(\frac{k}{n}\right)^4 \frac{1}{n}}$
 $= \frac{\int_0^1 x^2 dx \int_0^1 x^2 dx}{\int_0^1 x^4 dx} \quad \left(\because \frac{k}{n} = x, \frac{1}{n} = dx \right)$
 $= \frac{\frac{1}{3}[x^3]_0^1 \times \frac{1}{3}[x^3]_0^1}{\frac{1}{5}[x^5]_0^1} = \frac{5}{9}$

[다른 풀이]

n 제곱의 합 공식을 이용하면

$$\lim_{n \rightarrow \infty} \frac{(1+2^2+3^2+\dots+n^2)^2}{n(1+2^4+3^4+\dots+n^4)} = \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{3}n^3+\dots\right)^2}{n\left(\frac{1}{5}n^5+\dots\right)} = \frac{5}{9} \text{ 이다.}$$

18. $\lim_{n \rightarrow \infty} \frac{7}{\sqrt[3]{n^2}} \sum_{k=1}^n \frac{1}{\sqrt[3]{n+7k}}$
 $= \lim_{n \rightarrow \infty} \frac{7}{n} \sum_{k=1}^n \frac{1}{\sqrt[3]{1+\frac{7k}{n}}} \quad \left(\because \frac{7k}{n} = x, \frac{7}{n} = dx \right)$
 $= \int_0^7 \frac{1}{(1+x)^{\frac{1}{3}}} dx = \int_0^7 (1+x)^{-\frac{1}{3}} dx = \frac{3}{2} \left[(1+x)^{\frac{2}{3}} \right]_0^7 = \frac{9}{2}$

19. $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{3} + \sqrt{3+\frac{1}{n}} + \sqrt{3+\frac{2}{n}} + \dots + \sqrt{3+\frac{n-1}{n}} \right)$
 $= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \sqrt{3+\frac{k}{n}} \quad \left(\because \frac{k}{n} = x, \frac{1}{n} = dx \right)$
 $= \int_0^1 \sqrt{3+x} dx$
 $= \frac{2}{3} \left[(3+x)^{\frac{3}{2}} \right]_0^1 = \frac{2}{3} (8-3\sqrt{3})$

20. $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\cos \frac{2}{3n} + \cos \frac{4}{3n} + \cos \frac{6}{3n} + \dots + \cos \frac{6n}{3n} \right)$
 $= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^{3n} \cos \frac{2k}{3n} \quad \left(\because \frac{2k}{3n} = x, \frac{2}{3n} = dx \right)$
 $= \frac{3}{2} \int_0^2 \cos(x) dx = \frac{3}{2} [\sin(x)]_0^2 = \frac{3}{2} \sin 2$

21. $\lim_{n \rightarrow \infty} \frac{K_{2n+2}}{K_{2n}} = \lim_{n \rightarrow \infty} \frac{\int_0^{\frac{\pi}{2}} \sin^{2n+2} x dx}{\int_0^{\frac{\pi}{2}} \sin^{2n} x dx}$
 $= \lim_{n \rightarrow \infty} \frac{\frac{2n+1}{2n+2} \times \frac{2n-1}{2n} \times \dots \times \frac{1}{2} \times \frac{\pi}{2}}{\frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \dots \times \frac{1}{2} \times \frac{\pi}{2}} \quad (\because \text{왈리스 정리})$
 $= \lim_{n \rightarrow \infty} \frac{2n+1}{2n+2} = 1$

22. $\frac{\int_1^3 x^2 - 4x + 3 dx}{3-1} = \frac{\left[\frac{1}{3}x^3 - 2x^2 + 3x \right]_1^3}{2} = -\frac{2}{3}$ 이므로
 평균값이는 $\frac{2}{3}$ 이다.

23. $f(x) = \frac{1}{1+x^3}$ 은 구간 $[0,1]$ 에서 연속이므로

적분의 평균값 정리를 이용하면 다음이 성립한다.

$$\frac{\int_a^b \frac{1}{1+x^3} dx}{b-a} = \frac{1}{1+c^3}, \quad 0 \leq a < c \leq b < 1$$

또한, $0 < c < 1$ 이므로

$$\Rightarrow 0 < c^3 < 1 \Rightarrow 1 < 1+c^3 < 2 \Rightarrow 1 > \frac{1}{1+c^3} > \frac{1}{2} \text{ 이다.}$$

따라서 $\frac{1}{2} < \frac{\int_a^b \frac{1}{1+x^3} dx}{b-a} < 1$ 이다.

24. $F(x) = \int_0^{x^2} t \cos t^2 dt$ 에서 정적분과 도함수를 이용하면
 $F'(x) = x^2 \cos(x^4) \times 2x = 2x^3 \cos(x^4)$ 이다.

$$\begin{aligned} 25. f(x) &= \int_{-\sin^{-1}\sqrt{x}}^{\sin^{-1}\sqrt{x}} \sin \sqrt{|t|} dt \\ &= 2 \int_0^{\sin^{-1}\sqrt{x}} \sin \sqrt{|t|} dt \quad (\because \sin \sqrt{|t|} \text{는 우함수}) \\ &= 2 \int_0^{\sin^{-1}\sqrt{x}} \sin \sqrt{t} dt \quad (\because t \geq 0) \end{aligned}$$

여기서 정적분과 도함수를 이용하면

$$f'(x) = 2 \sin \sqrt{\sin^{-1}\sqrt{x}} \times \frac{1}{\sqrt{1-x}} \times \frac{1}{2\sqrt{x}} \text{이다.}$$

$$\therefore f'\left(\frac{1}{2}\right) = 2 \sin \frac{\sqrt{\pi}}{2}$$

$$26. H(x) = \frac{1}{x} \int_3^x [2t - 3H'(t)] dt$$

$$H'(x) = -\frac{1}{x^2} \int_3^x [2t - 3H'(t)] dt + \frac{1}{x} [2x - 3H'(x)] \text{이다.}$$

따라서 $H'(3) = 2 - H'(3) \Rightarrow H'(3) = 1$ 이다.

27. 적분식 안에 미분 변수 x 가 있으므로 라이프니츠 정리를 이용하자.

$$f(x) = \int_0^x (e^x - t) e^t dt$$

$$f'(x) = (e^x - x) e^x + \int_0^x e^x e^t dt$$

$$\therefore f'(0) = 1$$

[다른 풀이]

$$f(x) = \int_0^x (e^x - t) e^t dt = e^x \int_0^x e^t dt - \int_0^x t e^t dt \text{이므로}$$

양변을 x 에 대해 미분하면

$$f'(x) = e^x \int_0^x e^t dt + e^x \cdot e^x - x e^x \text{이다.}$$

즉, $f'(0) = 1$ 이다.

$$28. f(x) = \int_{2x}^{3x} \frac{1}{\sin(t-x)} dt \quad (\because t-x = \theta)$$

$$= \int_x^{2x} \frac{1}{\sin(\theta)} d\theta$$

$$f'(x) = \frac{1}{\sin 2x} \times 2 - \frac{1}{\sin x}$$

$$\therefore f'\left(\frac{\pi}{6}\right) = \frac{4}{\sqrt{3}} - 2$$

$$29. \lim_{x \rightarrow 0^+} \frac{\int_x^{3x} \frac{\sin 3t}{t} dt}{x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{\sin 9x}{3x} \times 3 - \frac{\sin 3x}{x} \times 1}{1} \quad (\because \text{로피탈 정리})$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin 9x}{9x} \times 9 - \frac{\sin 3x}{3x} \times 3 = 6$$

[다른 풀이]

$$f(t) = \frac{\sin 3t}{t} \text{는 구간 } [x, 3x] \text{에서 연속이므로}$$

적분의 평균값 정리를 이용하면 다음이 성립한다.

$$\frac{\int_x^{3x} \frac{\sin 3t}{t} dt}{2x} = \frac{\sin 3c}{c}, \quad x < c < 3x$$

이때, $x \rightarrow 0^+$ 이면 $c \rightarrow 0^+$ 이다.

$$\begin{aligned} \text{즉, } \lim_{x \rightarrow 0^+} \frac{\int_x^{3x} \frac{\sin 3t}{t} dt}{x} &= \lim_{x \rightarrow 0^+} \frac{2 \int_x^{3x} \frac{\sin 3t}{t} dt}{2x} \\ &= \lim_{x \rightarrow 0^+} 2 \frac{\sin 3c}{c} = 2 \cdot 3 = 6 \text{이다.} \end{aligned}$$

$$30. \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{4x^2 - \pi^2}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt + \sin x}{8x} \quad (\because \text{로피탈 정리})$$

$$= \frac{1}{4\pi}$$

[다른 풀이]

$$f(t) = \frac{\sin t}{t} \text{는 구간 } \left[\frac{\pi}{2}, x\right] \text{에서 연속이므로}$$

적분의 평균값 정리를 이용하면 다음이 성립한다.

$$\frac{\int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{x - \frac{\pi}{2}} = \frac{\sin c}{c}, \quad \frac{\pi}{2} < c < x$$

이때, $x \rightarrow \frac{\pi}{2}$ 이면 $c \rightarrow \frac{\pi}{2}$ 이다.

$$\begin{aligned} \text{즉, } \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{4x^2 - \pi^2} &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{(2x - \pi)(2x + \pi)} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \int_{\frac{\pi}{2}}^x \frac{\sin t}{t} dt}{2(x - \frac{\pi}{2})(2x + \pi)} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{x \frac{\sin c}{c}}{2(2x + \pi)} = \frac{\frac{\pi}{2} \cdot \frac{2}{\pi}}{2(2 \cdot \frac{\pi}{2} + \pi)} = 4\pi \text{이다.} \end{aligned}$$

$$31. f(t) = \frac{1}{2} (\sinh t \cosh t - 1) - \int_1^{\cosh t} \sqrt{x^2 - 1} dx$$

$$\begin{aligned} f'(t) &= \frac{1}{2} (\cosh^2 t + \sinh^2 t) - \sqrt{\cosh^2 t - 1} \times \sinh t \\ &= \frac{1}{2} (\cosh^2 t - \sinh^2 t) = \frac{1}{2} \end{aligned}$$

따라서 $f'(t) = \frac{1}{2}$ 이고 적분하면 $f(t) = \frac{1}{2}t + C$ 이다.

또한 $f(0) = -\frac{1}{2}$ 이므로 $f(t) = \frac{1}{2}t - \frac{1}{2}$ 이고 $f(2) = \frac{1}{2}$ 이다.

32. 극대점을 찾기 위해서 임계점을 구하자.

$$f(x) = \int_0^x \frac{\sin t}{t} dt$$

$$f'(x) = \frac{\sin x}{x} = 0 \quad \therefore x = \pi, 2\pi, 3\pi, \dots$$

$$f''(x) = \frac{\cos x \times x - \sin x}{x^2}$$

즉, $x = (2n-1)\pi$ 이면 $f''(x) < 0$ 이므로 극대값을 갖는다.

$$33. 6 + \int_a^x \frac{f(t)}{t^2} dt = 2\sqrt{x}$$

1) 양변에 $x = a$ 를 대입하면 $6 = 2\sqrt{a}$ 이다. 즉, $a = 9$

2) 양변을 미분하면 $\frac{f(x)}{x^2} = \frac{1}{\sqrt{x}} \Rightarrow f(x) = x^{\frac{3}{2}}$ 이다.

따라서 $f(a) = a^{\frac{3}{2}} = 3^3$ 이다.

34. $f(x) = \cos x + \int_0^{\frac{\pi}{2}} f(x) dx$ 에서 $\int_0^{\frac{\pi}{2}} f(x) dx$ 는 상수값이므로

$$\int_0^{\frac{\pi}{2}} f(x) dx = c \text{ 라 하면}$$

$$f(x) = \cos x + c \text{ 이다.}$$

$$\int_0^{\frac{\pi}{2}} f(x) dx = c$$

$$\Leftrightarrow \int_0^{\frac{\pi}{2}} (\cos x + c) dx = c$$

$$\Leftrightarrow [\sin x + cx]_0^{\frac{\pi}{2}} = c$$

$$\Leftrightarrow 1 + \frac{\pi}{2}c = c \quad \therefore c = \frac{2}{2-\pi}$$

$$\text{따라서 } f(x) = \cos x + \frac{2}{2-\pi} \text{ 이고 } f(0) = \frac{4-\pi}{2-\pi} \text{ 이다.}$$



장황수학

1. 3번	2. 1번	3. 2번	4. 3번	5. 4번
6. 1번	7. 2번	8. 1번	9. 4번	10. 2번
11. 4번	12. 4번	13. 4번	14. 1번	15. 2번
16. 4번	17. 4번	18. 3번		

1. $a_{n+2} = 2a_{n+1} + a_n$ 은 $\frac{a_{n+2}}{a_{n+1}} = 2 + \frac{a_n}{a_{n+1}}$ 이므로

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{a_{n+2}}{a_{n+1}} = \alpha \text{ 라 하면}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+2}}{a_{n+1}} = \lim_{n \rightarrow \infty} 2 + \frac{a_n}{a_{n+1}}$$

$$\Rightarrow \alpha = 2 + \frac{1}{\alpha}$$

$$\Rightarrow \alpha^2 - 2\alpha - 1 = 0$$

$$\Rightarrow \alpha = 1 + \sqrt{2} \text{ 또는 } 1 - \sqrt{2}$$

$$\Rightarrow a_3 = 2a_2 + a_1 = 3,$$

$$a_4 = 2a_3 + a_2 = 7,$$

$$a_5 = 2a_4 + a_3 = 17,$$

⋮

따라서 $\frac{a_4}{a_3} = \frac{7}{3}, \frac{a_5}{a_4} = \frac{17}{7}, \dots$ 이므로 $\lim_{n \rightarrow \infty} b_n$ 의 값은 $1 + \sqrt{2}$ 이다.

2. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{n(n+1)} = \lim_{k \rightarrow \infty} \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right)$

$$= \lim_{k \rightarrow \infty} \left(1 - \frac{1}{2} + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{k} - \frac{1}{k+1} \right) \right)$$

$$= \lim_{k \rightarrow \infty} 1 - \frac{1}{k+1} = 1$$

3. $\sum_{n=2}^{\infty} \ln \frac{(n-1)(n+1)}{n^2} = \lim_{k \rightarrow \infty} \sum_{n=2}^k \ln \frac{(n-1)(n+1)}{n^2}$

$$= \lim_{k \rightarrow \infty} \sum_{n=2}^k \ln \frac{n-1}{n} + \ln \frac{n+1}{n}$$

$$= \lim_{k \rightarrow \infty} \left(\ln \frac{1}{2} + \ln \frac{3}{2} \right) + \left(\ln \frac{2}{3} + \ln \frac{4}{3} \right) + \left(\ln \frac{3}{4} + \ln \frac{5}{4} \right)$$

$$+ \dots + \left(\ln \frac{k-1}{k} + \ln \frac{k+1}{k} \right)$$

$$= \lim_{k \rightarrow \infty} \ln \frac{1}{2} + \ln \frac{k+1}{k} = \ln \frac{1}{2}$$

4. $\lim_{n \rightarrow \infty} \left(\frac{8}{n} + \frac{8n+1}{n^2+1} + \frac{8n+2}{n^2+4} + \frac{8n+3}{n^2+9} + \dots + \frac{9n-1}{2n^2-2n+1} \right)$

$$= \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{8n+k}{n^2+k^2} = \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{8 + \frac{k}{n}}{\frac{n^2+k^2}{n^2}} = \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{8 + \frac{k}{n}}{1 + \left(\frac{k}{n} \right)^2} \cdot \frac{1}{n}$$

$$= \int_0^1 \frac{8+x}{1+x^2} dx = \left[8 \tan^{-1} x + \frac{1}{2} \ln(1+x^2) \right]_0^1 = 2\pi + \frac{1}{2} \ln 2$$

5. $\lim_{n \rightarrow \infty} \sum_{k=n+1}^{2n} \frac{1}{\sqrt{4n^2-k^2}} = \lim_{n \rightarrow \infty} \sum_{k=n+1}^{2n} \frac{1}{\sqrt{4n^2-k^2}} \cdot \frac{1}{n}$

$$= \lim_{n \rightarrow \infty} \sum_{k=n+1}^{2n} \frac{1}{\sqrt{4 - \left(\frac{k}{n} \right)^2}} \cdot \frac{1}{n} = \int_1^2 \frac{1}{\sqrt{4-x^2}} dx \left(\because \frac{k}{n} = x \right)$$

$$= \left[\sin^{-1} \frac{x}{2} \right]_1^2 = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$$

6. $\lim_{n \rightarrow \infty} \frac{\sqrt[n]{(n+1)(n+2) \cdots (2n-1)(2n)}}{n}$

$$= \lim_{n \rightarrow \infty} \left(\frac{(n+1)(n+2) \cdots (2n-1)(2n)}{n^n} \right)^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln \left(\frac{(n+1)(n+2) \cdots (2n-1)(2n)}{n^n} \right)^{\frac{1}{n}}}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln \left(\frac{(n+1)(n+2) \cdots (2n-1)(2n)}{n^n} \right)}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \left(\ln \frac{n+1}{n} + \ln \frac{n+2}{n} + \ln \frac{n+3}{n} + \dots + \ln \frac{n+n}{n} \right)}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln \frac{n+k}{n}} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{k}{n} \right)} = e^{\int_0^1 \ln(1+x) dx}$$

$$= e^{[(1+x) \ln(1+x) - (1+x)]_0^1} = e^{2 \ln 2 - 1} = \frac{4}{e}$$

7. $\lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{x^6} = \lim_{x \rightarrow 0} \frac{\sin \left(\frac{4x^4}{3} \right) \times 4x}{6x^5} = \lim_{x \rightarrow 0} \frac{\sin \left(\frac{4x^4}{3} \right) \times 2}{3x^4}$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{4x^4}{3} + \dots \right) \times 2}{3x^4} = \frac{8}{9}$$

[다른 풀이]

$$\lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{x^6} = \lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \frac{t^2}{3} - \frac{1}{3!} \left(\frac{t^2}{3} \right)^3 + \dots dt}{x^6}$$

$$= \lim_{x \rightarrow 0} \frac{\left[\frac{t^3}{9} - \dots \right]_0^{2x^2}}{x^6} = \lim_{x \rightarrow 0} \frac{\frac{8x^6}{9} - \dots}{x^6} = \frac{8}{9}$$

[참고]

해당 문제는 적분의 평균값정리로 풀 수 없다.

$f(t) = \sin \left(\frac{t^2}{3} \right)$ 는 구간 $[0, 2x^2]$ 에서 연속이므로

적분의 평균값 정리를 이용하면 다음이 성립한다.

$$\frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{2x^2 - 0} = \sin \left(\frac{c^2}{3} \right), \quad 0 < c < 2x^2$$

이때, $x \rightarrow 0$ 이면 $c \rightarrow 0$ 이다.

$$\text{즉, } \lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{x^6} = \lim_{x \rightarrow 0} \frac{\int_0^{2x^2} \sin \left(\frac{t^2}{3} \right) dt}{\frac{1}{2} x^4 \cdot 2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin \left(\frac{c^2}{3} \right)}{\frac{1}{2} x^4}$$

그러나 여기서 $c^2 \sim Kx^4$ 같은 비율 정보가 없기 때문에 극한을 결정할 수 없다.

8. $f(x) = \int_1^{2x+3} \frac{1}{t^4 - 2t^2 + 3} dt$ 일 때,

$$g'(0) = \frac{1}{f'(-1)} = \frac{1}{\frac{1}{(2x+3)^4 - 2(2x+3)^2 + 3} \times 2} \bigg|_{x=-1} = 1 \text{ 이다.}$$

9. $f(x) = (x^2 + 1)^2 - \int_0^x t^2 f'(t) dt$ 에서 양변을 미분하면

$$\Rightarrow f'(x) = 2(x^2 + 1)2x - x^2 f'(x)$$

$$\Rightarrow (1 + x^2)f'(x) = 4x(1 + x^2)$$

$$\Rightarrow f'(x) = 4x$$

$$\Rightarrow f(x) = 2x^2 + c \text{ 이고 } f(0) = 1 \text{ 이므로 } c = 1 \text{ 이다.}$$

$$\text{즉, } f(x) = 2x^2 + 1 \text{ 이므로 } f(2) = 9 \text{ 이다.}$$

10. $F(x) = \int_0^x f(t)(x-t)dt = x \int_0^x f(t)dt - \int_0^x tf(t)dt$ (양변을 미분)

$$\Rightarrow F'(x) = \int_0^x f(t)dt + xf(x) - x f(x) = \int_0^x f(t)dt$$

$$\Rightarrow F''(x) = f(x)$$

$$f > 0 \text{ 이므로 } F'(x) = \int_0^x f(t)dt > 0 \text{ 이고 } F''(x) = f(x) > 0 \text{ 이다.}$$

즉, $F(x)$ 는 증가함수이며 아래로 볼록인 함수이다.

11. (ㄱ) $f(a-x) = f(x)$ 일 때,

$$\int_0^a f(x)dx = \int_0^{\frac{a}{2}} f(x)dx + \int_{\frac{a}{2}}^a f(x)dx$$

$$= \int_0^{\frac{a}{2}} f(x)dx + \int_{\frac{a}{2}}^a f(a-x)dx$$

$$= \int_0^{\frac{a}{2}} f(x)dx - \int_{\frac{a}{2}}^0 f(t)dt \quad (\because a-x=t)$$

$$= \int_0^{\frac{a}{2}} f(x)dx + \int_0^{\frac{a}{2}} f(x)dx$$

$$= 2 \int_0^{\frac{a}{2}} f(x)dx \quad (\text{참})$$

[참고]

$f(a-x) = f(x)$ 를 만족하는 함수는 $x = \frac{a}{2}$ 축 대칭이므로

$$\int_0^a f(x)dx = 2 \int_0^{\frac{a}{2}} f(x)dx \text{가 항상 성립한다.}$$

$$(ㄴ) \int_{-a}^a f(x)dx = \int_{-a}^0 f(x)dx + \int_0^a f(x)dx$$

$$= - \int_a^0 f(-t)dt + \int_0^a f(x)dx \quad (\because x=-t)$$

$$= \int_0^a f(-x)dx + \int_0^a f(x)dx$$

$$= \int_0^a f(-x) + f(x)dx \quad (\text{참})$$

$$(ㄷ) \int_0^\pi xf(\sin x)dx = - \int_\pi^0 (\pi-t)f(\sin t)dt \quad (\because x=\pi-t)$$

$$= \int_0^\pi (\pi-x)f(\sin x)dx$$

$$= \int_0^\pi \pi f(\sin x)dx - \int_0^\pi x f(\sin x)dx$$

$$\text{이므로 } 2 \int_0^\pi xf(\sin x)dx = \int_0^\pi \pi f(\sin x)dx \text{ 이다.}$$

$$\text{즉, } \int_0^\pi xf(\sin x)dx = \frac{\pi}{2} \int_0^\pi f(\sin x)dx \quad (\text{참})$$

$$(ㄹ) \int_0^{\frac{\pi}{2}} f(\sin x)dx = - \int_{\frac{\pi}{2}}^0 f(\cos t)dt \quad (\because x=\frac{\pi}{2}-t)$$

$$= \int_0^{\frac{\pi}{2}} f(\cos x)dx \quad (\text{참})$$

12. $\int_0^{\frac{\pi}{2}} \frac{\sin^n x + \cos^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{2}$ 이므로

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx + \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{2} \text{ 이다.}$$

$$\text{이때, } \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx$$

$$= - \int_{\frac{\pi}{2}}^0 \frac{\cos^n t}{\cos^n t + \sin^n t} dt \quad (\because x = \frac{\pi}{2} - t)$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\cos^n x + \sin^n x} dx \text{ 을 알 수 있다.}$$

$$\text{즉, } \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4}$$

13. $\int_0^{\frac{\pi}{2}} \frac{\tan^n x + \cot^n x}{\tan^n x + \cot^n x} dx = \frac{\pi}{2}$ 이므로

$$\int_0^{\frac{\pi}{2}} \frac{\tan^n x}{\tan^n x + \cot^n x} dx + \int_0^{\frac{\pi}{2}} \frac{\cot^n x}{\tan^n x + \cot^n x} dx = \frac{\pi}{2} \text{ 이다.}$$

$$\text{이때, } \int_0^{\frac{\pi}{2}} \frac{\cot^n x}{\tan^n x + \cot^n x} dx$$

$$= - \int_{\frac{\pi}{2}}^0 \frac{\cot^n t}{\cot^n t + \tan^n t} dt \quad (\because x = \frac{\pi}{2} - t)$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cot^n x}{\cot^n x + \tan^n x} dx \text{ 을 알 수 있다.}$$

$$\text{즉, } \int_0^{\frac{\pi}{2}} \frac{\tan^n x}{\tan^n x + \cot^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\cot^n x}{\tan^n x + \cot^n x} dx = \frac{\pi}{4}$$

14. $\int_a^b [f(x)]^2 dx = 1$

$$\Rightarrow [xf^2(x)]_a^b - 2 \int_a^b xf(x)f'(x)dx = 1 \quad (\because \text{부분적분})$$

$$\Rightarrow bf^2(b) - af^2(a) - 2 \int_a^b xf(x)f'(x)dx = 1 \quad (\because \text{부분적분})$$

$$\Rightarrow -2 \int_a^b xf(x)f'(x)dx = 1$$

$$\text{즉, } \int_a^b xf(x)f'(x)dx = -\frac{1}{2}$$

15. $g(7) = 3, g(1) = 0 \Leftrightarrow f(3) = 7, f(0) = 1$ 이므로

$$\int_0^3 f(x)dx + \int_{f(0)}^{f(3)} g(x)dx = \int_0^3 f(x)dx + \int_1^7 g(x)dx$$

$$= 3 \times 7 - 0 \times 1 = 21 \text{ 이다.}$$

$$\text{즉, } \int_0^3 f(x)dx + 13 = 21 \text{ 이므로 } \int_0^3 f(x)dx = 8 \text{ 이다.}$$

16. $\int_1^2 f(x)dx + \int_{f(1)}^{f(2)} g(x)dx = \int_1^2 f(x)dx + \int_2^5 g(x)dx$

$$= 2 \times 5 - 1 \times 2 = 8 \text{ 이다.}$$

$$\text{이때, } \int_1^2 \frac{1}{3}x^3 + \frac{2}{3}x + 1dx + \int_2^5 g(x)dx = 8 \text{ 이므로}$$

$$\int_2^5 g(x)dx = 8 - \int_1^2 \frac{1}{3}x^3 + \frac{2}{3}x + 1dx$$

$$= 8 - \left[\frac{1}{12}x^4 + \frac{1}{3}x^2 + x \right]_1^2$$

$$= 8 - \left(\frac{4}{3} + \frac{4}{3} + 2 - \frac{1}{12} - \frac{1}{3} - 1 \right) = \frac{19}{4} \text{ 이다.}$$

[다른 풀이]

$$g(x) = y \text{ 라하면 } f(y) = x \text{ 이고,}$$

$$f(1) = 2, f(2) = 5 \Rightarrow g(2) = 1, g(5) = 2 \text{ 이므로}$$

$$\int_2^5 g(x)dx = \int_1^2 yf'(y)dy$$

$$= \int_1^2 y \left(y^2 + \frac{2}{3} \right) dy = \left[\frac{1}{4} y^4 + \frac{1}{3} y^2 \right]_1^2 = \frac{19}{4} \text{이다.}$$

$$\begin{aligned} 17. & \int_0^3 \frac{1}{f'(f^{-1}(x))(1+(f^{-1}(x))^2)} dx \\ &= \int_0^1 \frac{1}{f'(t)(1+t^2)} f'(t) dt \quad (\because f^{-1}(x) = t) \\ &= \int_0^1 \frac{1}{1+t^2} dt = [\tan^{-1} t]_0^1 = \frac{\pi}{4} \end{aligned}$$

[다른 풀이]

$f(f^{-1}(x)) = x$ 에서 양변을 미분하면

$$f'(f^{-1}(x)) \cdot (f^{-1})'(x) = 1 \Rightarrow (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \text{이다.}$$

$$\begin{aligned} \text{즉, } & \int_0^3 \frac{1}{f'(f^{-1}(x))(1+(f^{-1}(x))^2)} dx \\ &= \int_0^3 \frac{(f^{-1})'(x)}{(1+(f^{-1}(x))^2)} dx = [\tan^{-1}(f^{-1}(x))]_0^3 \\ &= \tan^{-1} 1 - \tan^{-1} 0 = \frac{\pi}{4} \text{이다.} \end{aligned}$$

$$18. \int_1^2 t f(t) dt \text{는 상수값이므로}$$

$$\int_1^2 t f(t) dt = C \text{라 하면 } f(x) = \ln x + C \text{이고}$$

$$\begin{aligned} C &= \int_1^2 t(\ln t + C) dt = \int_1^2 t \ln t + C t dt \\ &= \left[\frac{1}{2} t^2 \ln t - \frac{1}{2} \int t dt + \frac{1}{2} t^2 C \right]_1^2 \\ &= \left[\frac{1}{2} t^2 \ln t - \frac{1}{4} t^2 + \frac{1}{2} t^2 C \right]_1^2 = 2 \ln 2 - 1 + 2C + \frac{1}{4} - \frac{1}{2} C \end{aligned}$$

$$\Leftrightarrow -\frac{1}{2} C = 2 \ln 2 - \frac{3}{4}$$

$$\Leftrightarrow C = -4 \ln 2 + \frac{3}{2} \text{이다.}$$

$$\text{따라서 } f(x) = \ln x - 4 \ln 2 + \frac{3}{2} \text{이므로 } f(2) = -3 \ln 2 + \frac{3}{2} \text{이다.}$$



장황수학

1. 2번	2. 3번	3. 3번	4. 1번	5. 4번
6. 3번	7. 2번	8. $\frac{1}{\sqrt{2}}$	9. 3번	10. 1번
11. 1번	12. e^{-1}	13. 2번	14. 5번	15. 2번
16. 3번	17. 1번	18. 2번	19. 4번	

1. (1) 심프슨 공식

$$\frac{\Delta x}{3} \left(f(0) + 4f\left(\frac{\pi}{4}\right) + 2f\left(\frac{\pi}{2}\right) + 4f\left(\frac{3\pi}{4}\right) + f(\pi) \right)$$

$$= \frac{\pi}{12} (2\sqrt{2} + 2 + 2\sqrt{2}) = \frac{\pi}{6} (2\sqrt{2} + 1)$$

(2) 사다리꼴 공식

$$\frac{\Delta x}{2} \left(f(0) + 2f\left(\frac{\pi}{4}\right) + 2f\left(\frac{\pi}{2}\right) + 2f\left(\frac{3\pi}{4}\right) + f(\pi) \right)$$

$$= \frac{\pi}{8} (\sqrt{2} + 2 + \sqrt{2}) = \frac{\pi}{4} (\sqrt{2} + 1)$$

$$\Rightarrow 6A - 4B = \pi(2\sqrt{2} + 1) - \pi(\sqrt{2} + 1) = \sqrt{2}\pi$$

2. 가. $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$ (참)

나. $F(-x) = \int_x^{-x} f(t) dt = - \int_{-x}^x f(t) dt = -F(x)$ (참)

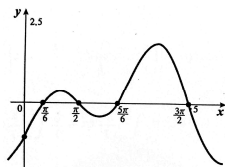
다. [반례]

$f(x) = 3x - 1$ 이라 하면, $\int_0^1 3x - 1 dx \leq 0$ 이지만

$x > \frac{1}{3}$ 일 때, $f(x) > 0$ 이다. (거짓)

라. 만약 모든 x 에서 $f(x) \leq 0$ 이면 적분값도 ≤ 0 이므로 모순이다.
따라서 어떤 점에선 반드시 $f(x) > 0$ 이다. (참)

3. $f(x) = \sin 2x - \cos x = 2\sin x \cos x - \cos x = \cos x (2\sin x - 1)$



$\int_a^b f(x) dx$ 가 최대가 되도록 하는 실수 a, b 에 대한

$b - a$ 의 최댓값은 $a = \frac{\pi}{6}, b = \frac{3\pi}{2}$ 이므로 $b - a$ 의 최댓값은 $\frac{4}{3}\pi$

4. $\int_1^n \frac{1}{x^p} dx \leq \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} \leq 1 + \int_1^n \frac{1}{x^p} dx$ 이므로

$$\lim_{n \rightarrow \infty} \int_1^n \frac{1}{x^p} dx \leq \lim_{n \rightarrow \infty} \left(\frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} \right)$$

$$\leq \lim_{n \rightarrow \infty} \left(1 + \int_1^n \frac{1}{x^p} dx \right)$$

$\therefore \int_1^\infty \frac{1}{x^p} dx \leq S_p \leq 1 + \int_1^\infty \frac{1}{x^p} dx$

$\therefore \frac{1}{p-1} \leq S_p \leq \frac{p}{p-1}$

5. $f(x) = \int_0^1 |t^2 - x^2| dt$

$= \int_0^1 (x^2 - t^2) dt \quad (\because 0 \leq t \leq 1, |x| \geq 1)$

$= \left[x^2 t - \frac{1}{3} t^3 \right]_0^1 = x^2 - \frac{1}{3}$

이때, $f(x) = x^2 - \frac{1}{3}, x \leq -1, x \geq 1$ 에서의

최솟값은 $f(-1) = f(1) = \frac{2}{3}$ 이다.

6. $g(e) + g\left(\frac{1}{e}\right) = \int_1^e \frac{\ln t}{1+t} dt + \int_1^{\frac{1}{e}} \frac{\ln t}{1+t} dt$

이때, $\int_1^{\frac{1}{e}} \frac{\ln t}{1+t} dt$ 에서 $t = \frac{1}{x}$ 로 치환하면

$\int_1^e \frac{\ln \frac{1}{x}}{1 + \frac{1}{x}} dx = \int_1^e \frac{-\ln x}{x(x+1)} dx$ 이다.

즉, $g(e) + g\left(\frac{1}{e}\right) = \int_1^e \frac{\ln t}{1+t} dt + \int_1^{\frac{1}{e}} \frac{\ln t}{1+t} dt$

$$= \int_1^e \frac{\ln t}{1+t} dt + \int_1^e \frac{\ln t}{t(1+t)} dt$$

$$= \int_1^e \frac{(1+t)\ln t}{t(1+t)} dt = \int_1^e \frac{\ln t}{t} dt = \left[\frac{1}{2} (\ln t)^2 \right]_1^e = \frac{1}{2}$$

7. $\theta = \pi - x$ 라 하면

$\int_0^\pi \theta f(\sin \theta) d\theta = \int_\pi^0 (\pi - x) f(\sin(\pi - x)) (-dx)$

$= \int_0^\pi (\pi - x) f(\sin x) dx$

$= \pi \int_0^\pi f(\sin x) dx - \int_0^\pi x f(\sin x) dx$

$= \pi \int_0^\pi f(\sin \theta) d\theta - \int_0^\pi \theta f(\sin \theta) d\theta$

$\therefore \int_0^\pi \theta f(\sin \theta) d\theta = \frac{\pi}{2} \int_0^\pi f(\sin \theta) d\theta$

8. $\int_0^\pi \frac{x \sin x}{1 + \sin^2 x} dx = \frac{\pi}{2} \int_0^\pi \frac{\sin t}{1 + \sin^2 t} dt$

$= \frac{\pi}{2} \int_0^\pi \frac{\sin t}{2 - \cos^2 t} dt = \frac{\pi}{2} \int_{-1}^1 \frac{1}{2 - \theta^2} d\theta \quad (\because \cos t = \theta)$

$= -\pi \int_0^1 \frac{1}{\theta^2 - 2} d\theta = \frac{-\pi}{2\sqrt{2}} \left[\ln \left| \frac{\theta - \sqrt{2}}{\theta + \sqrt{2}} \right| \right]_0^1$

$= \frac{\pi}{2\sqrt{2}} \ln \left| \frac{1 + \sqrt{2}}{1 - \sqrt{2}} \right| = \frac{\pi}{2\sqrt{2}} \ln \left| \frac{1 + \sqrt{2}}{1 - \sqrt{2}} \right|$

$= \frac{\pi}{2\sqrt{2}} \ln(\sqrt{2} + 1)^2 = \frac{\pi}{\sqrt{2}} \ln(\sqrt{2} + 1)$

즉, 주어진 문제의 정답은 : $\frac{1}{\sqrt{2}}$

9. $I_{2018} = \int_0^\pi \frac{\sin 2018x}{x} dx \quad (u = \pi - x \text{로 치환})$

$= \int_\pi^0 \frac{\sin 2018(\pi - u)}{\sin(\pi - u)} du$

$= \int_0^\pi \frac{\sin(2018\pi - 2018u)}{\sin u} du = \int_0^\pi \frac{-\sin 2018u}{\sin u} du$ 이므로

$\int_0^\pi \frac{\sin 2018x}{\sin x} dx = - \int_0^\pi \frac{\sin 2018x}{\sin x} dx$

$\Leftrightarrow 2 \int_0^\pi \frac{\sin 2018x}{\sin x} dx = 0 \Leftrightarrow \int_0^\pi \frac{\sin 2018x}{\sin x} dx = 0$

$I_{2017} = \int_0^\pi \frac{\sin 2017x}{\sin x} dx = \int_0^\pi \frac{\sin(2016x + x)}{\sin x} dx$

$= \int_0^\pi \frac{\sin 2016x \cos x + \cos 2016x \sin x}{\sin x} dx$

$$\begin{aligned}
&= \int_0^\pi \frac{\sin 2016x \cos x}{\sin x} dx + \int_0^\pi \cos 2016x dx \\
&= \int_0^\pi \frac{\frac{1}{2} \{\sin(2017x) + \sin(2015x)\}}{\sin x} dx + \frac{1}{2016} [\sin 2016x]_0^\pi \\
&= \frac{1}{2} \int_0^\pi \frac{\sin 2017x}{\sin x} dx + \frac{1}{2} \int_0^\pi \frac{\sin 2015x}{\sin x} dx \quad \text{이므로} \\
&\int_0^\pi \frac{\sin 2017x}{\sin x} dx = \frac{1}{2} \int_0^\pi \frac{\sin 2017x}{\sin x} dx + \frac{1}{2} \int_0^\pi \frac{\sin 2015x}{\sin x} dx \\
&\Leftrightarrow \frac{1}{2} \int_0^\pi \frac{\sin 2017x}{\sin x} dx = \frac{1}{2} \int_0^\pi \frac{\sin 2015x}{\sin x} dx \\
&\Leftrightarrow \int_0^\pi \frac{\sin 2017x}{\sin x} dx = \int_0^\pi \frac{\sin 2015x}{\sin x} dx \quad \text{이다.}
\end{aligned}$$

따라서

$$\int_0^\pi \frac{\sin 2017x}{\sin x} dx = \int_0^\pi \frac{\sin 2015x}{\sin x} dx = \dots = \int_0^\pi \frac{\sin x}{\sin x} dx = \pi$$

이다. 그러므로 $|I_{2018} - I_{2017}| = \pi$ 이다.

$$\begin{aligned}
10. \lim_{n \rightarrow \infty} \sum_{k=1}^n f_n(k) &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^2 - 3kn + n^2}{n^3} \\
&= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\left(\frac{k}{n} \right)^2 - \frac{3k}{n} + 1 \right] \frac{1}{n} \\
&= \int_0^1 x^2 - 3x + 1 dx = -\frac{1}{6}
\end{aligned}$$

$$\begin{aligned}
11. \lim_{n \rightarrow \infty} \sum_{k=n}^{3n} \frac{1}{n+k} &= \lim_{n \rightarrow \infty} \sum_{k=n}^{3n} \frac{1}{\frac{k}{n} + 1} \frac{1}{n} \\
&= \int_1^3 \frac{1}{1+x} dx = [\ln(1+x)]_1^3 = \ln 2
\end{aligned}$$

$$\begin{aligned}
12. \lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}} &= \lim_{n \rightarrow \infty} e^{\frac{\ln(n!)}{n}} \\
&= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln(n!)} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln(1 \cdot 2 \cdot 3 \cdots n)} \\
&= \lim_{n \rightarrow \infty} e^{\frac{1}{n} (\ln 1 + \ln 2 + \ln 3 + \dots + \ln n)} \\
&= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln k} = e^{\int_0^1 \ln x dx} = e^{-1}
\end{aligned}$$

[다른 풀이]

스털링 근사를 활용하면

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}} &\sim \lim_{n \rightarrow \infty} \left(\frac{\sqrt{2\pi n} n^n e^{-n}}{n^n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{(\sqrt{2\pi n})^{\frac{1}{n}}}{e} \right) = \frac{1}{e} \quad \text{이다.} \\
(\because (\sqrt{2\pi n})^{\frac{1}{n}} &= e^{\frac{\ln \sqrt{2\pi n}}{n}} = e^0 = 1)
\end{aligned}$$

$$\begin{aligned}
13. g_n'(x) &= \sum_{k=1}^n \left[f' \left(\frac{k}{n} x \right) \frac{k}{n} \frac{x^2}{n} + f \left(\frac{k}{n} x \right) \frac{2x}{n} \right] \\
&\Rightarrow \lim_{n \rightarrow \infty} g_n'(1) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[f' \left(\frac{k}{n} \right) \frac{k}{n} \frac{1}{n} + f \left(\frac{k}{n} \right) \frac{2}{n} \right] \\
&= \int_0^1 [x f'(x) + 2f(x)] dx \\
&= \int_0^1 (x e^x + 2e^x) dx = \int_0^1 (x+2)e^x dx \\
&= [(x+2)e^x]_0^1 - \int_0^1 e^x dx = 2e - 1
\end{aligned}$$

14. 정적분의 평균값 정리를 만족시키는 점 ξ 를 찾으려 한다.

$$f(\xi) = \frac{1}{\frac{3}{2}\pi - 0} \int_0^{\frac{3}{2}\pi} \sin 2x dx = \frac{2}{3\pi} = \sin 2\xi$$

따라서 $\sin 2a = \frac{2}{3\pi}$, $\left(0 < a < \frac{\pi}{4}\right)$ 라고 할 때,

$$\xi = a, \xi = \frac{\pi}{2} - a, \xi = \pi + a, \xi = \frac{3}{2}\pi - a$$

이므로 모든 ξ 값의 합은 $\frac{\pi}{2} + \frac{5}{2}\pi = 3\pi$ 이다.

$$15. \frac{\int_x^{f(x)} \frac{3+2t^2}{1+t^2} dt}{f(x)-x} = \frac{3+2c^2}{1+c^2}, \quad x < c < f(x)$$

$$\Leftrightarrow f(x) - x = \frac{1+c^2}{3+2c^2}$$

$$\therefore \lim_{x \rightarrow \infty} (f(x) - x) = \lim_{c \rightarrow \infty} \frac{1+c^2}{3+2c^2} = \frac{1}{2}$$

$$16. 넓이 = f(t) = \int_t^{2t} \sin x dx \text{의 최댓값}$$

$$\Rightarrow f'(t) = 2 \sin 2t - \sin t = 4 \sin t \cos t - \sin t = \sin t (4 \cos t - 1)$$

$$\Rightarrow t = \cos^{-1} \frac{1}{4} \quad \left(\because 0 < t < \frac{\pi}{2} \right)$$

$$\begin{aligned}
\Rightarrow f\left(\cos^{-1} \frac{1}{4}\right) &= \int_{\cos^{-1} \frac{1}{4}}^{2 \cos^{-1} \frac{1}{4}} \sin x dx \\
&= -\left[\cos 2\left(\cos^{-1} \frac{1}{4}\right) - \cos\left(\cos^{-1} \frac{1}{4}\right) \right] = \frac{9}{8}
\end{aligned}$$

$$17. A(t) = \frac{1}{2} \cosh t \sinh t - \int_1^{\cosh t} \sqrt{x^2 - 1} dx$$

$$A'(t) = \frac{1}{2} (\sinh^2 t + \cosh^2 t) - \sqrt{\cosh^2 t - 1} \sinh t$$

$$= \frac{1}{2} (\sinh^2 t + \cosh^2 t) - \sinh^2 t$$

$$= \frac{1}{2} (\cosh^2 t - \sinh^2 t) = \frac{1}{2}$$

$$\begin{aligned}
18. \lim_{x \rightarrow 1} \frac{\int_1^x (x-t)f(t)dt}{(x-1)^2} &= \lim_{x \rightarrow 1} \frac{x \int_1^x f(t)dt - \int_1^x t f(t)dt}{(x-1)^2} \\
&= \lim_{x \rightarrow 1} \frac{\int_1^x f(t)dt}{2(x-1)} = \lim_{x \rightarrow 1} \frac{f(x)}{2} = \frac{f(1)}{2} \quad (\because \text{로피탈 정리}) \\
&= \frac{1+1+1^2+\dots+1^{99}}{2} = 50
\end{aligned}$$

19. (i) $x > 0$ 일 때,

$$\int_{-x}^x (x^2 - t^2 + |t|)f(t)dt = x|x|$$

$$\Leftrightarrow \int_{-x}^x (x^2 - t^2 + |t|)f(t)dt = x^2$$

$$\Rightarrow x^2 \int_{-x}^x f(t)dt + \int_{-x}^x (-t^2 + |t|)f(t)dt = x^2$$

양변을 x 로 미분하면

$$\Rightarrow 2x \int_{-x}^x f(t)dt + x^2 \{f(x) + f(-x)\}$$

$$+ (-x^2 + |x|)f(x) + (-x^2 + |-x|)f(-x) = 2x$$

$$\Rightarrow 2x \int_{-x}^x f(t)dt + x \{f(x) + f(-x)\} = 2x \quad (\because x > 0)$$

$$\Rightarrow 2 \int_{-x}^x f(t)dt + \{f(x) + f(-x)\} = 2$$

양변을 x 로 미분하면

$$2\{f(x) + f(-x)\} + \{f'(x) - f'(-x)\} = 0$$

$$\Rightarrow \frac{f'(x) - f'(-x)}{f(x) + f(-x)} = -2$$

양변을 적분하면

$$\ln |f(x) + f(-x)| = -2x + C$$

$$\Rightarrow f(0) = 1 \text{ 이므로 } C = \ln 2$$

$$\Rightarrow \ln|f(x) + f(-x)| = -2x + \ln 2$$

$$\text{즉, } f(x) + f(-x) = e^{-2x + \ln 2} \text{이므로 } f(1) + f(-1) = 2e^{-2}$$

(ii) $x < 0$ 일 때도 같은 방법으로 구할 수 있다.

[다른 풀이]

$x > 0$ 일 때,

$$\int_{-x}^x (x^2 - t^2 + |t|)f(t)dt = x|x|$$

$$\Leftrightarrow \int_{-x}^x (x^2 - t^2 + |t|)f(t)dt = x^2 \text{에서}$$

라이프니츠 정리를 활용하면

$$|x|f(x) + |-x|f(-x) + \int_{-x}^x 2xf(t)dt = 2x$$

$$\Leftrightarrow x(f(x) + f(-x)) + \int_{-x}^x 2xf(t)dt = 2x \text{이다.}$$

$$\begin{aligned} \text{이때, } \int_{-x}^x 2xf(t)dt &= \int_{-x}^0 2xf(t)dt + \int_0^x 2xf(t)dt \\ &= \int_0^x 2xf(-\alpha)d\alpha + \int_0^x 2xf(t)dt \quad (\because t = -\alpha) \\ &\Rightarrow \int_0^x 2xf(-t)dt + \int_0^x 2xf(t)dt \\ &= 2x \int_0^x (f(t) + f(-t))dt \text{이므로} \end{aligned}$$

$$x(f(x) + f(-x)) + \int_{-x}^x 2xf(t)dt = 2x$$

$$\Leftrightarrow x(f(x) + f(-x)) + 2x \int_0^x (f(t) + f(-t))dt = 2x$$

$$\Leftrightarrow (f(x) + f(-x)) + 2 \int_0^x (f(t) + f(-t))dt = 2 \quad (\because x \neq 0)$$

이제 $f(x) + f(-x) = g(x)$ 라 하면

$$\Leftrightarrow g(x) + 2 \int_0^x g(t)dt = 2$$

$$\Leftrightarrow g'(x) + 2g(x) = 0 \quad (\because \text{양변 미분})$$

$$\Leftrightarrow \frac{g'(x)}{g(x)} = -2$$

$$\ln|g(x)| = -2x + C \text{에서}$$

$$f(0) + f(-0) = g(0) = 2 \text{이므로 } C = \ln 2 \text{이다.}$$

$$\text{즉, } g(x) = e^{-2x + \ln 2} \text{이므로}$$

$$\therefore f(1) + f(-1) = g(1) = 2e^{-2}$$



1. 2번	2. 3번	3. 2번	4. 2번	5. 1번
6. 1번	7. 1번	8. 5번	9. 3번	10. 2번
11. 4번	12. 4번	13. 1번	14. 4번	15. 4번
16. 2번	17. 3번	18. 1번	19. 4번	20. 4번
21. 4번	22. 1번	23. 4번	24. 4번	

$$1. \int_0^1 \ln x dx = [x \ln x - x]_0^1 = -1$$

$$2. \int_{-\frac{2}{3}}^{\infty} \frac{1}{(3x-1)^2+9} dx = \frac{1}{3} \times \frac{1}{3} \left[\tan^{-1} \frac{3x-1}{3} \right]_{-\frac{2}{3}}^{\infty} = \frac{\pi}{12}$$

$$3. \int_0^{\infty} e^{-x} \cos x dx = \left[\frac{e^{-x}}{2} (-\cos x + \sin x) \right]_0^{\infty} = \frac{1}{2}$$

$$4. \int_0^1 x \ln x dx = \left[\frac{1}{2} x^2 \ln x \right]_0^1 - \int_0^1 \frac{1}{2} x dx = 0 - \left[\frac{1}{4} x^2 \right]_0^1 = -\frac{1}{4}$$

$$5. \int_0^{\infty} \frac{1}{(x^2+1)^2} dx = \int_0^{\frac{\pi}{2}} \frac{\sec^2 \theta}{\sec^4 \theta} d\theta \quad (\because x = \tan \theta)$$

$$= \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta = \frac{1}{2} \times \frac{\pi}{2}$$

$$6. \int_{-\infty}^{\infty} \frac{1}{\pi} \frac{1}{1+x^2} dx = \frac{1}{\pi} [\tan^{-1} x]_{-\infty}^{\infty} = \frac{1}{\pi} \times \pi$$

$$7. \int_0^{\infty} \frac{1}{e^x+1} dx = \int_1^{\infty} \frac{1}{(t+1)t} dt \quad (\because e^x = t)$$

$$= \int_1^{\infty} \frac{1}{t} - \frac{1}{t+1} dt$$

$$= [\ln t - \ln(t+1)]_1^{\infty} = \left[\ln \frac{t}{t+1} \right]_1^{\infty} = -\ln \frac{1}{2} = \ln 2$$

$$8. \int_0^{\infty} \frac{20x}{(x^2+1)(3x+1)} dx = \int_0^{\infty} \frac{2x+6}{x^2+1} + \frac{-6}{3x+1} dx$$

$$= \int_0^{\infty} \frac{2x}{x^2+1} + \frac{6}{x^2+1} + \frac{-6}{3x+1} dx$$

$$= [\ln(x^2+1) + 6 \tan^{-1} x - 2 \ln(3x+1)]_0^{\infty}$$

$$= \left[\ln \frac{x^2+1}{(3x+1)^2} + 6 \tan^{-1} x \right]_0^{\infty} = \ln \frac{1}{9} + 6 \times \frac{\pi}{2}$$

$$9. \int_{e^3}^{\infty} \frac{dx}{x \ln x (\ln \ln x)^3} = \int_{\ln 3}^{\infty} \frac{1}{t^3} dt \quad (\because \ln \ln x = t, \frac{1}{x \ln x} dx = dt)$$

$$= \left[-\frac{1}{2t^2} \right]_{\ln 3}^{\infty} = \frac{1}{2(\ln 3)^2}$$

$$10. \int_0^{\infty} t^5 e^{-t^3} dt = \frac{1}{3} \int_0^{\infty} x e^{-x} dx \quad (\because t^3 = x) = \frac{1}{3} \times \Gamma(2) = \frac{1}{3} \times 1!$$

$$11. I = \int_{-1}^2 \frac{1}{x^2} dx \text{ 는 } p=2 > 1 \text{ 이므로 발산한다.}$$

$$12. \neg. \int_0^{\infty} e^{-x} dx = \int_0^{\infty} \frac{1}{e^x} dx : \text{수렴}$$

$$\neg. \int_0^{\infty} \frac{1}{1+e^x} dx \approx \int_0^{\infty} \frac{1}{e^x} dx : \text{수렴}$$

$$\neg. \int_{-\infty}^0 e^{-x} dx \quad (\because -x=t) = \int_0^{\infty} e^t dt : \text{발산}$$

$$\neg. \int_0^e \frac{1}{x} dx : \text{발산 } (p=1)$$

$$13. \textcircled{1} \int_2^{\infty} \frac{dx}{(x-1)^{3/2}} \approx \int_2^{\infty} \frac{1}{x^{3/2}} dx : \text{수렴 } (p > 1)$$

$$\textcircled{2} \int_0^1 \frac{1}{x} dx : \text{발산 } (p=1)$$

$$\textcircled{3} \int_1^{\infty} \frac{x}{x^2+1} dx \approx \int_1^{\infty} \frac{1}{x} dx : \text{발산 } (p=1)$$

$$\textcircled{4} \int_0^{\infty} \frac{dx}{\sqrt{x+1}} \approx \int_1^{\infty} \frac{1}{\sqrt{x}} dx : \text{발산 } (p < 1)$$

$$\textcircled{5} \int_1^3 \frac{1}{x-2} dx : \text{발산 } (p=1)$$

$$14. \textcircled{1} \int_1^{\infty} \frac{1}{x} dx : \text{발산 } (p=1)$$

$$\textcircled{2} \int_0^2 \frac{1}{(x-1)^2} dx : \text{발산 } (p > 1)$$

$$\textcircled{3} \int_0^{\frac{\pi}{2}} \sec x dx = [\ln(\sec x + \tan x)]_0^{\frac{\pi}{2}} = \infty : \text{발산}$$

$$\textcircled{4} \int_{-\infty}^0 x e^x dx \quad (\because -x=t)$$

$$= \int_{\infty}^0 t e^{-t} dt = - \int_0^{\infty} \frac{t}{e^t} dt \approx - \int_0^{\infty} \frac{1}{e^t} dt : \text{수렴}$$

$$15. \textcircled{1} \int_0^{\infty} \frac{1}{1+x^2} dx \text{ 이 수렴}$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = 2 \int_0^{\infty} \frac{1}{1+x^2} dx : \text{수렴 } (p > 1)$$

$$\textcircled{2} \int_2^5 \frac{1}{\sqrt{x-2}} dx : \text{수렴 } (p < 1)$$

$$\textcircled{3} \int_1^{\infty} e^{-x^2} dx = \int_1^{\infty} \frac{1}{e^{x^2}} dx : \text{수렴}$$

$$\textcircled{4} \int_1^{\infty} \frac{1+e^{-x}}{x} dx \approx \int_1^{\infty} \frac{1}{x} dx : \text{발산 } (p=1)$$

$$16. \neg. \int_1^{\infty} \frac{\cos^4 x}{x^3+1} dx \approx \int_1^{\infty} \frac{1}{x^3} dx : \text{수렴 } (p > 1)$$

$$\neg. \int_0^1 \frac{e^{-x}}{x^2} dx \approx \int_0^1 \frac{1}{x^2} dx : \text{발산 } (p > 1)$$

$$\neg. \int_2^{\infty} \frac{x^2}{\sqrt{x^5-1}} dx \approx \int_2^{\infty} \frac{1}{\sqrt{x}} dx : \text{발산 } (p < 1)$$

$$17. \textcircled{1} \int_1^{\infty} \frac{1}{x+\sqrt{x}} dx \approx \int_1^{\infty} \frac{1}{x} dx : \text{발산 } (p=1)$$

$$\textcircled{2} \int_0^1 \frac{\ln x}{x} dx : \text{발산 } (p=1)$$

$$\textcircled{3} \int_1^{\infty} \frac{1}{x+e^x} dx \approx \int_1^{\infty} \frac{1}{e^x} dx : \text{수렴}$$

$$\textcircled{4} \int_0^1 \frac{1}{x^2} dx : \text{발산 } (p > 1)$$

18. 가. $\int_0^{\infty} \frac{1}{\sqrt{x}} dx = \int_0^1 \frac{1}{\sqrt{x}} dx + \int_1^{\infty} \frac{1}{\sqrt{x}} dx$: 수렴 + 발산 = 발산

나. $\int_0^2 \frac{1}{x-1} dx$: 발산 ($p=1$)

다. $\int_0^{\frac{\pi}{2}} \sec x dx$: 발산

라. $\int_2^4 \frac{1}{\sqrt[3]{x-2}} dx = \int_2^4 \frac{1}{(x-2)^{\frac{1}{3}}} dx$: 수렴 ($p < 1$)

19. ① $\int_0^{\infty} \frac{1}{1+x^2} dx \approx \int_1^{\infty} \frac{1}{x^2} dx$: 수렴 ($p > 1$)

② $\int_1^{\infty} e^{-x} dx = \int_1^{\infty} \frac{1}{e^x} dx$: 수렴

③ $\int_0^1 \ln x dx = -1$: 수렴

④ $\int_0^1 \frac{\sqrt{x}}{1-x} dx \approx \int_0^1 \frac{1}{1-x} dx$: 발산 ($p=1$)

20. ① $\int_1^{\infty} \frac{1}{x+x^2} dx \approx \int_1^{\infty} \frac{1}{x^2} dx$: 수렴 ($p > 1$)

② $\int_0^{\infty} \frac{1}{\sqrt{x}} e^{-3x} dx$
 $= \int_0^1 \frac{1}{\sqrt{x}} e^{-3x} dx + \int_1^{\infty} \frac{1}{\sqrt{x} e^{3x}} dx$
 $\approx \int_0^1 \frac{1}{\sqrt{x}} dx + \int_1^{\infty} \frac{1}{e^{3x}} dx$: 수렴 + 수렴 = 수렴

③ $\int_0^1 \frac{\sin x}{x^{3/2}} dx \approx \int_0^1 \frac{x}{x^{3/2}} dx = \int_0^1 \frac{1}{x^{1/2}} dx$: 수렴 ($p < 1$)

④ $\int_0^1 \frac{\cos \pi x}{1-x} dx \approx \int_0^1 \frac{-1}{1-x} dx$: 발산 ($p=1$)

21. ㄱ. $\int_1^{\infty} (\sqrt{x^4+x+1} - \sqrt{x^4+1}) dx$
 $= \int_1^{\infty} \frac{(\sqrt{x^4+x+1} - \sqrt{x^4+1})(\sqrt{x^4+x+1} + \sqrt{x^4+1})}{(\sqrt{x^4+x+1} + \sqrt{x^4+1})} dx$
 $= \int_1^{\infty} \frac{x}{\sqrt{x^4+x+1} + \sqrt{x^4+1}} dx$
 $\approx \int_1^{\infty} \frac{x}{2x^2} dx = \frac{1}{2} \int_1^{\infty} \frac{1}{x} dx$: 발산 ($p=1$)

ㄴ. $\int_2^{\infty} \frac{\sin x}{x \log x} dx$: 수렴 ($\because$ 디리클레 판정법)

ㄷ. $\int_0^1 x \sin\left(\frac{1}{x}\right) dx$ ($\because \frac{1}{x} = t$)
 $= \int_1^{\infty} \frac{\sin t}{t^3} dt$: 수렴 ($\because$ 디리클레 판정법)

22. ① $\int_1^2 \frac{dx}{x^2-1} = \int_1^2 \frac{1}{(x-1)(x+1)} dx$
 $\approx \frac{1}{2} \int_1^2 \frac{1}{(x-1)} dx$: 발산 ($p=1$)

② $\int_2^{\infty} \frac{\ln x}{x^2} dx \approx \int_2^{\infty} \frac{1}{x^2} dx$: 수렴 ($p > 1$)

③ $\int_1^2 \frac{1}{\sqrt{x-1}} dx$: 수렴 ($p < 1$)

④ $\int_2^{\infty} \frac{dx}{x(\ln x)^2}$: 수렴 ($p > 1$) (아래 공식 참고)

23. $\int_e^{\infty} \frac{1}{x(\ln x)^p} dx = \int_1^{\infty} \frac{1}{t^p} dt$ ($\because \ln x = t$) 이므로 $p > 1$ 일 때 수렴한다.

24. $\int_0^1 x^p \ln x dx = \int_0^1 \frac{\ln x}{x^{-p}} dx$ 이므로 $-p < 1 \Leftrightarrow p > -1$ 일 때 수렴한다.

1. 1번	2. 1번	3. 3번	4. 3번	5. 5번
6. 3번	7. 4번	8. 1번	9. 5번	10. 4번
11. 3번	12. 3번			

$$\begin{aligned}
 1. I &= \int_0^\infty \left(\frac{1}{\sqrt{x^2+9}} - \frac{a}{x+1} \right) dx \\
 &= \left[\sinh^{-1} \left(\frac{x}{3} \right) - a \ln(x+1) \right]_0^\infty \\
 &= \left[\ln \left(\frac{x}{3} + \sqrt{\frac{x^2}{9} + 1} \right) - a \ln(x+1) \right]_0^\infty \\
 &= \left[\ln \left(\frac{\frac{x}{3} + \sqrt{\frac{x^2}{9} + 1}}{(x+1)^a} \right) \right]_0^\infty \quad (a=1) \\
 &= \left[\ln \left(\frac{\frac{x}{3} + \sqrt{\frac{x^2}{9} + 1}}{(x+1)} \right) \right]_0^\infty = \ln \left(\frac{2}{3} \right) \text{이다.} \\
 \text{즉, } a=1, b=1, c=2, d=3 \text{이고 } a+b+c-d=1 \text{이다.}
 \end{aligned}$$

$$\begin{aligned}
 2. (가) \int_0^1 \frac{\cos x}{2x} dx &\approx \int_0^1 \frac{1}{2x} dx \text{ 이므로 발산한다.} \\
 (나) \int_{-\infty}^{-1} \frac{1}{\sqrt{3-x}} dx &= \int_4^\infty \frac{1}{\sqrt{t}} dt \quad (\because 3-x=t) \text{ 이므로 발산한다.} \\
 (다) \int_0^1 \frac{e^x}{\sqrt{2x}} dx &\approx \int_0^1 \frac{1}{\sqrt{2x}} dx \text{ 이므로 수렴한다.} \\
 (라) \int_0^1 \frac{\ln x}{1+x^3} dx &\approx \int_0^1 \ln x dx = -1 \text{ 이므로 수렴한다.}
 \end{aligned}$$

$$\begin{aligned}
 3. \neg. \int_0^1 \frac{x}{1-x} dx &\approx \int_0^1 \frac{1}{1-x} dx \text{ 이므로 발산한다.} \\
 \neg. \int_{-1}^1 \frac{1}{x^2} dx &\text{은 발산한다.} \\
 \neg. \int_{-2}^2 \frac{1}{\sqrt{4-x^2}} dx &= \left[\sin^{-1} \frac{x}{2} \right]_{-2}^2 = \pi \text{이므로 수렴한다.}
 \end{aligned}$$

[다른 풀이]

$$\begin{aligned}
 \int_{-2}^2 \frac{1}{\sqrt{4-x^2}} dx &= \int_{-2}^0 \frac{1}{\sqrt{(2-x)(2+x)}} dx + \int_0^2 \frac{1}{\sqrt{(2-x)(2+x)}} dx \\
 &\approx \int_{-2}^0 \frac{1}{\sqrt{4(2+x)}} dx + \int_0^2 \frac{1}{\sqrt{4(2-x)}} dx \\
 &= \text{수렴} + \text{수렴} = \text{수렴}
 \end{aligned}$$

$$\begin{aligned}
 \neg. \int_{-\infty}^\infty \frac{1}{(x^2+1)^2} dx &= \int_{-\infty}^0 \frac{1}{(x^2+1)^2} dx + \int_0^\infty \frac{1}{(x^2+1)^2} dx \text{ 이므로} \\
 \int_0^\infty \frac{1}{(x^2+1)^2} dx &\approx \int_0^\infty \frac{1}{x^4} dx \text{은 수렴한다.} \\
 \text{즉, 우함수이므로 } \int_{-\infty}^\infty \frac{1}{(x^2+1)^2} dx &= 2 \int_0^\infty \frac{1}{(x^2+1)^2} dx \text{은} \\
 \text{수렴한다.}
 \end{aligned}$$

$$\begin{aligned}
 4. (\neg) \int_{-\infty}^\infty \frac{x}{1+x^4} dx &= \int_{-\infty}^0 \frac{x}{1+x^4} dx + \int_0^\infty \frac{x}{1+x^4} dx \text{ 이므로} \\
 \int_0^\infty \frac{x}{1+x^4} dx &\approx \int_0^\infty \frac{1}{x^3} dx \text{은 수렴한다.} \\
 \text{즉, 수렴하는 기함수이므로 } \int_{-\infty}^\infty \frac{x}{1+x^4} dx &= 0 \text{으로 수렴한다.}
 \end{aligned}$$

$$\begin{aligned}
 (\neg) \int_1^\infty \frac{1}{\ln x} dx &\text{은 } p=1 \text{이므로 발산한다.} \\
 (\neg) \int_{-1}^1 \frac{1}{x} dx &\text{은 발산한다.} \\
 (\neg) \int_3^\infty \frac{1}{x(\ln x)^2} dx &= \int_3^\infty \frac{1}{x(\ln x)^p} dx = \int_3^\infty \frac{1}{t^p} dt \quad (\because \ln x = t) \\
 \text{에서 } p=2 > 1 \text{이므로 수렴한다.}
 \end{aligned}$$

$$\begin{aligned}
 5. (\neg) \int_1^\infty \frac{\ln x}{x^3} dx &\approx \int_1^\infty \frac{1}{x^3} dx \text{은 수렴한다.} \\
 (\neg) \int_1^\infty \frac{\sin x}{x} dx &\text{은 디리클레 판정법에 의하여 수렴한다.} \\
 (\neg) \int_0^1 (\ln x)^2 dx &= (-1)^2 \times 2! \text{이므로 수렴한다.}
 \end{aligned}$$

[다른 풀이]

$$\int_0^1 (\ln x)^2 dx = \int_0^1 \frac{1}{(\ln x)^{-2}} dx \text{는 } p=-2 < 1 \text{이므로 수렴한다.}$$

[참고]

$$\int_0^1 (\ln x)^n dx = (-1)^n n!$$

$$\begin{aligned}
 6. (가) \int_{-\infty}^0 x^2 e^x dx &= - \int_\infty^0 t^2 e^{-t} dt \quad (\because x=-t) \\
 &= - \int_0^\infty \frac{t^2}{e^t} dt \approx - \int_0^\infty \frac{1}{e^t} dt \text{이므로 수렴한다.} \\
 (나) \int_1^\infty \frac{2+e^{-2x^2}}{1+\ln x} dx &\approx \int_1^\infty \frac{2}{\ln x} dx \quad (\text{특이점이 존재하지 않는다.}) \\
 \text{이때, } \ln x = t \text{로 놓으면 } \int_1^\infty \frac{2e^t}{t} dt &\text{이므로 발산한다.}
 \end{aligned}$$

$$\begin{aligned}
 (다) \int_0^{\frac{\pi}{3}} \frac{1}{x^2 \sqrt{\sin x}} dx &= \int_0^{\frac{\pi}{3}} \frac{1}{x^2 \sqrt{x - \frac{1}{3!}x^3 + \dots}} dx \\
 &= \int_0^{\frac{\pi}{3}} \frac{1}{x^2 \sqrt{1 - \frac{1}{3!}x^2 + \dots}} dx \\
 &\approx \int_0^{\frac{\pi}{3}} \frac{1}{x^2 \sqrt{x}} dx = \int_0^{\frac{\pi}{3}} \frac{1}{x^{\frac{5}{2}}} dx \text{이므로 발산한다.}
 \end{aligned}$$

$$(라) \int_0^{\frac{2}{3}} \frac{1}{\sqrt{|3x-1|}} dx = \int_0^{\frac{2}{3}} \frac{1}{|3x-1|^{\frac{1}{2}}} dx \text{은 수렴한다.}$$

$$\begin{aligned}
 7. (가) \int_1^\infty \frac{e^{-x^2}}{|x-3|^{1/2}} dx &= \int_1^4 \frac{e^{-x^2}}{|x-3|^{1/2}} dx + \int_4^\infty \frac{e^{-x^2}}{|x-3|^{1/2}} dx \\
 &\approx \int_1^4 \frac{e^{-9}}{|x-3|^{1/2}} dx + \int_4^\infty e^{-x^2} dx \text{일 때,} \\
 \int_1^4 \frac{e^{-9}}{|x-3|^{1/2}} dx &\text{은 수렴하고, } \int_4^\infty \frac{1}{e^{x^2}} dx \text{은 수렴하므로} \\
 \int_1^\infty \frac{e^{-x^2}}{|x-3|^{1/2}} dx &\text{은 수렴한다.} \\
 (나) \int_{-\infty}^\infty \frac{x}{(x^2+2)\ln(x^2+2)} dx &= \int_{-\infty}^0 \frac{x}{(x^2+2)\ln(x^2+2)} dx + \int_0^\infty \frac{x}{(x^2+2)\ln(x^2+2)} dx \text{일 때,} \\
 \int_0^\infty \frac{x}{(x^2+2)\ln(x^2+2)} dx &= \int_0^\infty \frac{x}{(x^2+2)\ln(x^2+2)} dx \quad (\because \text{특이점이 존재하지 않는다.})
 \end{aligned}$$

$$\approx \int_1^{\infty} \frac{x}{x^2 \ln(x^2)} dx = \int_1^{\infty} \frac{1}{2x \ln x} dx \text{은 발산한다.}$$

$$\text{따라서 } \int_{-\infty}^{\infty} \frac{x}{(x^2+2)\ln(x^2+2)} dx \text{은 발산한다.}$$

$$\text{다. } \int_{-\infty}^{\infty} x^4 e^{-x^2} dx = \int_{-\infty}^0 x^4 e^{-x^2} dx + \int_0^{\infty} x^4 e^{-x^2} dx \text{일 때,}$$

$$\int_0^{\infty} x^4 e^{-x^2} dx = \int_0^{\infty} \frac{x^4}{e^{x^2}} dx \approx \int_0^{\infty} \frac{1}{e^{x^2}} dx \text{은 수렴한다.}$$

$$\text{이때, 수렴하는 우함수이므로 } \int_{-\infty}^{\infty} x^4 e^{-x^2} dx = 2 \int_0^{\infty} x^4 e^{-x^2} dx \text{은 수렴한다.}$$

$$\text{라. } \int_{-\infty}^{\infty} x^{100} e^{-x^2} dx = \int_{-\infty}^0 x^{100} e^{-x^2} dx + \int_0^{\infty} x^{100} e^{-x^2} dx \text{일 때,}$$

$$\int_0^{\infty} x^{100} e^{-x^2} dx = \int_0^{\infty} \frac{x^{100}}{e^{x^2}} dx \approx \int_0^{\infty} \frac{1}{e^{x^2}} dx \text{이므로 수렴하고,}$$

$$\int_{-\infty}^0 x^{100} e^{-x^2} dx = - \int_0^{\infty} t^{100} e^{-t^2} dt = \int_0^{\infty} \frac{t^{100}}{e^{t^2}} dt \approx \int_0^{\infty} \frac{1}{e^{t^2}} dt$$

이므로 수렴한다.

$$\text{즉, } \int_{-\infty}^{\infty} x^{100} e^{-x^2} dx \text{은 수렴+수렴이므로 수렴한다.}$$

$$8. \text{ 가. } \int_0^1 \frac{x}{\ln x} dx \approx \int_0^1 \frac{1}{\ln x} dx \text{은 발산한다.}$$

$$\text{나. } \int_0^1 \frac{1}{\ln(2x)} dx = \frac{1}{2} \int_0^2 \frac{1}{\ln t} dt \quad (\because 2x=t) \text{은 발산한다.}$$

$$\text{다. } \int_0^1 \frac{1}{x \ln x} dx = \int_0^1 \frac{1}{e^{-t} t} e^{-t} dt \quad (\because \ln x = -t) \\ = - \int_0^1 \frac{1}{t} dt = - \left(\int_0^1 \frac{1}{t} dt + \int_1^{\infty} \frac{1}{t} dt \right) \text{은 발산한다.}$$

$$\text{라. } \int_0^1 \frac{1}{x(\ln(2x))^2} dx = \int_0^2 \frac{1}{t(\ln t)^2} dt \quad (\because 2x=t)$$

$$= \int_{-\infty}^{-\ln 2} \frac{1}{e^{-\theta}(-\theta)^2} e^{-\theta} d\theta \quad (\because \ln t = -\theta)$$

$$= \int_{-\ln 2}^{\infty} \frac{1}{\theta^2} d\theta = \int_{-\ln 2}^1 \frac{1}{\theta^2} d\theta + \int_1^{\infty} \frac{1}{\theta^2} d\theta$$

은 발산+수렴이므로 발산한다.

$$9. \text{ 가. } \int_1^{\infty} \frac{\sin x}{x} dx \text{은 디리클레 판정법에 의하여 수렴한다.}$$

$$\text{나. } \int_0^1 \frac{1}{\sqrt{x}} \cos \frac{1}{x} dx = \int_{\infty}^1 \sqrt{t} \times \cos t \times -\frac{1}{t^2} dt \quad (\because \frac{1}{x}=t) \\ = \int_1^{\infty} \frac{\cos t}{t^{\frac{3}{2}}} dt \text{이므로 디리클레 판정법에 의하여 수렴한다.}$$

$$\text{다. } \int_{\pi}^{\infty} \sin(x^2) dx \text{은 } p > 1 \text{이므로 수렴한다.}$$

$$\text{라. } \int_0^{\infty} \frac{\sqrt{x}}{x+x^2} dx = \int_0^1 \frac{\sqrt{x}}{x+x^2} dx + \int_1^{\infty} \frac{\sqrt{x}}{x+x^2} dx \text{일 때,}$$

$$\int_0^1 \frac{\sqrt{x}}{x+x^2} dx = \int_0^1 \frac{\sqrt{x}}{x(1+x)} dx \approx \int_0^1 \frac{\sqrt{x}}{x} dx = \int_0^1 \frac{1}{\sqrt{x}} dx$$

이므로 수렴하고,

$$\int_1^{\infty} \frac{\sqrt{x}}{x+x^2} dx \approx \int_1^{\infty} \frac{\sqrt{x}}{x^2} dx = \int_1^{\infty} \frac{1}{x^{\frac{3}{2}}} dx \text{이므로 수렴한다.}$$

$$\text{즉, } \int_0^{\infty} \frac{\sqrt{x}}{x+x^2} dx \text{은 수렴+수렴이므로 수렴한다.}$$

$$10. \text{ a. } \int_1^{\infty} \frac{|\sin x|}{x} dx \approx \int_1^{\infty} \frac{1}{x} dx \text{은 발산한다.}$$

$$\text{b. } \int_1^9 \frac{1}{\sqrt[3]{x-1}} dx = \int_1^9 \frac{1}{(x-1)^{\frac{1}{3}}} dx \text{은 수렴한다.}$$

$$\text{c. } \int_0^{\infty} \frac{\arctan(e^x)}{2+e^x} dx \approx \int_0^{\infty} \frac{1}{e^x} dx \text{은 수렴한다.}$$

$$\text{d. } \int_0^1 \frac{\ln x}{x} dx \text{은 발산한다.}$$

$$11. \text{ a. } \int_1^2 \frac{x+1}{x\sqrt{x-1}} dx \approx \int_1^2 \frac{2}{(x-1)^{\frac{1}{2}}} dx \text{은 수렴한다.}$$

$$\text{b. } \int_0^1 \frac{1}{\sqrt{x^2+2x}} dx = \int_0^1 \frac{1}{\sqrt{x(x+2)}} dx \approx \int_0^1 \frac{1}{\sqrt{2x}} dx \text{은 수렴한다.}$$

$$\text{c. } \int_1^{\infty} \frac{1}{x \ln x} dx \text{은 발산한다.}$$

$$\text{d. } \int_{-\infty}^{\infty} \operatorname{sech} x dx = \int_{-\infty}^{\infty} \frac{2}{e^x + e^{-x}} dx \\ = \int_{-\infty}^0 \frac{2}{e^x + e^{-x}} dx + \int_0^{\infty} \frac{2}{e^x + e^{-x}} dx \text{일 때,} \\ \int_0^{\infty} \frac{2}{e^x + e^{-x}} dx \approx \int_0^{\infty} \frac{2}{e^x} dx \text{이므로 수렴한다.}$$

따라서 수렴하는 우함수이므로

$$\int_{-\infty}^{\infty} \frac{2}{e^x + e^{-x}} dx = 2 \int_0^{\infty} \frac{2}{e^x + e^{-x}} dx \text{은 수렴한다.}$$

$$12. \text{ a. } \int_2^{\infty} e^{-\sqrt{\ln x}} dx = \int_{\ln 2}^{\infty} e^{-\sqrt{t}} e^t dt \quad (\because \ln x = t)$$

$$= \int_{\ln 2}^{\infty} e^{t-\sqrt{t}} dt \approx \int_{\ln 2}^{\infty} e^t dt \text{은 발산한다.}$$

$$\text{b. } \int_{-1}^0 \frac{e^x}{x^3} dx = - \int_1^{\infty} t e^{-t} dt \quad (\because \frac{1}{x} = -t) \\ = - \int_1^{\infty} \frac{t}{e^t} dt \approx - \int_1^{\infty} \frac{1}{e^t} dt \text{은 수렴한다.}$$

$$\text{c. } \int_0^{\pi} \frac{x}{1-\cos x} dx = \int_0^{\pi} \frac{x}{1-\left(1-\frac{1}{2!}x^2+\frac{1}{4!}x^4-\dots\right)} dx \\ = \int_0^{\pi} \frac{x}{\frac{1}{2!}x^2-\frac{1}{4!}x^4+\dots} dx = \int_0^{\pi} \frac{x}{x^2\left(\frac{1}{2!}-\frac{1}{4!}x^2+\dots\right)} dx$$

$$\approx \frac{1}{\frac{1}{2!}} \int_0^{\pi} \frac{x}{x^2} dx = 2! \int_0^{\pi} \frac{1}{x} dx \text{은 발산한다.}$$

[다른 풀이]

$$\int_0^{\pi} \frac{x}{1-\cos x} dx = \int_0^{\pi} \frac{x^2}{1-\cos x} \cdot \frac{1}{x} dx$$

$$\approx 2 \int_0^{\pi} \frac{1}{x} dx \text{는 발산한다. } \left(\because \lim_{x \rightarrow 0} \frac{1-\cos x}{x^2} = \frac{1}{2} \right)$$

$$\text{d. } \int_0^{\frac{\pi}{2}} \frac{x}{\sqrt{\cos^3 x}} dx = - \int_{\frac{\pi}{2}}^0 \frac{\frac{\pi}{2}-t}{\sqrt{\sin^3 t}} dt \quad (\because \frac{\pi}{2}-x=t)$$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}-t}{\sqrt{\sin^3 t}} dt$$

$$\approx \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{\sqrt{\sin^3 t}} dt = \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{\sqrt{\left(t-\frac{1}{3!}t^3+\dots\right)^3}} dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{\sqrt{t^3\left(1-\frac{1}{3!}t^2+\dots\right)^3}} dt \approx \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{\sqrt{t^3}} dt = \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{t^{\frac{3}{2}}} dt \text{이므로}$$

발산한다.

1. $p < 1, -\frac{1}{(1-p)^2}$	2. 1번	3. 4번	4. 2번
5. 3번	6. 3번	7. 2번	8. 2번
9. 3번			

$$1. I = \int_1^e \frac{\ln(\ln x)}{x(\ln x)^p} dx = \int_0^1 \frac{\ln t}{t^p} dt \quad (\because \ln x = t)$$

이므로 수렴하기 위한 $p < 1$ 이다.

$$\therefore I = \int_0^1 \frac{\ln t}{t^p} dt = \left[\frac{1}{1-p} t^{1-p} \ln t - \frac{1}{(1-p)^2} t^{1-p} \right]_0^1$$

$$= -\frac{1}{(1-p)^2}$$

$$2. \int_0^\infty \left(\frac{1}{\sqrt{x^2+4}} - \frac{k}{x+2} \right) dx$$

$$= \left[\sinh^{-1} \frac{x}{2} - k \ln(x+2) \right]_0^\infty$$

$$= \left[\ln \left(\frac{x}{2} + \sqrt{\frac{x^2}{4} + 1} \right) - \ln(x+2)^k \right]_0^\infty$$

$$= \left[\ln \frac{\frac{x}{2} + \sqrt{\frac{x^2}{4} + 1}}{(x+2)^k} \right]_0^\infty \quad (k=1)$$

$$= \left[\ln \frac{\frac{x}{2} + \sqrt{\frac{x^2}{4} + 1}}{(x+2)} \right]_0^\infty = -\ln \frac{1}{2} = \ln 2$$

$$3. \int_0^\infty \frac{e^{\beta x^2}}{x^\alpha} dx = \int_0^1 \frac{e^{\beta x^2}}{x^\alpha} dx + \int_1^\infty \frac{e^{\beta x^2}}{x^\alpha} dx \quad \text{일 때,}$$

$$\int_0^1 \frac{e^{\beta x^2}}{x^\alpha} dx \approx \int_0^1 \frac{1}{x^\alpha} dx \quad \text{은 } \alpha < 1 \text{ 이면 수렴하며,}$$

$$\int_1^\infty \frac{e^{\beta x^2}}{x^\alpha} dx \int_1^\infty e^{\beta x^2} dx \quad \text{은 } \beta < 0 \text{ 이면 수렴한다.}$$

따라서 $\alpha < 1, \beta < 0$ 이다. 즉, (가), (나)는 참이다.

(다) [반례]

$$\alpha = 0.5, \beta = -0.1 \text{ 이면 } \int_0^\infty \frac{e^{\beta x^2}}{x^\alpha} dx \text{ 는 수렴하지만}$$

$$\alpha + \beta > 0 \text{ 이다. 따라서 (다)는 거짓이다.}$$

$$4. \int_1^2 \frac{x^x - x}{(x-1)^p} dx \quad (x-1=t \text{ 로 치환})$$

$$= \int_0^1 \frac{(t+1)^{t+1} - (t+1)}{t^p} dt$$

$$= \int_0^1 \frac{(t+1)\{(t+1)^t - 1\}}{t^p} dt$$

$$= \int_0^1 \frac{(t+1)^t - 1}{t^p} dt$$

$$\text{이때, } (t+1)^t = e^{t \ln(t+1)}$$

$$= e^{t \left(t - \frac{1}{2}t^2 + \frac{1}{3}t^3 + \dots \right)}$$

$$= e^{t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots}$$

$$= 1 + \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right) + \frac{1}{2!} \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right)^2 + \dots$$

이다.

$$\int_0^1 \frac{(t+1)^t - 1}{t^p} dt$$

$$= \int_0^1 \frac{1}{t^p} \left(1 + \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right) + \frac{1}{2!} \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right)^2 + \dots - 1 \right) dt$$

$$= \int_0^1 \frac{1}{t^p} \left(\left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right) + \frac{1}{2!} \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right)^2 + \dots \right) dt$$

$$= \int_0^1 \frac{t^2}{t^p} \left(1 - \frac{1}{2}t + \dots \right) dt$$

$$\approx \int_0^1 \frac{t^2}{t^p} dt$$

$$= \int_0^1 \frac{1}{t^{p-2}} dt$$

따라서 $p-2 < 1 \Leftrightarrow p < 3$ 일 때 수렴하므로 가장 큰 자연수는 2이다.

$$5. \Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t} dt = \int_0^\infty \frac{1}{\sqrt{t}} e^{-t} dt$$

$$= \int_0^\infty \frac{1}{u} e^{-u^2} 2u du \quad (\because \sqrt{t} = u)$$

$$= 2 \int_0^\infty e^{-u^2} du$$

$$= \sqrt{\pi} \quad \left(\because \int_0^\infty e^{-u^2} du = \frac{\sqrt{\pi}}{2} \right)$$

$$6. \textcircled{1} \Gamma(n+1) = n\Gamma(n) : \text{참}$$

$$\textcircled{2} n \text{ 은 자연수이면 } \Gamma(n+1) = n! : \text{참}$$

$$\textcircled{3} \Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt = 2 \int_0^\infty s^{2x-1} e^{-s^2} ds \quad (\because t = s^2) : \text{거짓}$$

$$\textcircled{4} \frac{\Gamma(5/2)}{\Gamma(1/2)} = \frac{(3/2)\Gamma(3/2)}{\Gamma(1/2)} = \frac{(3/2)(1/2)\Gamma(1/2)}{\Gamma(1/2)} = \frac{3}{4} : \text{참}$$

$$7. \textcircled{1} f(1) = \int_0^\infty e^{-x} dx = \Gamma(1) = 0! = 1 : \text{참}$$

$$\textcircled{2} f(2) = \int_0^\infty x e^{-x} dx = \Gamma(2) = 1! = 1 : \text{거짓}$$

$$\textcircled{3} \lim_{\alpha \rightarrow 0+} f(\alpha) = \infty : \text{참} (\because \text{감마함수 그래프 참고})$$

$$\textcircled{4} \lim_{\alpha \rightarrow \infty} f(\alpha) = \infty : \text{참} (\because \text{감마함수 그래프 참고})$$

$$8. A(n) = (n-1)A(n-1) \text{ 이므로}$$

$$\frac{A(n)}{A(n-1)} + \frac{A(n-1)}{A(n-2)} + \frac{A(n-2)}{A(n-3)} + \dots + \frac{A(2)}{A(1)}$$

$$= \frac{(n-1)A(n-1)}{A(n-1)} + \frac{(n-2)A(n-2)}{A(n-2)} + \dots + \frac{1A(1)}{A(1)}$$

$$= (n-1) + (n-2) + (n-3) + \dots + 1$$

$$= \frac{n(n-1)}{2}$$

$$9. \int_0^1 (-\ln t)^5 dt = - \int_0^1 (\ln t)^5 dt$$

$$= -(-1)^5 \times 5! = 120$$

[참고] 자연수 n 에 대하여 $\int_0^1 (\ln x)^n dx = (-1)^n n!$ 이 성립한다.

1. 1번	2. 4번	3. 4번	4. 1번	5. 4번
6. 4번	7. 1번	8. 5번	9. 3번	10. 4번
11. 2번	12. 3번	13. 1번	14. 2번	15. 1번
16. 1번				

1. ① 제 2사분면
② 제 3사분면
③ 제 4사분면
④ 제 1사분면

2. 직교좌표 : $(-1, 1) \Leftrightarrow$ 극좌표 : $(\sqrt{2}, \frac{3\pi}{4})$

3. 가. (참)
나. (참)
다. (참)
라. (참)

4. $c^2 = 1^2 + (\sqrt{2})^2 - 2 \times 1 \times \sqrt{2} \times \cos \frac{\pi}{6}$ ($\therefore$ 제 2코사인법칙)
 $\Leftrightarrow c^2 = 3 - \sqrt{6}$ 이므로 $c = \sqrt{3 - \sqrt{6}}$ 이다.

5. $r = 2\sin\theta \Leftrightarrow$ 반지름이 1이고 중심이 $(0, 1)$ 인 원

6. ① 반지름이 2인 원

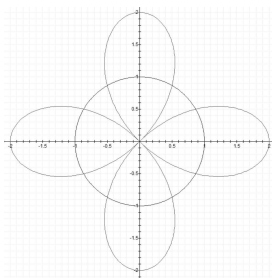
② $r = 4\sin\frac{\theta}{2} \cos\frac{\theta}{2} = 2\sin\theta$ 이므로 반지름이 1인 원

③ $r = 4\left(\sin^2\frac{\theta}{2} - \cos^2\frac{\theta}{2}\right) = -4\cos\theta$ 이므로 반지름이 2인 원

④ $r = 4(\sin\theta + \cos\theta) = 4\sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right)$ 이므로 반지름이 $2\sqrt{2}$ 인 원

7. ① $r = \sin\theta$ 의 그래프는 y 축 대칭이다.

8.



따라서 교점은 8개다.

9. $r = \frac{2}{2 - \cos\theta} \Leftrightarrow r(2 - \cos\theta) = 2$
 $\Leftrightarrow 2r - r\cos\theta = 2 \Leftrightarrow 2\sqrt{x^2 + y^2} - x = 2$
 $\Leftrightarrow 2\sqrt{x^2 + y^2} = x + 2 \Leftrightarrow 4(x^2 + y^2) = (x + 2)^2$
 $\Leftrightarrow 3x^2 - 4x + 4y^2 = 4 \Leftrightarrow 3\left(x^2 - \frac{4}{3}x\right) + 4y^2 = 4$
 $\Leftrightarrow 3\left(x^2 - \frac{4}{3}x + \frac{4}{9}\right) + 4y^2 = 4 + \frac{4}{3}$
 $\Leftrightarrow 3\left(x - \frac{2}{3}\right)^2 + 4y^2 = \frac{16}{3}$

10. $y^2 = x^2 \frac{1+x}{1-x} \Leftrightarrow r^2 \sin^2\theta = r^2 \cos^2\theta \frac{1+r\cos\theta}{1-r\cos\theta}$
 $\Leftrightarrow \sin^2\theta = \cos^2\theta \frac{1+r\cos\theta}{1-r\cos\theta}$
 $\Leftrightarrow \sin^2\theta(1-r\cos\theta) = \cos^2\theta(1+r\cos\theta)$
 $\Leftrightarrow \sin^2\theta - r\sin^2\theta\cos\theta = \cos^2\theta + r\cos^3\theta$
 $\Leftrightarrow r(\cos^3\theta + \sin^2\theta\cos\theta) = \sin^2\theta - \cos^2\theta$
 $\Leftrightarrow r\cos\theta = -\cos 2\theta$
 $\Leftrightarrow r = -\frac{\cos 2\theta}{\cos\theta}$

11. 1) $(r, \theta) = \left(\sqrt{2}, \frac{\pi}{4}\right) \Leftrightarrow (x, y) = (1, 1)$

2) $r = \frac{4}{2\cos\theta - \sin\theta} \Leftrightarrow r(2\cos\theta - \sin\theta) = 4$
 $\Leftrightarrow 2x - y - 4 = 0$

$\therefore$ 최단거리 = $\frac{|2 - 1 - 4|}{\sqrt{4 + 1}} = \frac{3}{\sqrt{5}}$

12. 1) $(r, \theta) = (-r, \pi + \theta)$ 에 대하여

$-r = 1 + \cos\left(\frac{\pi + \theta}{2}\right) \Leftrightarrow r = -1 + \sin\left(\frac{\theta}{2}\right)$

2) $(r, \theta) = (r, 2\pi + \theta)$ 에 대하여

$r = 1 + \cos\left(\frac{2\pi + \theta}{2}\right) \Leftrightarrow r = 1 - \cos\left(\frac{\theta}{2}\right)$

3) $(r, \theta) = (-r, 3\pi + \theta)$ 에 대하여

$-r = 1 + \cos\left(\frac{3\pi + \theta}{2}\right) \Leftrightarrow r = -1 - \sin\left(\frac{\theta}{2}\right)$

13. $x = \sin 3\theta \cos\theta, y = \sin 3\theta \sin\theta$ 일 때,

$\frac{dy}{dx} = \frac{3\cos 3\theta \sin\theta + \sin 3\theta \cos\theta}{3\cos 3\theta \cos\theta - \sin 3\theta \sin\theta}$ 이므로

$\theta = \frac{\pi}{6}$ 에서 $\frac{dy}{dx} = \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}} = -\sqrt{3}$ 이다.

14. $x = \sin^2\theta \cos\theta, y = \sin^3\theta$ 일 때,

$\frac{dy}{dx} = \frac{3\sin^2\theta \cos\theta}{2\sin\theta \cos^2\theta - \sin^3\theta} = \frac{3\sin\theta \cos\theta}{2\cos^2\theta - \sin^2\theta}$

이므로 $\theta = \frac{\pi}{3}$ 에서 $\frac{dy}{dx} = \frac{\frac{3\sqrt{3}}{4}}{\frac{1}{2} - \frac{3}{4}} = -3\sqrt{3}$ 이다.

15. $\tan\phi = \frac{r}{r'} \Big|_{\theta=\frac{\pi}{3}} \Leftrightarrow \tan\phi = \frac{2\cos\theta}{-2\sin\theta} \Big|_{\theta=\frac{\pi}{3}}$

$\Leftrightarrow \tan\phi = -\frac{1}{\sqrt{3}}$

$\Leftrightarrow \phi = \left| \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) \right| = \frac{\pi}{6}$

16. $\tan\frac{\pi}{6} = \frac{r}{r'} \Big|_{\theta=\alpha} \Leftrightarrow \tan\frac{\pi}{6} = \frac{2\sin\theta}{2\cos\theta} \Big|_{\theta=\alpha}$

$\Leftrightarrow \tan\frac{\pi}{6} = \tan\alpha \Leftrightarrow \alpha = \frac{\pi}{6}$

즉, $r = 2\sin\theta$ 이므로 $(r, \theta) = \left(1, \frac{\pi}{6}\right)$ 이다.

1. 3번

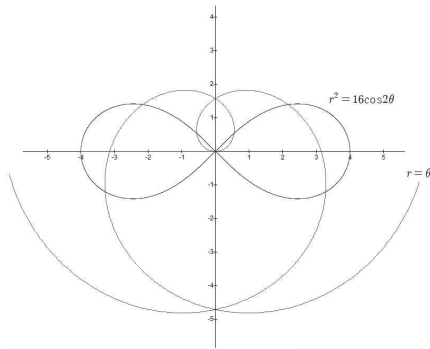
2. 3번

3. 4번

4. 1번

5. 4번

1. 그래프를 그려보면 아래와 같다.



즉, 교점의 개수는 7개다.

$$2. r = \frac{3}{1+2\sin\theta}$$

$$\Leftrightarrow r + 2r\sin\theta = 3$$

$$\Leftrightarrow \sqrt{x^2+y^2} + 2y = 3$$

$$\Leftrightarrow x^2 + y^2 = (3-2y)^2$$

$$\Leftrightarrow x^2 - 3y^2 + 12y - 9 = 0$$

이므로 $y = ax + b$ 를 사점근선이라 가정하면

$$x^2 - 3(ax+b)^2 + 12(ax+b) - 9 = 0$$

$$\Leftrightarrow (1-3a^2)x^2 + (-6ab+12a)x + \dots = 0 \text{ 이다.}$$

즉, 점근선의 기울기는 $a = \pm \frac{1}{\sqrt{3}}$ 이므로 m_1, m_2 의 절댓값은 $\frac{1}{3}$ 이다.

$$3. (r, \theta) = \left(4, \frac{\pi}{3}\right) \Leftrightarrow (x, y) = (2, 2\sqrt{3})$$

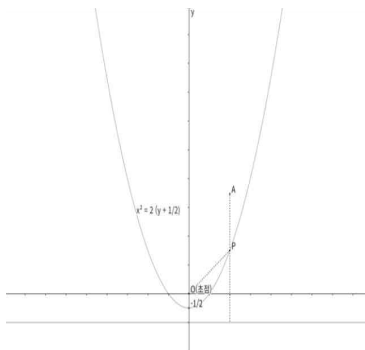
$$r = \frac{1}{1-\sin\theta}$$

$$\Leftrightarrow r - r\sin\theta = 1$$

$$\Leftrightarrow \sqrt{x^2+y^2} - y = 1$$

$$\Leftrightarrow x^2 + y^2 = (1+y)^2$$

$$\Leftrightarrow x^2 = 2\left(y + \frac{1}{2}\right)$$



포물선의 초점과 준선을 이용하면

$\overline{OP} + \overline{PA}$ 의 최솟값은 $2\sqrt{3} - (-1) = 2\sqrt{3} + 1$ 이다.

$$4. x = r\cos\theta = (1+\sin\theta)\cos\theta$$

$$y = r\sin\theta = (1+\sin\theta)\sin\theta$$

도함수를 구하면

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\cos\theta\sin\theta + (1+\sin\theta)\cos\theta}{\cos^2\theta - (1+\sin\theta)\sin\theta} = \frac{\cos\theta(1+2\sin\theta)}{1-\sin\theta-2\sin^2\theta} \text{ 이다.}$$

수평 접선을 갖으려면 $\cos\theta(1+2\sin\theta) = 0$ 이므로

$$\cos\theta = 0 \text{ 또는 } \sin\theta = -\frac{1}{2} \text{ 이다.}$$

$$\text{따라서 } \theta = \pm\frac{\pi}{2}, -\frac{\pi}{6}, -\frac{5}{6}\pi \text{ 이다.}$$

이때, $\theta = -\frac{\pi}{2}$ 일 때는 $\frac{dx}{d\theta} = 0$ 이므로

$$\lim_{x \rightarrow -\frac{\pi}{2}} \frac{\cos\theta(1+2\sin\theta)}{1-\sin\theta-2\sin^2\theta} = \lim_{x \rightarrow -\frac{\pi}{2}} \frac{\cos\theta + \sin 2\theta}{1-\sin\theta-2\sin^2\theta}$$

$$= \lim_{x \rightarrow -\frac{\pi}{2}} \frac{-\sin\theta + 2\cos 2\theta}{-\cos\theta - 4\sin\theta\cos\theta} = \infty \text{ 이므로 수평접선을 만족하지 않는다.}$$

즉, 수평접선을 갖는 θ 들의 합은 $\frac{\pi}{2} - \frac{\pi}{6} - \frac{5}{6}\pi = -\frac{\pi}{2}$ 이다.

$$5. x = r\cos\theta = (\sqrt{3}+\cos\theta)\cos\theta$$

$$y = r\sin\theta = (\sqrt{3}+\cos\theta)\sin\theta$$

도함수를 구하면

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{-\sin^2\theta + (\sqrt{3}+\cos\theta)\cos\theta}{-\sin\theta\cos\theta - (\sqrt{3}+\cos\theta)\sin\theta} \\ &= \frac{-\sin^2\theta + \cos^2\theta + \sqrt{3}\cos\theta}{-\sin\theta(2\cos\theta + \sqrt{3})} \text{ 이다.} \end{aligned}$$

수직 접선을 갖으려면 $-\sin\theta(2\cos\theta + \sqrt{3}) = 0$ 이므로

$$\sin\theta = 0 \text{ 또는 } \cos\theta = -\frac{\sqrt{3}}{2} \text{ 이다.}$$

즉, 수직접선을 갖는 θ 들의 합은 $0 + \pi + \frac{5}{6}\pi + \frac{7}{6}\pi = 3\pi$ 이다.

1. 5번	2. 2번	3. 1번	4. 4번	5. 3번
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1. $\begin{cases} x = r \cos \theta = (1 + \sin \theta) \cos \theta \\ y = r \sin \theta = (1 + \sin \theta) \sin \theta \end{cases}$ 이므로

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{2 \cos \theta \sin \theta + \cos \theta}{\cos^2 \theta - \sin^2 \theta - \sin \theta} = \frac{\sin 2\theta + \cos \theta}{\cos 2\theta - \sin \theta}$$

이다. 수직접선을 찾으려면

$$0 = \cos 2\theta - \sin \theta = 1 - 2\sin^2 \theta - \sin \theta \quad \text{이므로}$$

$$\sin \theta = \frac{1}{2} \quad \text{또는} \quad \sin \theta = -1 \quad \text{이다.}$$

$$\text{따라서 } \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2} \quad \text{이다.}$$

$$\text{이때, } \theta = \frac{3\pi}{2} \quad \text{일 때에는 } \frac{dy}{d\theta} = 0 \quad \text{이므로,}$$

$$\lim_{\theta \rightarrow \frac{3\pi}{2}} \frac{\sin 2\theta + \cos \theta}{\cos 2\theta - \sin \theta}$$

$$= \lim_{\theta \rightarrow \frac{3\pi}{2}} \frac{2 \cos 2\theta - \sin \theta}{-2 \sin 2\theta - \cos \theta} \quad (\because \text{로피탈 정리})$$

$$= \infty \quad \text{이므로 수직접선을 갖는다.}$$

$$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

$$\text{즉, 수직접선을 갖는 } \theta \text{들의 합은 } \frac{5\pi}{2} \quad \text{이다.}$$

2. $\begin{cases} x = r \cos \theta = \left(\frac{7}{2} + 3 \sin \theta - 2 \cos^2 \theta\right) \cos \theta \\ y = r \sin \theta = \left(\frac{7}{2} + 3 \sin \theta - 2 \cos^2 \theta\right) \sin \theta \end{cases}$ 이므로

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = 0 \quad \text{이다.}$$

$$\frac{dy}{d\theta} = (3 \cos \theta + 4 \cos \theta \sin \theta) \sin \theta + \left(\frac{7}{2} + 3 \sin \theta - 2 \cos^2 \theta\right) \cos \theta = 0$$

$$\Rightarrow \cos \theta \left(6 \sin^2 \theta + 6 \sin \theta + \frac{3}{2}\right) = 0$$

$$\Rightarrow \frac{3}{2} \cos \theta (4 \sin^2 \theta + 4 \sin \theta + 1) = 0$$

$$\Rightarrow \frac{3}{2} \cos \theta (2 \sin \theta + 1)^2 = 0$$

$$\therefore \theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore (x, y) = \left(-\frac{\sqrt{3}}{4}, -\frac{1}{4}\right)$$

3. $r = 1 - \cos \theta$ 에 대해 $\begin{cases} x = (1 - \cos \theta) \cos \theta \\ y = (1 - \cos \theta) \sin \theta \end{cases}$ 이고,

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\sin^2 \theta + (1 - \cos \theta) \cos \theta}{\sin \theta \cos \theta - (1 - \cos \theta) \sin \theta}$$

$$\text{이므로 } \theta = \frac{\pi}{2} \quad \text{일 때, 접선의 기울기는 } -1 \quad \text{이다.}$$

$$\text{또한, 극좌표상의 점 } \left(1, \frac{\pi}{2}\right) \text{은 직교좌표계로 } (0, 1) \text{이다.}$$

$$\text{그러므로 } P_1 \text{의 방정식은 직교좌표계로}$$

$$y - 1 = -(x - 0) \Leftrightarrow y = -x + 1 \quad \text{이다.}$$

$$\text{마찬가지로 } r = 2 \sin \theta \text{에 대해 } \begin{cases} x = 2 \sin \theta \cos \theta = \sin 2\theta \\ y = 2 \sin \theta \sin \theta = 2 \sin^2 \theta \end{cases} \quad \text{이고}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{4 \sin \theta \cos \theta}{2 \cos 2\theta} \quad \text{이므로 } \theta = \frac{\pi}{4} \quad \text{일 때, 접선의 기울기는 } \infty \quad \text{이다.}$$

$$\text{또한, 극좌표상의 점 } \left(\sqrt{2}, \frac{\pi}{4}\right) \text{은 직교좌표계로 } (1, 1) \text{이다.}$$

$$\text{그러므로 } P_2 \text{의 방정식은 직교좌표계로 } x = 1 \quad \text{이다.}$$

$$\text{따라서 위의 두 선분의 교점은 직교좌표계로 } (1, 0) \text{이다.}$$

$$\text{이것을 극좌표로 나타내면 } r = \sqrt{1^2 + 0^2} = 1, \quad \theta = \tan^{-1} \frac{0}{1} = 0$$

$$\text{즉, 극좌표로 } (1, 0) \text{이다.}$$

4. 두 극방정식의 교점을 구하자.

$$1 - \cos 2\theta = 1 + \cos \theta$$

$$\Leftrightarrow \cos 2\theta + \cos \theta = 0$$

$$\Leftrightarrow 2 \cos^2 \theta + \cos \theta - 1 = 0$$

$$\Leftrightarrow (2 \cos \theta - 1)(\cos \theta + 1) = 0$$

$$\Leftrightarrow 0 \leq \theta \leq \frac{\pi}{2}$$

$$\text{이므로 } \cos \theta = \frac{1}{2} \Leftrightarrow \theta = \frac{\pi}{3} \quad \text{이다.}$$

두 곡선 각각의 접선과 동경벡터 사이의 각도를 ϕ_1, ϕ_2 이라 하면

$$\tan \phi_1 = \frac{r}{\frac{dr}{d\theta}} = \frac{1 - \cos 2\theta}{2 \sin 2\theta} \Big|_{\theta = \frac{\pi}{3}} = \frac{\sqrt{3}}{2}$$

$$\tan \phi_2 = \frac{r}{\frac{dr}{d\theta}} = \frac{1 + \cos \theta}{-\sin \theta} \Big|_{\theta = \frac{\pi}{3}} = -\sqrt{3}$$

이다. 따라서 구하는 교각을

$$\tan \psi = \tan |\phi_1 - \phi_2| = \left| \frac{\tan \phi_1 - \tan \phi_2}{1 + \tan \phi_1 \tan \phi_2} \right| = 3\sqrt{3}$$

$$\therefore \psi = \tan^{-1} |3\sqrt{3}|$$

5. $\lim_{n \rightarrow \infty} \sum_{k=1}^{n-1} \left[\left\{ f\left(\frac{\pi k}{2n}\right) \right\}^2 + \left\{ g\left(\frac{\pi k}{2n}\right) \right\}^2 \right] \frac{\pi}{2n}$

$$= \int_0^{\frac{\pi}{2}} \{f(x)\}^2 + \{g(x)\}^2 dx$$

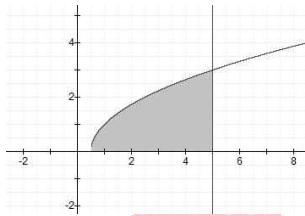
$$= \int_0^{\frac{\pi}{2}} \{f(\theta)\}^2 + \{g(\theta)\}^2 d\theta$$

$$= \int_0^{\frac{\pi}{2}} r^2 d\theta = \int_0^{\frac{\pi}{2}} 16 \cos^2 \theta d\theta$$

$$= 4\pi$$

1. 4번	2. 2번	3. 2번	4. 3번	5. 2번
6. 2번	7. 4번	8. 3번	9. 2번	10. 1번
11. 4번	12. 4번	13. 1번	14. 1번	15. 1번
16. 3번	17. 2번	18. 3번	19. 3번	20. 4번
21. 3번	22. 4번	23. 3번	24. 2번	25. 4번
26. 3번	27. 3번	28. 4번	29. 3번	30. 3번
31. 4번	32. 2번	33. 3번	34. 5번	35. 2번
36. 5번	37. 1번	38. 4번	39. 2번	40. 4번
41. 1번	42. 2번	43. 2번	44. 2번	45. 1번
46. 2번	47. 1번	48. 4번	49. 1번	50. 4번
51. 4번	52. 3번	53. $\sqrt{2}$		

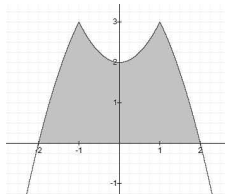
1.



곡선 $y = \sqrt{2x-1}$ 은 $(\frac{1}{2}, 0)$ 을 지나므로 적분 구간은 $[\frac{1}{2}, 5]$ 이다.

$$\text{즉, } \int_{\frac{1}{2}}^5 \sqrt{2x-1} dx = \frac{1}{2} \times \frac{2}{3} \left[(2x-1)^{\frac{3}{2}} \right]_{\frac{1}{2}}^5 = 9$$

2.

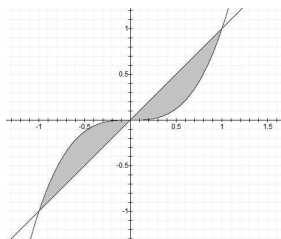


$y = \begin{cases} x^2 + 2 & (-1 < x < 1) \\ -x^2 + 4 & (x < -1, x > 1) \end{cases}$ 이고 $(-2, 0), (2, 0)$ 을 지나므로

$y = 3 - |x^2 - 1|$ 와 $y = 0$ 으로 둘러싸인 도형의 넓이는

$$\begin{aligned} \int_{-2}^2 |y-0| dx &= \int_{-2}^2 3 - |x^2 - 1| dx \quad (\because \text{구간 } [-2, 2] \text{ 에서 } y \text{ 는 항상 양수}) \\ &= 2 \left(\int_0^1 x^2 + 2 dx + \int_1^2 -x^2 + 4 dx \right) \quad (\because \text{우함수}) \\ &= 2 \left(\left[\frac{1}{3} x^3 + 2x \right]_0^1 + \left[-\frac{1}{3} x^3 + 4x \right]_1^2 \right) = 8 \end{aligned}$$

3.



$f(x)$ 와 $g(x)$ 는 $(-1, -1), (0, 0), (1, 1)$ 에서 만나고
구간 $[-1, 0]$ 에서 $f(x) > g(x)$, 구간 $[0, 1]$ 에서 $g(x) > f(x)$ 이므로

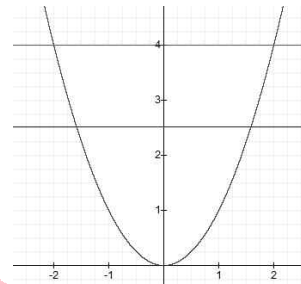
영역의 넓이는

$$\begin{aligned} &\int_{-1}^1 |f(x) - g(x)| dx \\ &= \int_{-1}^0 f(x) - g(x) dx + \int_0^1 g(x) - f(x) dx \\ &= \int_{-1}^0 x^3 - x dx + \int_0^1 x - x^3 dx \\ &= \int_0^1 t - t^3 dt + \int_0^1 x - x^3 dx \quad (\because x = -t \text{ 치환}) \\ &= 2 \int_0^1 x - x^3 dx = 2 \left[\frac{1}{2} x^2 - \frac{1}{4} x^4 \right]_0^1 = \frac{1}{2} \end{aligned}$$

[참고]

둘러싸인 영역의 면적은 원점을 기준으로 대칭이므로
한쪽 영역의 넓이 $\times 2$ 를 해도 된다.

4.

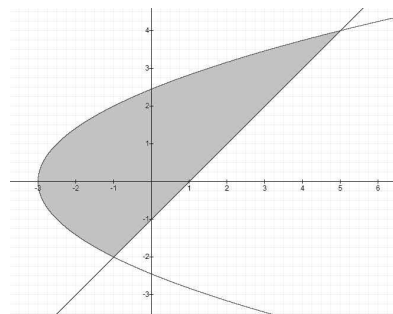


곡선 $y = x^2$ 은 $(-2, 4), (2, 4)$ 를 지나므로 구간은 $[-2, 2]$ 이고,
우함수이므로 구간 $[0, 2]$ 에서만 확인해도 된다.

이때, 곡선이 $(\sqrt{b}, b)$ 를 지나므로

$$\begin{aligned} \int_0^2 4 - x^2 dx &= 2 \int_0^{\sqrt{b}} b - x^2 dx \\ \Leftrightarrow \frac{16}{3} &= \frac{4}{3} b \sqrt{b} \text{ 이므로 } b = 4^{\frac{2}{3}} \text{ 이다.} \end{aligned}$$

5.



두 함수가 만나는 점을 찾기 위해 직선 $y = x - 1$ 을

포물선 $y^2 = 2x + 6$ 에 대입하면

$$(x-1)^2 = 2x+6$$

$$\Leftrightarrow x^2 - 4x - 5 = 0$$

$$\Leftrightarrow (x-5)(x+1) = 0 \text{ 에서 } x = -1 \text{ 또는 } 5 \text{ 이다.}$$

따라서 두 함수가 만나는 점은 $(-1, -2), (5, 4)$ 이다.

이제 편의를 위해 $x = y + 1$, $x = \frac{y^2 - 6}{2}$ 으로 바꾸면

둘러싸인 영역의 넓이는

$$\begin{aligned} \int_{-2}^4 (y+1) - \left(\frac{1}{2} y^2 - 3 \right) dy &= \int_{-2}^4 -\frac{1}{2} y^2 + y + 4 dy \\ &= \left[-\frac{1}{6} y^3 + \frac{1}{2} y^2 + 4y \right]_{-2}^4 = 18 \end{aligned}$$

6. 곡선 $x = \cos^3 t, y = \sin^2 t$ 는

$t = \pi$ 일 때 $(-1, 0)$, $t = 0$ 일 때 $(1, 0)$ 을 지나므로

곡선과 x 축으로 둘러싸인 부분의 넓이는

$$\begin{aligned} \int_{-1}^1 |y| dx &= \int_{\pi}^0 |\sin^2 t| 3\cos^2 t \times (-\sin t) dt \\ &= 3 \int_0^{\pi} \cos^2 t \sin^3 t dt = 3 \int_0^{\pi} \sin^3 t - \sin^5 t dt \\ &= 3 \left(2 \times \frac{2}{3} \times 1 - 2 \times \frac{4}{5} \times \frac{2}{3} \times 1 \right) = \frac{4}{5} \quad (\because \text{Wallis 정리}) \end{aligned}$$

7. 싸이클로이드의 넓이는 $3\pi a^2$ 이므로 27π 이다.

[다른 풀이]

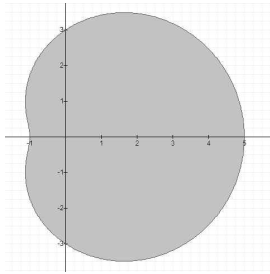
곡선 $x = 3(\theta - \sin \theta), y = 3(1 - \cos \theta)$ 는 $\theta = 0$ 일 때 $(0, 0)$

$\theta = 2\pi$ 일 때 $(6\pi, 0)$ 을 지나므로

곡선과 x 축으로 둘러싸인 부분의 넓이는

$$\begin{aligned} \int_0^{6\pi} |y| dx &= \int_0^{2\pi} |3(1 - \cos \theta)| \times 3(1 - \cos \theta) d\theta \\ &= 9 \int_0^{2\pi} (1 - \cos \theta)^2 d\theta \quad (\because 1 - \cos \theta \text{는 } 0 \leq \theta \leq 2\pi \text{에서 항상 양수}) \\ &= 9 \int_0^{2\pi} 1 - 2\cos \theta + \cos^2 \theta d\theta \\ &= 9 \times 2\pi + 0 + 9 \times 4 \times \frac{1}{2} \times \frac{\pi}{2} \quad (\because \text{Wallis 정리}) \\ &= 27\pi \end{aligned}$$

8.



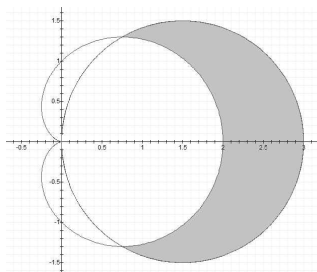
$$\begin{aligned} \frac{1}{2} \int_0^{2\pi} r^2 d\theta &= \frac{1}{2} \int_0^{2\pi} (2\cos \theta + 3)^2 d\theta \\ &= 2 \times \frac{1}{2} \int_0^{\pi} (2\cos \theta + 3)^2 d\theta \quad (\because x\text{-축 대칭}) \\ &= \int_0^{\pi} 4\cos^2 \theta + 12\cos \theta + 9 d\theta \\ &= 4 \times 2 \times \frac{1}{2} \times \frac{\pi}{2} + 0 + 9\pi = 11\pi \end{aligned}$$

[다른 풀이]

$a = 3$, $b = 2$ 이므로 영역의 넓이는 공식에 의하여

$$\frac{\pi}{2} (2a^2 + b^2) = \frac{\pi}{2} (18 + 4) = 11\pi$$

9.



$$3\cos \theta = 1 + \cos \theta \Leftrightarrow \theta = \frac{\pi}{3}, \frac{5}{3}\pi \text{ 이므로 넓이는}$$

$$\frac{1}{2} \int_0^{\frac{\pi}{3}} r_1^2 - r_2^2 d\theta + \frac{1}{2} \int_{\frac{5}{3}\pi}^{2\pi} r_1^2 - r_2^2 d\theta$$

$$= 2 \times \frac{1}{2} \int_0^{\frac{\pi}{3}} r_1^2 - r_2^2 d\theta \quad (\because \text{대칭성})$$

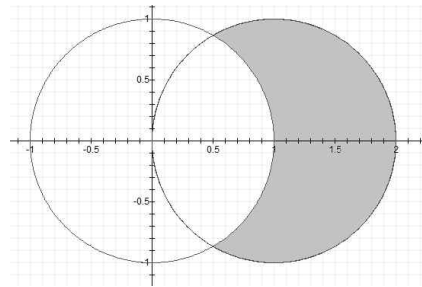
$$= 2 \times \frac{1}{2} \int_0^{\frac{\pi}{3}} 9\cos^2 \theta - (1 + \cos \theta)^2 d\theta = \int_0^{\frac{\pi}{3}} -1 - 2\cos \theta + 8\cos^2 \theta d\theta$$

$$= \int_0^{\frac{\pi}{3}} 3 - 2\cos \theta + 4\cos 2\theta d\theta = [3\theta - 2\sin \theta + 2\sin 2\theta]_0^{\frac{\pi}{3}} = \pi$$

[다른 풀이]

$a = 1$ 이므로 $r = 3\cos \theta$ 의 내부와 $r = 1 + \cos \theta$ 의 외부로 둘러싸인 영역의 넓이는 공식 πa^2 에 의하여 π 이다.

10.

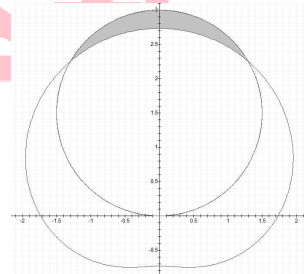


$$S = 2 \times \frac{1}{2} \int_0^{\frac{\pi}{3}} 4\cos^2 \theta - 1 d\theta$$

$$= \int_0^{\frac{\pi}{3}} 1 + 2\cos 2\theta d\theta$$

$$= [\theta + \sin 2\theta]_0^{\frac{\pi}{3}} = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

11.

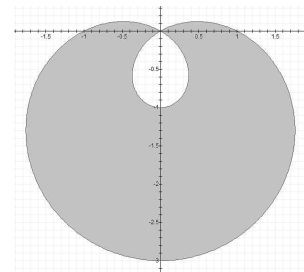


$$S = 2 \times \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 9\sin^2 \theta - (\sqrt{3} + \sin \theta)^2 d\theta$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 8\sin^2 \theta - 3 - 2\sqrt{3}\sin \theta d\theta = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 1 - 4\cos 2\theta - 2\sqrt{3}\sin \theta d\theta$$

$$= [\theta - 2\sin 2\theta + 2\sqrt{3}\cos \theta]_{\frac{\pi}{3}}^{\frac{\pi}{2}} = \frac{\pi}{6}$$

12.



극방정식 $r = 1 - 2\sin \theta$ 과 $r = 1 - 2\cos \theta$ 의 내부곡선과 외부곡선으로 둘러싸인 부분의 면적은 같다.

$$\text{이때, 내부 곡선의 넓이는 } 2 \times \frac{1}{2} \int_0^{\frac{\pi}{3}} (1 - 2\cos \theta)^2 d\theta$$

$$\text{외부 곡선의 넓이는 } 2 \times \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} (1 - 2\cos \theta)^2 d\theta \text{ 이므로}$$

둘러싸인 부분의 면적은

$$\begin{aligned}
 & 2 \times \left[\frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} (1 - 2\cos\theta)^2 d\theta - \frac{1}{2} \int_0^{\frac{\pi}{3}} (1 - 2\cos\theta)^2 d\theta \right] \\
 &= \int_{\frac{\pi}{3}}^{\pi} (1 - 2\cos\theta)^2 d\theta - \int_0^{\frac{\pi}{3}} (1 - 2\cos\theta)^2 d\theta \\
 &= \int_{\frac{\pi}{3}}^{\pi} (3 - 4\cos\theta + 2\cos 2\theta) d\theta - \int_0^{\frac{\pi}{3}} (3 - 4\cos\theta + 2\cos 2\theta) d\theta \\
 &= [3\theta - 4\sin\theta + \sin 2\theta]_{\frac{\pi}{3}}^{\pi} - [3\theta - 4\sin\theta + \sin 2\theta]_0^{\frac{\pi}{3}} \\
 &= \pi + 3\sqrt{3}
 \end{aligned}$$

[다른 풀이]

$r = 1 - 2\sin\theta$ 과 $r = 1 + 2\cos\theta$ 의 내부곡선과 외부곡선으로

둘러싸인 부분의 면적은 같다.

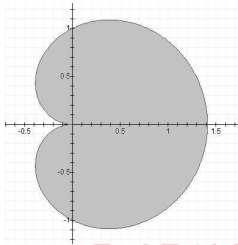
따라서 공식에 의하면 $r = a(1 + 2\cos\theta)$ 의 내부곡선과 외부곡선으로

둘러싸인 부분의 면적은 $(\pi + 3\sqrt{3})a^2$ 이므로 $S = \pi + 3\sqrt{3}$ 이다.

13. $1 + 2\sin\theta = 0 \Leftrightarrow \sin\theta = -\frac{1}{2}$ 이므로 $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$ 이다.

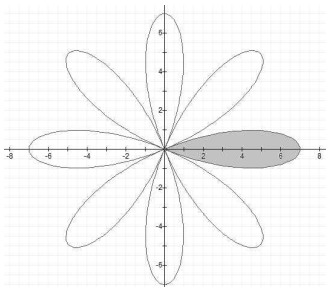
즉, $\frac{1}{2} \int_{\frac{7\pi}{6}}^{\frac{11\pi}{6}} (1 + 2\sin\theta)^2 d\theta$

14.



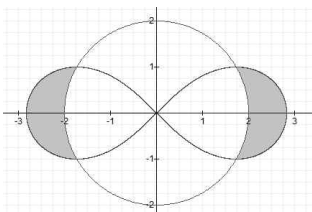
$$\frac{1}{2} \int_0^{2\pi} 1 + \cos\theta d\theta = \frac{1}{2} [\theta + \sin\theta]_0^{2\pi} = \pi$$

15.



$$\begin{aligned}
 & 2 \times \frac{1}{2} \int_0^{\frac{\pi}{8}} 49 \cos^2 4\theta d\theta = 49 \int_0^{\frac{\pi}{8}} \frac{1 + \cos 8\theta}{2} d\theta \\
 &= \frac{49}{2} \left[\theta + \frac{1}{8} \sin 8\theta \right]_0^{\frac{\pi}{8}} = \frac{49}{16} \pi
 \end{aligned}$$

16.



$$4 \times \frac{1}{2} \int_0^{\frac{\pi}{6}} 8 \cos 2\theta - 4 d\theta = 2 [4 \sin 2\theta - 4\theta]_0^{\frac{\pi}{6}} = 4\sqrt{3} - \frac{4}{3}\pi$$

$$\begin{aligned}
 17. \int_1^2 \sqrt{1 + \left(\frac{1}{2}x^3 - \frac{1}{2x^3}\right)^2} dx &= \int_1^2 \sqrt{\left(\frac{1}{2}x^3 + \frac{1}{2x^3}\right)^2} dx \\
 &= \int_1^2 \frac{1}{2}x^3 + \frac{1}{2x^3} dx = \left[\frac{1}{8}x^4 - \frac{1}{4x^2} \right]_1^2 = \frac{33}{16}
 \end{aligned}$$

$$\begin{aligned}
 18. \int_1^9 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy &= \int_1^9 \sqrt{1 + \left(\frac{\sqrt{y}}{2} - \frac{1}{2\sqrt{y}}\right)^2} dy \\
 &= \int_1^9 \sqrt{\left(\frac{\sqrt{y}}{2} + \frac{1}{2\sqrt{y}}\right)^2} dy = \int_1^9 \frac{\sqrt{y}}{2} + \frac{1}{2\sqrt{y}} dy \\
 &= \int_1^9 \frac{\sqrt{y}}{2} + \frac{1}{2\sqrt{y}} dy = \left[\frac{1}{3}y^{\frac{3}{2}} + \sqrt{y} \right]_1^9 = \frac{32}{3}
 \end{aligned}$$

19. $y' = \sqrt{\cos 2x}$ 이므로 곡선의 길이는

$$\begin{aligned}
 \int_0^{\frac{\pi}{4}} \sqrt{1 + \cos 2x} dx &= \int_0^{\frac{\pi}{4}} \sqrt{2 \cos^2 x} dx \\
 &= \sqrt{2} \int_0^{\frac{\pi}{4}} \cos x dx = \sqrt{2} [\sin x]_0^{\frac{\pi}{4}} = 1
 \end{aligned}$$

$$\begin{aligned}
 20. \int_0^1 \sqrt{(\sin t + t \cos t)^2 + (\cos t - t \sin t)^2} dt \\
 &= \int_0^1 \sqrt{1 + t^2} dt = \int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta \\
 &= \frac{1}{2} [\sec \theta \tan \theta + \ln(\sec \theta + \tan \theta)]_0^{\frac{\pi}{4}} \\
 &= \frac{1}{2} [\sqrt{2} + \ln(\sqrt{2} + 1)]
 \end{aligned}$$

$$\begin{aligned}
 21. \int_0^{\frac{\pi}{6}} \sqrt{(-3 \cos^2 t \sin t)^2 + (3 \sin^2 t \cos t)^2 + (4 \sin t \cos t)^2} dt \\
 &= \int_0^{\frac{\pi}{6}} \sqrt{25 \sin^2 t \cos^2 t} dt = \int_0^{\frac{\pi}{6}} 5 \sin t \cos t dt = \frac{5}{2} [\sin^2 t]_0^{\frac{\pi}{6}} = \frac{5}{8}
 \end{aligned}$$

22. 사이클로이드의 한 호의 길이는 $8a$ 이므로 곡선의 길이는 16이다.

[다른 풀이]

곡선의 길이는

$$\begin{aligned}
 & \int_0^{2\pi} \sqrt{(2(1 - \cos\theta))^2 + (2\sin\theta)^2} d\theta \\
 &= \int_0^{2\pi} \sqrt{8 - 8\cos\theta} d\theta \\
 &= \int_0^{2\pi} 4 \sqrt{\frac{1 - \cos\theta}{2}} d\theta \\
 &= 4 \int_0^{2\pi} \left| \sin \frac{\theta}{2} \right| d\theta \\
 &= 4 \int_0^{2\pi} \sin \frac{\theta}{2} d\theta \\
 &= \left[8 \cos \frac{\theta}{2} \right]_0^{2\pi} = 16
 \end{aligned}$$

23. 성마형의 전체의 길이는 $6a$ 이므로 곡선의 길이는 6이다.

[다른 풀이]

$x = \cos^3 t$, $y = \sin^3 t$ 이므로 곡선의 길이는

$$\begin{aligned}
 & \int_0^{2\pi} \sqrt{(-3 \cos^2 t \sin t)^2 + (3 \sin^2 t \cos t)^2} dt \\
 &= \int_0^{2\pi} \sqrt{9 \cos^2 t \sin^2 t (\cos^2 t + \sin^2 t)} dt \\
 &= \int_0^{2\pi} 3 |\cos t \sin t| dt \\
 &= \frac{3}{2} \int_0^{2\pi} |\sin 2t| dt \\
 &= 4 \times \frac{3}{2} \int_0^{\frac{\pi}{2}} \sin 2t dt = 6 \left[-\frac{1}{2} \cos 2t \right]_0^{\frac{\pi}{2}} = 6
 \end{aligned}$$

24. $r = 1 - \sin \theta \left(0 \leq \theta \leq \frac{\pi}{2} \right)$ 의 길이는

$r = 1 - \cos \theta \left(0 \leq \theta \leq \frac{\pi}{2} \right)$ 의 길이와 같다.

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sqrt{(1 - \cos \theta)^2 + \sin^2 \theta} d\theta &= \int_0^{\frac{\pi}{2}} \sqrt{2 - 2\cos \theta} d\theta \\ &= \int_0^{\frac{\pi}{2}} 2\sin \frac{\theta}{2} d\theta = \left[-4\cos \frac{\theta}{2} \right]_0^{\frac{\pi}{2}} = 4 - 2\sqrt{2} \end{aligned}$$

25. $r = 1 - \cos \theta$ 와 $r = 1 + \cos \theta$ 의 곡선의 길이는 같고, 심장형 $r = a(1 + \cos \theta)$ 의 전체 길이는 $8a$ 이므로 곡선의 길이는 8이다.

[다른 풀이]

곡선의 길이는

$$\begin{aligned} \int_0^{2\pi} \sqrt{(1 - \cos \theta)^2 + \sin^2 \theta} d\theta &= \int_0^{2\pi} \sqrt{2 - 2\cos \theta} d\theta \\ &= 2 \int_0^{2\pi} \sqrt{\frac{1 - \cos \theta}{2}} d\theta \\ &= 2 \int_0^{2\pi} \left| \sin \frac{\theta}{2} \right| d\theta \\ &= 2 \int_0^{2\pi} \sin \frac{\theta}{2} d\theta \\ &= 4 \left[-\cos \frac{\theta}{2} \right]_0^{2\pi} = 8 \end{aligned}$$

26. $\int_0^1 \sqrt{\theta^4 + 4\theta^2} d\theta = \int_0^1 \theta \sqrt{\theta^2 + 4} d\theta = \left[\frac{1}{3}(\theta^2 + 4)^{\frac{3}{2}} \right]_0^1 = \frac{1}{3}(5\sqrt{5} - 8)$

27. $\int_0^\pi \sqrt{\sin^6 \left(\frac{\theta}{3} \right) + \sin^4 \left(\frac{\theta}{3} \right) \cos^2 \left(\frac{\theta}{3} \right)} d\theta$
 $= \int_0^\pi \sqrt{\sin^4 \left(\frac{\theta}{3} \right)} d\theta = \int_0^\pi \sin^2 \left(\frac{\theta}{3} \right) d\theta = \frac{1}{2} \int_0^\pi 1 - \cos \left(\frac{2\theta}{3} \right) d\theta$
 $= \frac{1}{2} \left[\theta - \frac{3}{2} \sin \left(\frac{2\theta}{3} \right) \right]_0^\pi = \frac{1}{2} \left(\pi - \frac{3}{4} \sqrt{3} \right)$

28. Pappus 정리에 의하여 회전체의 부피는 $2\pi \times 1 \times \pi \times 1^2$ 이므로 $2\pi^2$ 이다.

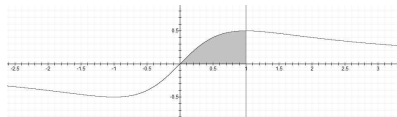
29. Pappus 정리에 의하여 회전체의 부피는 $2\pi \times 3 \times \pi \times 2^2$ 이므로 $24\pi^2$ 이다.

30. Pappus 정리에 의하여 회전체의 겉넓이는 $2\pi \times 4 \times \pi \times 2^2$ 이므로 $32\pi^2$ 이다.

31. Pappus 정리에 의하여 회전체의 부피는 $2\pi \times \frac{3}{\sqrt{2}} \times \frac{1}{2} \times 2 \times 3$ 이므로 $9\sqrt{2}\pi$ 이다.

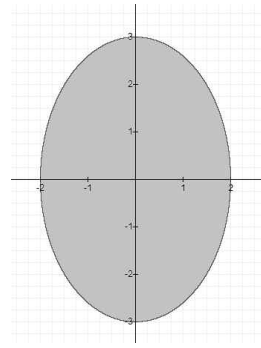
32. Pappus 정리에 의하여 회전체의 부피는 $2\pi \times \frac{3}{\sqrt{5}} \times \frac{1}{2}$ 이므로 $\frac{3}{\sqrt{5}}\pi$ 이다.

33.



$$\begin{aligned} V_x &= \pi \int_0^1 y^2 dx = \pi \int_0^1 \frac{x^2}{(x^2+1)^2} dx = \pi \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{\sec^4 \theta} \sec^2 \theta d\theta \\ &= \pi \int_0^{\frac{\pi}{4}} \sin^2 \theta d\theta = \frac{\pi}{2} \int_0^{\frac{\pi}{4}} 1 - \cos 2\theta d\theta = \frac{\pi}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{8}(\pi - 2) \end{aligned}$$

34.



$$\begin{aligned} V_x &= 2 \times \pi \int_0^2 y^2 dx = 2\pi \times 9 \int_0^2 \left(1 - \frac{x^2}{4} \right) dx \\ &= 18\pi \left[x - \frac{x^3}{12} \right]_0^2 = 24\pi \end{aligned}$$

[다른 풀이]

타원 $\frac{x^2}{4} + \frac{y^2}{9} = 1$ 로 둘러싸인 영역을 x 축의 둘레로 회전시키면

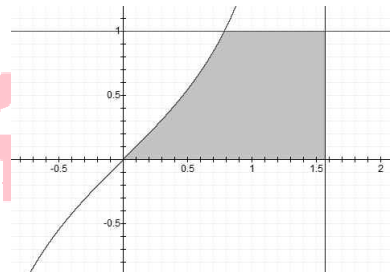
회전체는 타원체가 된다.

이때 타원의 반지름은 x 축 방향으로 2, y 축 방향으로 3이므로, 회전하여 얻어지는 타원체의 반지름은 각각 3, 3, 2이다.

따라서 타원체의 부피는

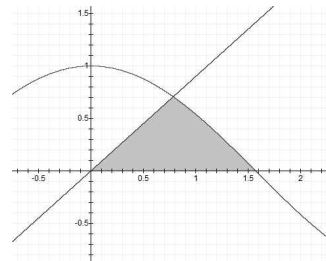
$$V = \frac{4}{3}\pi abc = \frac{4}{3}\pi \times 3 \times 3 \times 2 = 24\pi \text{이다.}$$

35.



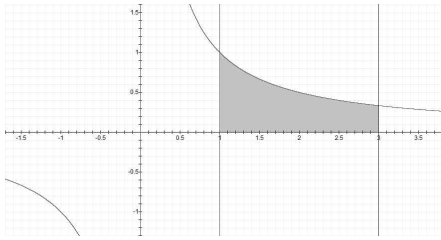
$$\begin{aligned} V_x &= \pi \int_0^{\frac{\pi}{4}} \tan^2 x dx + \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 1 dx \\ &= \pi \int_0^{\frac{\pi}{4}} \sec^2 x - 1 dx + \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 1 dx = \pi [\tan x - x]_0^{\frac{\pi}{4}} + \frac{\pi^2}{4} \\ &= \pi \left(1 - \frac{\pi}{4} \right) + \frac{\pi^2}{4} = \pi \end{aligned}$$

36.



$$\begin{aligned} V_x &= \pi \int_0^{\frac{\pi}{4}} \frac{8}{x^2} x^2 dx + \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 x dx \\ &= \frac{8}{\pi} \int_0^{\frac{\pi}{4}} x^2 dx + \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1 + \cos 2x}{2} dx \\ &= \frac{8}{\pi} \left[\frac{1}{3} x^3 \right]_0^{\frac{\pi}{4}} + \frac{\pi}{2} \left[x + \frac{1}{2} \sin 2x \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\pi^2}{6} - \frac{\pi}{4} \end{aligned}$$

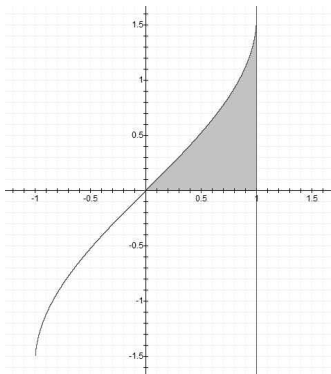
37.



$$V_x = \pi \int_1^3 \left(\frac{1}{x} + 1 \right)^2 - 1 dx = \pi \int_1^3 \frac{1}{x^2} + \frac{2}{x} dx$$

$$= \pi \left[-\frac{1}{x} + 2 \ln x \right]_1^3 = 2\pi \left(\frac{1}{3} + \ln 3 \right)$$

38.

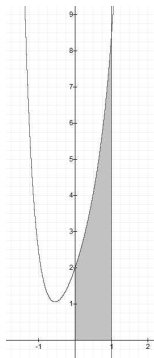


$$V = 2\pi \int_0^1 xy dx = 2\pi \int_0^1 x \sin^{-1} x dx$$

$$= 2\pi \left[\frac{1}{2} x^2 \sin^{-1} x \right]_0^1 - \pi \int_0^1 \frac{x^2}{\sqrt{1-x^2}} dx$$

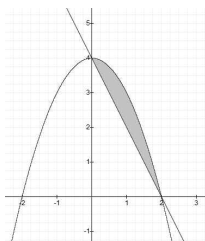
$$= \frac{\pi^2}{2} - \pi \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta = \frac{\pi^2}{4}$$

39.



$$V = 2\pi \int_0^1 xy dx = 2\pi \int_0^1 x(3x + 2e^x) dx = 2\pi [x^3 + e^{x^2}]_0^1 = 2\pi e$$

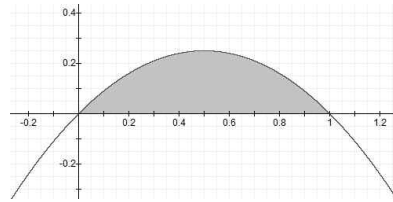
40.



$$V = 2\pi \int_0^2 x [(-x^2 + 4) - (-2x + 4)] dx$$

$$= 2\pi \int_0^2 -x^3 + 2x^2 dx = 2\pi \left[-\frac{1}{4} x^4 + \frac{2}{3} x^3 \right]_0^2 = \frac{8}{3} \pi$$

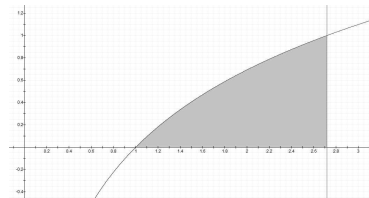
41.



$$V = 2\pi \int_0^1 (3-x)(x-x^2) dx = 2\pi \int_0^1 x^3 - 4x^2 + 3x dx$$

$$= 2\pi \left[\frac{1}{4} x^4 - \frac{4}{3} x^3 + \frac{3}{2} x^2 \right]_0^1 = \frac{5}{6} \pi$$

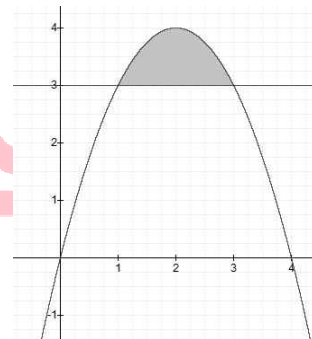
42.



$$V = 2\pi \int_1^e (x+1) \ln x dx = 2\pi \left[\left(\frac{1}{2} x^2 + x \right) \ln x - \int \left(\frac{1}{2} x + 1 \right) dx \right]_1^e$$

$$= 2\pi \left[\left(\frac{1}{2} x^2 + x \right) \ln x - \left(\frac{1}{4} x^2 + x \right) \right]_1^e = \frac{\pi}{2} (e^2 + 5)$$

43.

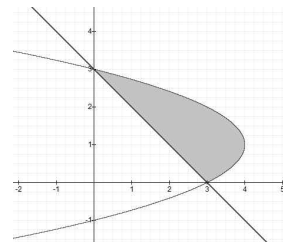


$$V = 2\pi \int_1^3 (x-1) [x(4-x) - 3] dx$$

$$= 2\pi \int_1^3 -x^3 + 5x^2 - 7x + 3 dx$$

$$= 2\pi \left[-\frac{1}{4} x^4 + \frac{5}{3} x^3 - \frac{7}{2} x^2 + 3x \right]_1^3 = \frac{8}{3} \pi$$

44.



$$V = 2\pi \int_0^3 x [(-x^2 + 2x + 3) - (3 - x)] dx$$

$$= 2\pi \int_0^3 -x^3 + 3x^2 dx = 2\pi \left[-\frac{1}{4} x^4 + x^3 \right]_0^3 = \frac{27}{2} \pi$$

$$45. V = \int_0^1 \text{단면의 넓이} dx = \int_0^1 \frac{\sqrt{3}}{4} y^2 dx = \frac{\sqrt{3}}{4} \int_0^1 x^6 dx = \frac{\sqrt{3}}{28}$$

$$46. V = \int_0^2 \text{단면의 넓이} dy = \int_0^2 4x^2 dy = 4 \int_0^2 2 - y dy = 4 \left[2y - \frac{1}{2} y^2 \right]_0^2 = 8$$

$$47. S_x = 2\pi \int_0^1 y \sqrt{1+(y')^2} dx = 2\pi \int_0^1 x^3 \sqrt{1+9x^4} dx$$

$$= 2\pi \left[\frac{1}{36} \times \frac{2}{3} (1+9x^4)^{\frac{3}{2}} \right]_0^1 = \frac{\pi}{27} (10\sqrt{10} - 1)$$

$$48. S_x = 2\pi \int_0^1 y \sqrt{1+(y')^2} dx = 2\pi \int_0^1 \sqrt{x} \sqrt{1+\frac{1}{4x}} dx$$

$$= 2\pi \int_0^1 \sqrt{x+\frac{1}{4}} dx = 2\pi \left[\frac{2}{3} \left(x+\frac{1}{4}\right)^{\frac{3}{2}} \right]_0^1 = \frac{\pi}{6} (5\sqrt{5} - 1)$$

$$49. L = \int_1^{16} \sqrt{1+\sqrt{x-1}} dx = \int_1^{16} x^{\frac{1}{4}} dx = \frac{4}{5} \left[x^{\frac{5}{4}} \right]_1^{16} = \frac{124}{5}$$

$$S_y = 2\pi \int_1^{16} x \sqrt{1+(y')^2} dx = 2\pi \int_1^{16} x^{\frac{5}{4}} dx = 2\pi \times \frac{4}{9} \left[x^{\frac{9}{4}} \right]_1^{16} = \frac{4088}{9} \pi$$

$$50. S_y = 2\pi \int_0^{\sqrt{2}} x \sqrt{1+(y')^2} dx = 2\pi \int_0^{\sqrt{2}} x \sqrt{1+x^2} dx$$

$$= \frac{2\pi}{3} \left[(1+x^2)^{\frac{3}{2}} \right]_0^{\sqrt{2}} = \frac{2\pi}{3} (3\sqrt{3} - 1)$$

$$51. S_x = 2\pi \int_a^b y \sqrt{(x')^2 + (y')^2} dt$$

$$= 2\pi \int_0^{\frac{\pi}{2}} \cos^2 t \sqrt{4\sin^2 t \cos^2 t + 4\cos^2 t \sin^2 t} dt$$

$$= 4\sqrt{2}\pi \int_0^{\frac{\pi}{2}} \sin t \cos^3 t dt = -\sqrt{2}\pi [\cos^4 t]_0^{\frac{\pi}{2}} = \sqrt{2}\pi$$

[다른 풀이]

주어진 매개방정식은 $x+y=\sin^2 t+\cos^2 t=1$ 을 만족하므로

$0 \leq t \leq \frac{\pi}{2}$ 일 때, $x+y=1$ ($0 \leq x \leq 1$)로 나타낼 수 있다.

이때, $x+y=1$ ($0 \leq x \leq 1$)에 대한 선분의 중심좌표는 $(\frac{1}{2}, \frac{1}{2})$ 이므로

중심과 축과의 거리 $k=\frac{1}{2}$ 이다. 또한 선분의 길이는 $\sqrt{2}$ 이다.

즉, 파푸스 정리에 의하여

$$S=2\pi k \times \text{길이} = 2\pi \times \frac{1}{2} \times \sqrt{2} = \sqrt{2}\pi \text{이다.}$$

$$52. \int_0^{\pi} |v(t)| dt = \int_0^{\pi} |2\sin(3t)| dt$$

$$= \frac{2}{3} \int_0^{3\pi} |\sin x| dx = 3 \times \frac{2}{3} \int_0^{\pi} \sin x dx = 4$$

$$53. \int_0^{\frac{\pi}{2}} |v(t)| dt = \int_0^{\frac{\pi}{2}} \sqrt{(x')^2 + (y')^2} dt$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{4\sin^2 t \cos^2 t + 4\cos^2 t \sin^2 t} dt$$

$$= 2\sqrt{2} \int_0^{\frac{\pi}{2}} \sin t \cos t dt = \sqrt{2} [\sin^2 t]_0^{\frac{\pi}{2}} = \sqrt{2}$$

[다른 풀이]

주어진 매개방정식은 $x+y=\sin^2 t+\cos^2 t=1$ 을 만족하므로

$0 \leq t \leq \frac{\pi}{2}$ 일 때, $x+y=1$ ($0 \leq x \leq 1$)로 나타낼 수 있다.

따라서 직선 $x+y=1$ ($0 \leq x \leq 1$)이 움직이는 거리는 선분의 길이인 $\sqrt{2}$ 이다.

1. 3번	2. 4번	3. 1번	4. 2번	5. 3번
6. 1번	7. 1번	8. 4번	9. 4번	10. 2번
11. 1번	12. 1번	13. 2번	14. 3번	15. 1번
16. 3번	17. 1번	18. 2번	19. 2번	20. 3번
21. 4번	22. 1번	23. 2번	24. 3번	25. 2번

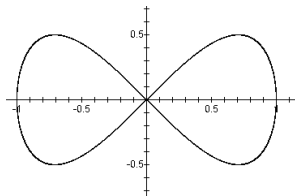
1. 두 그래프의 교점은 0과 $\frac{4}{3}$ 이므로 넓이는

$$S = \frac{|a-a'|}{6}(\beta-\alpha)^3 = \frac{\left|\frac{1}{2}-(-1)\right|}{6}\left(\frac{4}{3}-0\right)^3 = \frac{16}{27} \text{ 이다.}$$

[다른 풀이]

$$\begin{aligned} \int_0^{\frac{4}{3}} (-x^2+2x) - \frac{1}{2}x^2 dx &= \left[-\frac{1}{3}x^3+x^2-\frac{1}{6}x^3\right]_0^{\frac{4}{3}} \\ &= -\frac{1}{3}\times\left(\frac{4}{3}\right)^3 + \left(\frac{4}{3}\right)^2 - \frac{1}{6}\times\left(\frac{4}{3}\right)^3 = \frac{16}{27} \end{aligned}$$

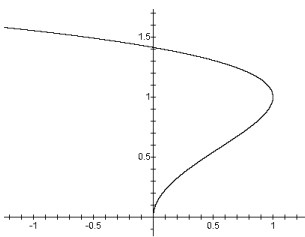
2.



따라서 $y^2 = x^2 - x^4 \Leftrightarrow y = \pm \sqrt{x^2 - x^4}$ 의 그래프 중 1사분면의 넓이에 4배를 하면 둘러싸인 영역의 면적이 된다.

$$\begin{aligned} S &= 4 \int_0^1 \sqrt{x^2 - x^4} dx = 4 \int_0^1 x \sqrt{1 - x^2} dx \\ &= 4 \left[-\frac{1}{3}(1-x^2)^{\frac{3}{2}} \right]_0^1 = \frac{4}{3} \end{aligned}$$

3.



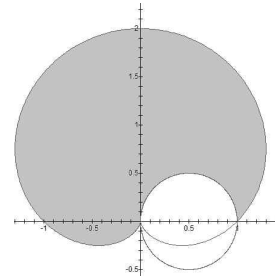
$$\begin{aligned} S &= \int_0^{\sqrt{2}} x dy = \int_0^2 (2t-t^2) \frac{1}{2\sqrt{t}} dt \\ &= \int_0^2 \sqrt{t} - \frac{1}{2}t^{\frac{3}{2}} dt = \left[\frac{2}{3}t^{\frac{3}{2}} - \frac{1}{5}t^{\frac{5}{2}} \right]_0^2 \\ &= \frac{2}{3} \times 2^{\frac{3}{2}} - \frac{1}{5} \times 2^{\frac{5}{2}} = \frac{8}{15} \sqrt{2} \end{aligned}$$

[다른 풀이]

$x = 2t - t^2, y = \sqrt{t} \ (t \geq 0) \Leftrightarrow x = 2y^2 - y^4 \ (y \geq 0)$ 이므로 그림보면 위와 같다.

$$\begin{aligned} S &= \int_0^{\sqrt{2}} x dy = \int_0^{\sqrt{2}} (2y^2 - y^4) dy = \left[\frac{2}{3}y^3 - \frac{1}{5}y^5 \right]_0^{\sqrt{2}} \\ &= \frac{2}{3} \times 2^{\frac{3}{2}} - \frac{1}{5} \times 2^{\frac{5}{2}} = \frac{8}{15} \sqrt{2} \end{aligned}$$

4.



$$\begin{aligned} S &= \frac{1}{2} \int_0^{\frac{3\pi}{2}} (1 + \sin\theta)^2 d\theta - \pi \times \left(\frac{1}{2}\right)^2 \times \frac{1}{2} \\ &= \frac{1}{2} \int_0^{\frac{3\pi}{2}} 1 + 2\sin\theta + \sin^2\theta d\theta - \frac{\pi}{8} \\ &= \frac{1}{2} \left(\frac{3\pi}{2} + 2 + 3 \times \frac{1}{2} \times \frac{\pi}{2} \right) - \frac{\pi}{8} \\ &= \pi + 1 \end{aligned}$$

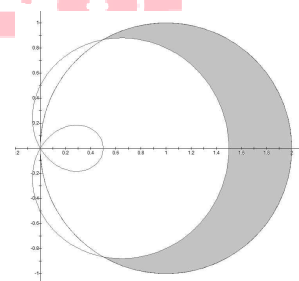
5. $r = a \cos\theta + b$ 가 $(r, \theta) = \left(\frac{3}{2}, 0\right)$ 을 지나므로 대입하면 $\frac{3}{2} = a + b$ 이다.

$r = 2 \cos\theta$ 와 $r = 1$ 의 교점은 $(r, \theta) = \left(1, \frac{\pi}{3}\right)$ 이므로 $r = a \cos\theta + b$ 가

지나므로 대입하면 $1 = \frac{1}{2}a + b$ 이다.

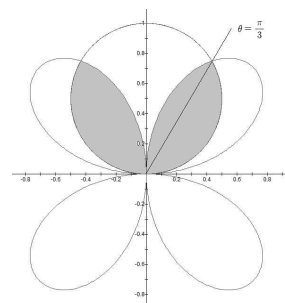
따라서 $a = 1, b = \frac{1}{2}$ 이므로 $r = \cos\theta + \frac{1}{2}$ 이고,

$r = \frac{1}{2} + \cos\theta$ 의 외부와 $r = 2 \cos\theta$ 의 내부를 그려보면 아래와 같다.



$$\begin{aligned} S &= 2 \times \frac{1}{2} \int_0^{\frac{\pi}{3}} \left[(2\cos\theta)^2 - \left(\cos\theta + \frac{1}{2}\right)^2 \right] d\theta \\ &= \int_0^{\frac{\pi}{3}} \left(3\cos^2\theta - \cos\theta - \frac{1}{4} \right) d\theta \\ &= \int_0^{\frac{\pi}{3}} \left(\frac{3}{2}(1 + \cos 2\theta) - \cos\theta - \frac{1}{4} \right) d\theta \\ &= \left[\frac{5}{4}\theta + \frac{3}{4}\sin 2\theta - \sin\theta \right]_0^{\frac{\pi}{3}} = \frac{5}{12}\pi - \frac{\sqrt{3}}{8} \end{aligned}$$

6.

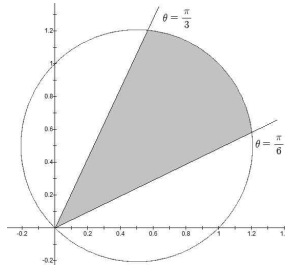


$$\begin{aligned}
S &= 2 \times \left[\frac{1}{2} \int_0^{\frac{\pi}{3}} \sin^2 \theta d\theta + \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin^2 2\theta d\theta \right] \\
&= \int_0^{\frac{\pi}{3}} \frac{1 - \cos 2\theta}{2} d\theta + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1 - \cos 4\theta}{2} d\theta \\
&= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{3}} + \frac{1}{2} \left[\theta - \frac{1}{4} \sin 4\theta \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} \\
&= \frac{1}{2} \left(\frac{\pi}{3} - \frac{1}{2} \times \frac{\sqrt{3}}{2} \right) + \frac{1}{2} \left(\frac{\pi}{2} - \frac{\pi}{3} - \frac{1}{4} \times \frac{\sqrt{3}}{2} \right) \\
&= \frac{\pi}{4} - \frac{3\sqrt{3}}{16}
\end{aligned}$$

7. $x^2 + y^2 = x + y$

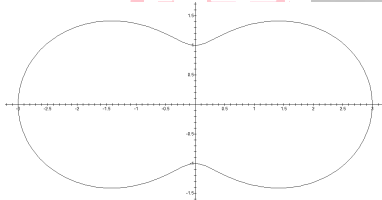
$$\Rightarrow r^2 = r \cos \theta + r \sin \theta \Rightarrow r = \cos \theta + \sin \theta$$

이므로 그림을 그려보면 아래와 같다.



$$\begin{aligned}
S &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\cos \theta + \sin \theta)^2 d\theta = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (1 + 2 \sin \theta \cos \theta) d\theta \\
&= \frac{1}{2} \left[\theta + \sin^2 \theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \frac{\pi}{12} + \frac{1}{4}
\end{aligned}$$

8. $r = 2 + \cos^2 \theta - \sin^2 \theta = 2 + \cos 2\theta$ 의 그래프는 다음과 같다.



$$\begin{aligned}
S &= 4 \times \frac{1}{2} \int_0^{\frac{\pi}{2}} (2 + \cos 2\theta)^2 d\theta \\
&= 2 \int_0^{\frac{\pi}{2}} (4 + 4 \cos 2\theta + \cos^2 2\theta) d\theta \\
&= 2 \left[4\theta + 2 \sin 2\theta \right]_0^{\frac{\pi}{2}} + \int_0^{\pi} \cos^2 t dt \quad (\because 2\theta = t) \\
&= 4\pi + 2 \times \frac{1}{2} \times \frac{\pi}{2} = \frac{9}{2} \pi
\end{aligned}$$

9. $r = 2 + \sin \theta + \cos \theta = 2 + \sqrt{2} \sin \left(\theta + \frac{\pi}{4} \right)$ 의 그래프는

$r = 2 + \sqrt{2} \sin \theta$ 의 그래프를 시계방향으로 $\frac{\pi}{4}$ 만큼 회전한 모양이다.

따라서 넓이는 $r = 2 + \sqrt{2} \sin \theta$ 와 동일하며

$$S = \frac{\pi}{2} (2a^2 + b^2) = \frac{\pi}{2} (2 \times 4 + 2) = 5\pi \text{ 이다.}$$

[다른 풀이]

$$\begin{aligned}
S &= 2 \times \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (2 + \sqrt{2} \sin \theta)^2 d\theta \\
&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4 + 4\sqrt{2} \sin \theta + 2 \sin^2 \theta d\theta \\
&= 4\pi + 0 + 2 \times 2 \times \frac{1}{2} \times \frac{\pi}{2} \\
&= 5\pi
\end{aligned}$$

$$\begin{aligned}
10. l &= \int_1^{e^4} \sqrt{1 + (y')^2} dx = \int_1^{e^4} \sqrt{1 + \left(2x - \frac{1}{8x} \right)^2} dx \\
&= \int_1^{e^4} \sqrt{4x^2 + \frac{1}{2} + \frac{1}{64x^2}} dx = \int_1^{e^4} \sqrt{\left(2x + \frac{1}{8x} \right)^2} dx \\
&= \int_1^{e^4} 2x + \frac{1}{8x} dx = \left[x^2 + \frac{1}{8} \ln x \right]_1^{e^4} = e^8 - \frac{1}{2}
\end{aligned}$$

$$\begin{aligned}
11. l &= \int_0^1 \sqrt{(x')^2 + (y')^2 + (z')^2} dt \\
&= \int_0^1 \sqrt{(2t)^2 + (-\sin t)^2 + (\cos t)^2} dt \\
&= \int_0^1 \sqrt{4t^2 + 1} dt \\
&= \frac{1}{2} \int_0^{\tan^{-1} 2} \sec^3 \theta d\theta \quad (\because 2t = \tan \theta) \\
&= \frac{1}{2} \times \frac{1}{2} [\sec \theta \tan \theta + \ln(\sec \theta + \tan \theta)]_0^{\tan^{-1} 2} \\
&= \frac{1}{4} [2\sqrt{5} + \ln(\sqrt{5} + 2)]
\end{aligned}$$

12. $0 \leq \theta \leq \frac{\pi}{2}$ 일 때, 극곡선 $r = 1 - \sin \theta$ 의 길이는 $(4 - 2\sqrt{2})a = (4 - 2\sqrt{2}) \times 1$ 이다.

[다른 풀이1]

$$\begin{aligned}
l &= \int_0^{\frac{\pi}{2}} \sqrt{r^2 + (r')^2} d\theta = \int_0^{\frac{\pi}{2}} \sqrt{(1 - \sin \theta)^2 + \cos^2 \theta} d\theta \\
&= \int_0^{\frac{\pi}{2}} \sqrt{2 - 2 \sin \theta} d\theta \\
&= \int_0^{\frac{\pi}{2}} \sqrt{2 - 2 \cos t} dt \quad (\because \theta = \frac{\pi}{2} - t) \\
&= \int_0^{\frac{\pi}{2}} \sqrt{2^2 \sin^2 \frac{t}{2}} dt \\
&= \int_0^{\frac{\pi}{2}} 2 \sin \frac{t}{2} dt = \left[-4 \cos \frac{t}{2} \right]_0^{\frac{\pi}{2}} = 4 - 2\sqrt{2}
\end{aligned}$$

[다른 풀이2]

$$\begin{aligned}
0 \leq \theta \leq \frac{\pi}{2} \text{ 일 때, 극곡선 } r &= 1 - \sin \theta \text{ 의 길이는} \\
0 \leq \theta \leq \frac{\pi}{2} \text{ 일 때, 극곡선 } r &= 1 - \cos \theta \text{ 의 길이와 동일하다.} \\
l &= \int_0^{\frac{\pi}{2}} \sqrt{r^2 + (r')^2} d\theta = \int_0^{\frac{\pi}{2}} \sqrt{(1 - \cos \theta)^2 + \sin^2 \theta} d\theta \\
&= \int_0^{\frac{\pi}{2}} \sqrt{2 - 2 \cos \theta} d\theta = \int_0^{\frac{\pi}{2}} \sqrt{2^2 \sin^2 \frac{\theta}{2}} d\theta \\
&= \int_0^{\frac{\pi}{2}} 2 \sin \frac{\theta}{2} d\theta = \left[-4 \cos \frac{\theta}{2} \right]_0^{\frac{\pi}{2}} = 4 - 2\sqrt{2}
\end{aligned}$$

13. $x^2 + y^2 = \sqrt{x^2 + y^2} + x$

$$\Rightarrow r^2 = r + r \cos \theta$$

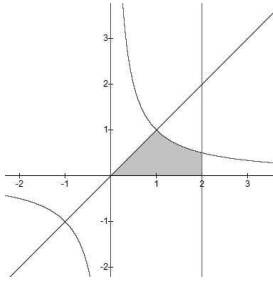
$$\Rightarrow r = 1 + \cos \theta$$

이므로 곡선의 길이는 $2\sqrt{2}a = 2\sqrt{2} \times 1$ 이다.

[다른 풀이]

$$\begin{aligned}
l &= \int_0^{\frac{\pi}{2}} \sqrt{r^2 + (r')^2} d\theta = \int_0^{\frac{\pi}{2}} \sqrt{(1 + \cos \theta)^2 + \sin^2 \theta} d\theta \\
&= \int_0^{\frac{\pi}{2}} \sqrt{2 + 2 \cos \theta} d\theta = \int_0^{\frac{\pi}{2}} \sqrt{2^2 \cos^2 \frac{\theta}{2}} d\theta \\
&= \int_0^{\frac{\pi}{2}} 2 \cos \frac{\theta}{2} d\theta = \left[4 \sin \frac{\theta}{2} \right]_0^{\frac{\pi}{2}} = 2\sqrt{2}
\end{aligned}$$

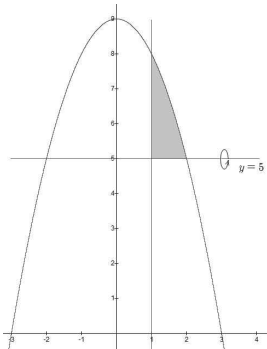
14.



$$V_x = \pi \int_0^1 x^2 dx + \pi \int_1^2 \frac{1}{x^2} dx$$

$$= \pi \times \frac{1}{3} + \pi \left[-\frac{1}{x} \right]_1^2 = \frac{5\pi}{6}$$

15.

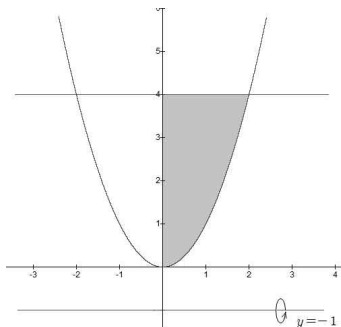


$$V = \pi \int_1^2 (y-5)^2 dx = \pi \int_1^2 (9-x^2-5)^2 dx$$

$$= \pi \int_1^2 (4-x^2)^2 dx = \pi \int_1^2 16-8x^2+x^4 dx$$

$$= \pi \left[16x - \frac{8}{3}x^3 + \frac{1}{5}x^5 \right]_1^2 = \frac{53}{15}\pi$$

16.

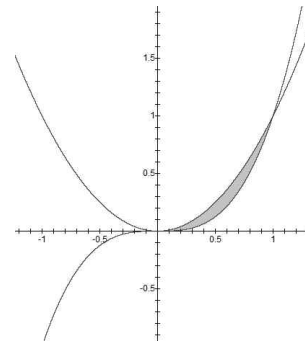


$$V = \pi \int_0^2 5^2 - (y+1)^2 dx = \pi \int_0^2 25 - (x^2+1)^2 dx$$

$$= \pi \int_0^2 24 - 2x^2 - x^4 dx = \pi \left[24x - \frac{2}{3}x^3 - \frac{1}{5}x^5 \right]_0^2$$

$$= \pi \left(48 - \frac{16}{3} - \frac{32}{5} \right) = \frac{544}{15}\pi$$

17.



y축 중심으로 회전한 부피는 다음과 같다.

$$V = 2\pi \int_0^1 x(y_1 - y_2) dx = 2\pi \int_0^1 x(x^2 - x^3) dx$$

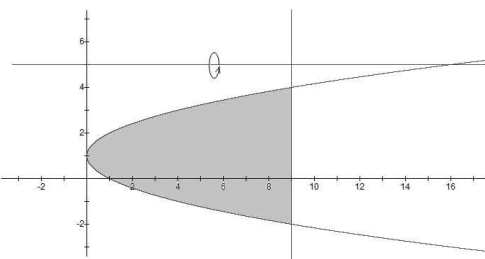
$$= 2\pi \int_0^1 x^3 - x^4 dx = 2\pi \left[\frac{1}{4}x^4 - \frac{1}{5}x^5 \right]_0^1 = 2\pi \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{\pi}{10}$$

x축 중심으로 회전한 부피는 다음과 같다.

$$V_x = \pi \int_0^1 y_1^2 - y_2^2 dx = \pi \int_0^1 x^4 - x^6 dx = \pi \left[\frac{1}{5}x^5 - \frac{1}{7}x^7 \right]_0^1$$

$$= \pi \left(\frac{1}{5} - \frac{1}{7} \right) = \frac{2\pi}{35}$$

18.



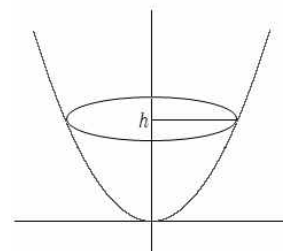
영역의 중심좌표는 $M(*, 1)$ 이므로 축과의 거리는 $k=4$ 이다.

또한 영역의 넓이는 $\frac{|a|}{6}(\beta - \alpha)^3 = \frac{1}{6}(4 - (-2))^3 = 36$ 이다.

파푸스 정리에 의하여 회전체의 부피는

$$V = 2\pi k \times \text{넓이} = 2\pi \times 4 \times 36 = 288\pi \text{ 이다.}$$

19.



$$\frac{dV}{dt} = 2, \quad h = 9 \text{ 일 때}$$

$$V = \pi \int_0^h x^2 dy = \pi \int_0^h y dy \text{ 이므로 } \frac{dV}{dt} = \pi h \times \frac{dh}{dt} \text{ 이다.}$$

$$\text{즉, } 2 = 9\pi \frac{dh}{dt} \text{ 이므로 } \frac{dh}{dt} = \frac{2}{9\pi} (\text{cm/sec}) \text{ 이다.}$$

$$20. V = \int_{-1}^1 \text{단면의 넓이} dy = \int_{-1}^1 \pi x^2 dy$$

$$= \pi \int_{-1}^1 \frac{1}{4}(1-y^2)^2 dy = \frac{\pi}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 \theta d\theta \quad (\because y = \sin \theta)$$

$$= \frac{\pi}{4} \times 2 \times \frac{4}{5} \times \frac{2}{3} = \frac{4}{15}\pi$$

$$\begin{aligned}
 21. S_x &= 2\pi \int_1^2 y \sqrt{1+(x')^2} dy = 2\pi \int_1^2 y \sqrt{1+\left(\frac{1}{2}y^3 - \frac{1}{2}y^{-3}\right)^2} dy \\
 &= 2\pi \int_1^2 y \sqrt{\frac{1}{4}y^6 + \frac{1}{2} + \frac{1}{4}y^{-6}} dy = 2\pi \int_1^2 y \sqrt{\left(\frac{1}{2}y^3 + \frac{1}{2}y^{-3}\right)^2} dy \\
 &= \pi \int_1^2 y^4 + y^{-2} dy = \pi \left[\frac{1}{5}y^5 - \frac{1}{y} \right]_1^2 = \frac{67}{10}\pi
 \end{aligned}$$

22. $x+y=1$ ($0 \leq x \leq 1$)에 대한 선분의 중심좌표는 $\left(\frac{1}{2}, \frac{1}{2}\right)$ 이므로

중심과 축과의 거리 $k = \frac{1}{2}$ 이다. 또한 선분의 길이는 $\sqrt{2}$ 이다.

즉, 파푸스 정리에 의하여

$$S = 2\pi k \times \text{길이} = 2\pi \times \frac{1}{2} \times \sqrt{2} = \sqrt{2}\pi \text{이다.}$$

[다른 풀이]

$$\begin{aligned}
 S_x &= 2\pi \int_0^{\frac{\pi}{2}} y \sqrt{(x')^2 + (y')^2} dt \\
 &= 2\pi \int_0^{\frac{\pi}{2}} \sin^2 t \sqrt{(-2\cos t \sin t)^2 + (2\sin t \cos t)^2} dt \\
 &= 4\sqrt{2}\pi \int_0^{\frac{\pi}{2}} \sin^3 t \cos t dt \\
 &= 4\sqrt{2}\pi \times \frac{1}{4} [\sin^4 t]_0^{\frac{\pi}{2}} = \sqrt{2}\pi
 \end{aligned}$$

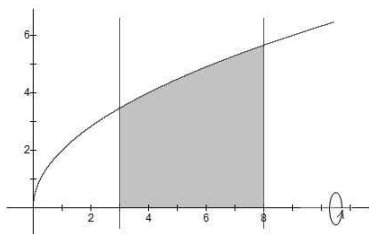
$$\begin{aligned}
 23. S_x &= 2\pi \int_0^{\frac{\pi}{2}} y \sqrt{r^2 + (r')^2} d\theta = 2\pi \int_0^{\frac{\pi}{2}} r \sin \theta \sqrt{r^2 + (r')^2} d\theta \\
 &= 2\pi \int_0^{\frac{\pi}{2}} e^{\theta} \sin \theta \sqrt{e^{2\theta} + e^{2\theta}} d\theta \\
 &= 2\sqrt{2}\pi \int_0^{\frac{\pi}{2}} e^{2\theta} \sin \theta d\theta \\
 &= 2\sqrt{2}\pi \left[-\frac{e^{2\theta}}{5} (2\sin \theta - \cos \theta) \right]_0^{\frac{\pi}{2}} \\
 &= \frac{2\sqrt{2}}{5}\pi (2e^{\pi} + 1)
 \end{aligned}$$

$$\begin{aligned}
 24. S &= 4\pi r^2 \times \frac{\text{구간의 길이}}{\text{지름}} \\
 &= 4\pi \times 9 \times \frac{2 - (-1)}{6} = 18\pi
 \end{aligned}$$

[다른 풀이]

$$\begin{aligned}
 S_x &= 2\pi \int_{-1}^2 y \sqrt{1+(y')^2} dx \\
 &= 2\pi \int_{-1}^2 \sqrt{9-x^2} \sqrt{1+\left(\frac{-x}{\sqrt{9-x^2}}\right)^2} dx \\
 &= 2\pi \int_{-1}^2 3 dx = 2\pi \times 3 \times 3 = 18\pi
 \end{aligned}$$

25.



곡선을 회전하여 나온 겉넓이는 다음과 같다.

$$\begin{aligned}
 S_1 &= 2\pi \int_3^8 y \sqrt{1+(y')^2} dx \\
 &= 2\pi \int_3^8 2\sqrt{x} \sqrt{1+\frac{1}{x}} dx \\
 &= 4\pi \int_3^8 \sqrt{x+1} dx = 4\pi \left[\frac{2}{3}(x+1)^{\frac{3}{2}} \right]_3^8 \\
 &= \frac{152}{3}\pi
 \end{aligned}$$

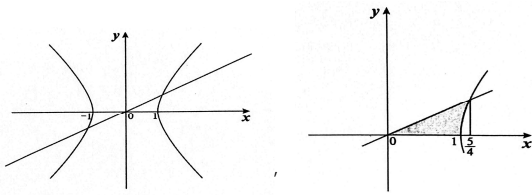
양 끝의 표면은 원의 넓이와 같다.

$$S_2 = \pi \times (2\sqrt{3})^2 = 12\pi, \quad S_3 = \pi \times (2\sqrt{8})^2 = 32\pi$$

즉, 영역을 회전하여 나온 회전체의 겉넓이는 $S_1 + S_2 + S_3 = \frac{284}{3}\pi$ 이다.

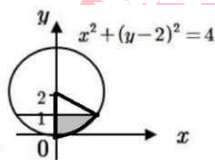
1. 1번	2. 2번	3. 5번	4. 1번	5. 2번
6. 1번	7. 4번	8. 2번	9. 2번	10. 3번
11. 3번	12. 4번	13. 2번	14. 4번	15. 1번
16. 3번	17. 4번			

1. $\begin{cases} x^2 - y^2 = 1 \\ 5y = 3x \end{cases}$ 의 두 그래프를 그려보면,



$$\begin{aligned}
 A &= \frac{1}{2} \times \frac{3}{4} \times \frac{5}{4} - \int_1^{\frac{5}{4}} y dx \\
 &= \frac{15}{32} - \int_0^{\ln 2} \sinh^2 t dt \quad (\because x = \cosh t, y = \sinh t) \\
 &= \frac{15}{32} - \int_0^{\ln 2} \frac{1}{4} (e^{2t} - 2 + e^{-2t}) dt \\
 &= \frac{15}{32} - \frac{1}{4} \left[\frac{1}{2} e^{2t} - 2t - \frac{1}{2} e^{-2t} \right]_0^{\ln 2} \\
 &= \frac{15}{32} - \frac{1}{4} \left(\frac{1}{2} e^{2\ln 2} - 2\ln 2 - \frac{1}{2} e^{-2\ln 2} \right) = \frac{1}{2} \ln 2
 \end{aligned}$$

2.



주어진 영역의 면적은 중심각이 $\frac{\pi}{3}$ 인 부채꼴의 면적에서 작은 삼각형의 면적을 빼면 된다.

$$\therefore \frac{1}{2} \times 2^2 \times \frac{\pi}{3} - \sqrt{3} \times \frac{1}{2} \times \frac{1}{2} = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

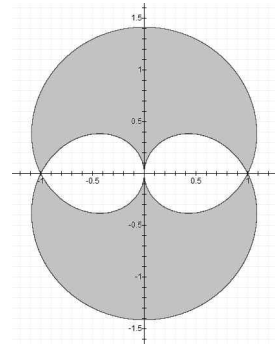
$$\begin{aligned}
 3. \quad 2 \int_{\frac{\pi}{4}}^0 |\cos 2t| \tan t (-2 \sin 2t) dt &= 4 \int_0^{\frac{\pi}{4}} \sin 2t \cos 2t \tan t dt \\
 &= 4 \int_0^{\frac{\pi}{4}} (2 \sin t \cos t) (1 - 2 \sin^2 t) \times \frac{\sin t}{\cos t} dt \\
 &= 8 \int_0^{\frac{\pi}{4}} \sin^2 t (1 - 2 \sin^2 t) dt = 8 \int_0^{\frac{\pi}{4}} (\sin^2 t - 2 \sin^4 t) dt
 \end{aligned}$$

$$1) \int_0^{\frac{\pi}{4}} \sin^2 t dt = \left[\frac{t}{2} - \frac{1}{4} \sin 2t \right]_0^{\frac{\pi}{4}} = \frac{\pi}{8} - \frac{1}{4}$$

$$\begin{aligned}
 2) \int_0^{\frac{\pi}{4}} \sin^4 t dt &= \int_0^{\frac{\pi}{4}} \left(\frac{1 - \cos 2t}{2} \right)^2 dt = \int_0^{\frac{\pi}{4}} \frac{1}{4} (1 - 2 \cos 2t + \cos^2 2t) dt \\
 &= \frac{1}{4} \left[t - \sin 2t + \frac{t}{2} + \frac{1}{8} \sin 4t \right] \\
 &= \frac{1}{4} \left(\frac{\pi}{4} - 1 + \frac{\pi}{8} \right) = \frac{1}{4} \left(\frac{3}{8} \pi - 1 \right)
 \end{aligned}$$

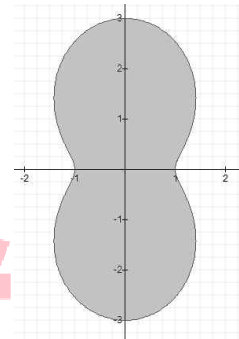
$$\text{즉, } A = 8 \left\{ \frac{\pi}{8} - \frac{1}{4} - 2 \frac{1}{4} \left(\frac{3}{8} \pi - 1 \right) \right\} = 2 - \frac{\pi}{2}$$

4.



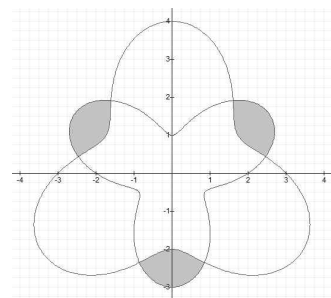
$$\begin{aligned}
 4 \times \left(\frac{1}{2} \int_0^{\frac{\pi}{2}} 1 + \sin \theta d\theta - \frac{1}{2} \int_{\pi}^{\frac{3\pi}{2}} 1 + \sin \theta d\theta \right) \\
 = 2 \left([\theta - \cos \theta]_0^{\frac{\pi}{2}} - [\theta - \cos \theta]_{\pi}^{\frac{3\pi}{2}} \right) = 4
 \end{aligned}$$

5.



$$\begin{aligned}
 \frac{1}{2} \int_0^{2\pi} (2 - \cos 2\theta)^2 d\theta \\
 = \frac{1}{2} \int_0^{2\pi} 4 - 4 \cos 2\theta + \cos^2 2\theta d\theta \\
 = \frac{1}{2} \int_0^{2\pi} \frac{9}{2} - 4 \cos 2\theta + \frac{1}{2} \cos 4\theta d\theta \\
 = \frac{1}{2} \left[\frac{9}{2} \theta - 2 \sin 2\theta + \frac{1}{8} \sin 4\theta \right]_0^{2\pi} = \frac{9\pi}{2}
 \end{aligned}$$

6.



$$\begin{aligned}
 \left[\frac{1}{2} \int_{\frac{\pi}{18}}^{\frac{5}{18}\pi} \{ (2 + \sin 3\theta)^2 - (3 - \sin 3\theta)^2 \} d\theta \right] \times 3 \\
 = \frac{3}{2} \int_{\frac{\pi}{18}}^{\frac{5}{18}\pi} \{ 4 + 4 \sin 3\theta + \sin^2 3\theta - (9 - 6 \sin 3\theta + \sin^2 3\theta) \} d\theta \\
 = \frac{3}{2} \int_{\frac{\pi}{18}}^{\frac{5}{18}\pi} (-5 + 10 \sin 3\theta) d\theta = \frac{3}{2} \left[-5\theta - \frac{10}{3} \cos 3\theta \right]_{\frac{\pi}{18}}^{\frac{5}{18}\pi} \\
 = 5\sqrt{3} - \frac{5}{3}\pi
 \end{aligned}$$

$$7. y = \sqrt[3]{2-x^3} \Leftrightarrow r = \frac{\sqrt[3]{2}}{(\cos^3\theta + \sin^3\theta)^{\frac{1}{3}}} \text{ 이므로}$$

$$s(t) = \frac{1}{2} \int_0^{\tan^{-1}t} \frac{2^{\frac{2}{3}}}{(\cos^3\theta + \sin^3\theta)^{\frac{2}{3}}} d\theta \text{ 이다.}$$

$$s'(t) = \frac{1}{\sqrt[3]{2}} \frac{1}{\{\cos^3(\tan^{-1}t) + \sin^3(\tan^{-1}t)\}^{\frac{2}{3}}} \times \frac{1}{1+t^2}$$

$$\therefore s'(1) = \frac{1}{2}$$

$$8. r = \frac{2}{2+\cos\theta} \Leftrightarrow 2\sqrt{x^2+y^2} = 2-x$$

$$\Leftrightarrow 3x^2+4x+4y^2=4 \Leftrightarrow \frac{\left(x+\frac{2}{3}\right)^2}{\left(\frac{4}{3}\right)^2} + \frac{y^2}{\left(\frac{2}{\sqrt{3}}\right)^2} = 1$$

$$\text{타원의 방정식의 면적 : } \frac{2}{\sqrt{3}} \times \frac{4}{3} \times \pi = \frac{8\pi}{3\sqrt{3}}$$

$$9. L = \int_0^{\frac{\pi}{3}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$= \int_0^{\frac{\pi}{3}} \sqrt{(-\sin t)^2 + (-2\sin 2t)^2} dt$$

$$= \int_0^{\frac{\pi}{3}} \sin t \sqrt{1+16\cos^2 t} dt = \int_1^{\frac{1}{2}} \sqrt{1+16x^2} dx$$

$$\text{이므로 하합 : } \frac{\sqrt{5}}{2}, \text{ 상합 : } \frac{\sqrt{17}}{2} \text{ 이다.}$$

$$\therefore \frac{\sqrt{5}}{2} \leq L \leq \frac{\sqrt{17}}{2}$$

$$10. x^2 + (y-a)^2 \leq 1 - \frac{1}{4}a \Leftrightarrow \text{중심 : } (0, a), \text{ 반지름 : } \sqrt{1 - \frac{1}{4}a}$$

$$\text{Pappus 정리를 이용하면 } V_a = \left(1 - \frac{1}{4}a\right) \times 2\pi \times a = 2\pi \left(a - \frac{1}{4}a^2\right) \text{ 이다.}$$

$$V'_a = 2\pi \left(1 - \frac{1}{2}a\right) \text{ 이므로 } a=2 \text{ 일 때, 부피는 최대값을 갖는다.}$$

$$11. \text{Pappus 정리를 이용하면 } V = 2\pi \times \pi r \times 3\pi r^2 = 6\pi^3 r^3 \text{ 이다.}$$

$$12. V_1 = \int_0^1 \pi \{1^2 - (\sqrt[4]{x})^2\} dx = \pi \int_0^1 1 - \sqrt{x} dx = \frac{\pi}{3}$$

$$V_2 = \pi \int_0^1 \{y^2 - (y^4)^2\} dy = \frac{2}{9}\pi$$

$$\text{이므로 } V_1 : V_2 = \frac{\pi}{3} : \frac{2\pi}{9} = 3 : 2 \text{ 이다.}$$

$$13. V_h = \pi \int_0^1 r^2 dh = \pi \int_0^1 (1-h)^4 dh = \frac{\pi}{5}$$

$$\therefore \frac{\pi}{3} - \frac{\pi}{5} = \frac{2\pi}{15}$$

$$14. 1 \leq x \leq 3 \text{ 에 대하여 } x^2 + y^2 = 16 \text{ 을 } x \text{ 축 중심으로 회전한 부피와 같다.}$$

$$V_x = \pi \int_1^3 y^2 dx = \pi \int_1^3 16 - x^2 dx$$

$$= \pi \left[16x - \frac{1}{3}x^3 \right]_1^3 = \frac{70}{3}\pi$$

$$15. V = \pi \int_c^d x^2 dy = \pi \int_0^4 t^3 \times \frac{1}{2} t^{-\frac{1}{2}} dt$$

$$= \frac{\pi}{2} \int_0^4 t^{\frac{5}{2}} dt = \frac{\pi}{2} \times \frac{2}{7} \left[t^{\frac{7}{2}} \right]_0^4 = \frac{2^7}{7}\pi$$

$$16. r = 2\sin\theta, -\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4} \text{ 에 대하여 } x \text{ 축 회전한 겉넓이와 같다.}$$

$$S_x = 2\pi \int_c^d r \sin\theta \sqrt{r^2 + (r')^2} d\theta$$

$$= 2\pi \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 2\sin^2\theta \sqrt{4\sin^2\theta + 4\cos^2\theta} d\theta$$

$$= 16\pi \int_0^{\frac{\pi}{4}} \sin^2\theta d\theta = 8\pi \int_0^{\frac{\pi}{4}} 1 - \cos 2\theta d\theta$$

$$= 8\pi \left[\theta - \frac{1}{2}\sin 2\theta \right]_0^{\frac{\pi}{4}} = 2\pi(\pi - 2)$$

$$17. S_x = 2\pi \int_c^d r \sin\theta \sqrt{r^2 + (r')^2} d\theta$$

$$= 2\pi \int_0^{\frac{\pi}{2}} (1 + \cos\theta) \sin\theta \sqrt{(1 + \cos\theta)^2 + \sin^2\theta} d\theta$$

$$= 2\sqrt{2}\pi \int_0^{\frac{\pi}{2}} \sin\theta (1 + \cos\theta)^{\frac{3}{2}} d\theta$$

$$= -2\sqrt{2}\pi \times \frac{2}{5} \left[(1 + \cos\theta)^{\frac{5}{2}} \right]_0^{\frac{\pi}{2}} = \frac{4\sqrt{2}}{5}\pi(4\sqrt{2} - 1)$$